A modern data and AI platform

An end-to-end overview of the data and AI life cycle, based on a financial-sector use case

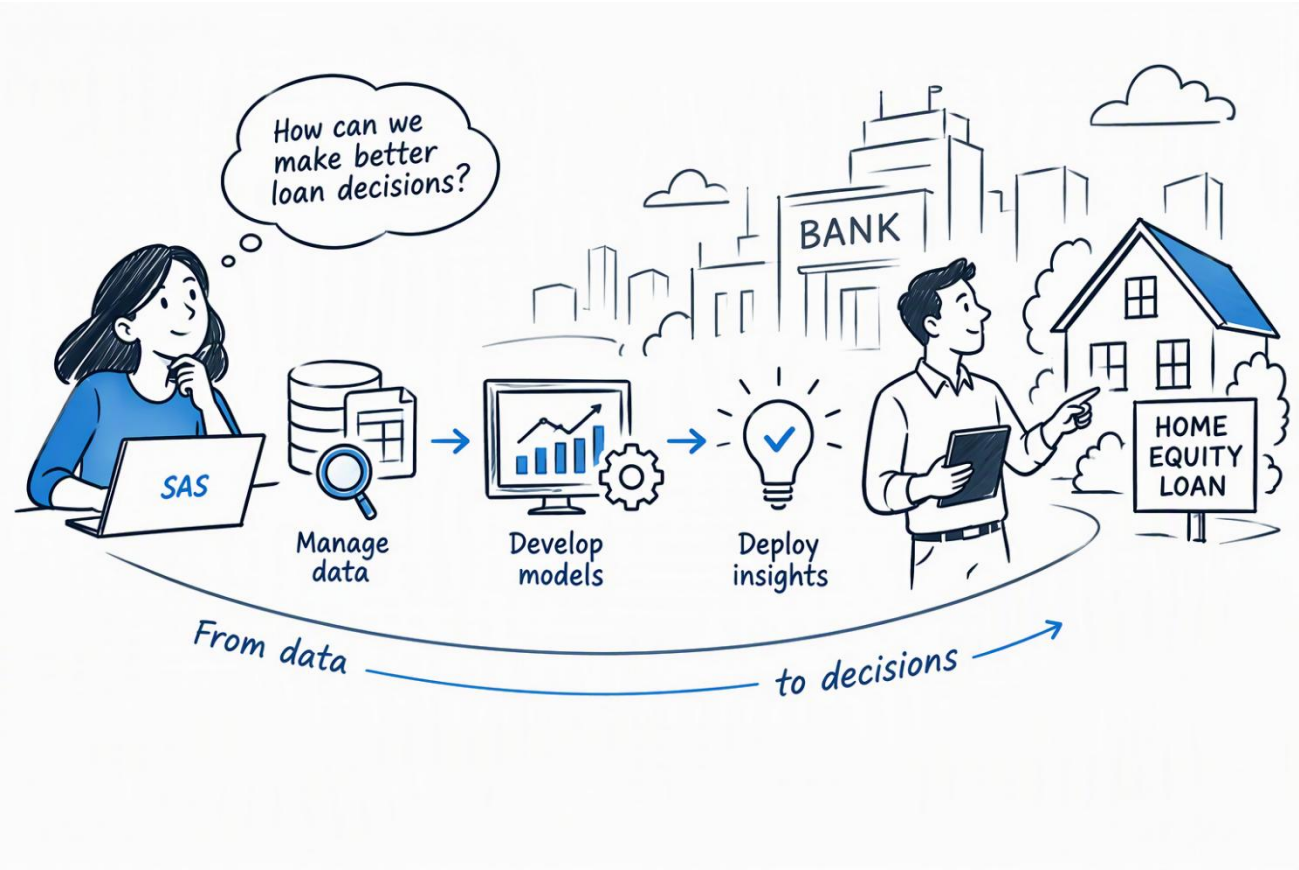


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This document was written by Jos van der Velden.

This guide provides step-by-step instructions that introduce the end-to-end data and AI life cycle through a real-world use case from the financial sector. You can complete the tasks during a hands-on workshop or classroom session using SAS Viya for Learners.

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Introduction to the data and AI lifecycle

This section briefly describes the data and AI life cycle and its three stages. The data and AI life cycle, as shown in Figure 1, illustrates the process of starting with a question to reach an actionable decision and has three stages.

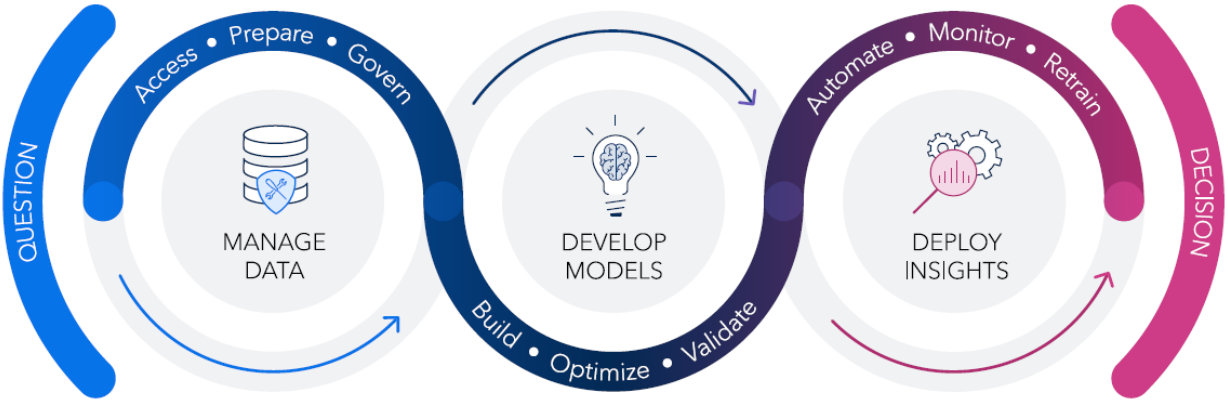


Figure 1: The data and AI life cycle

The three main stages of the data and AI life cycle

The three stages are manage data, develop models and deploy insights. The tools and platforms used by data and AI practitioners help them move through these stages.

SAS Viya for Learners (VFL) gives educators and students one environment in which they can work through all three stages. This use case introduces each stage. A workshop or classroom session might not provide enough time to explore every feature in depth, so the guide also links to courses and activities for further practice.

For an introductory video series about the data and AI life cycle, watch the SAS Viya Quick Start Tutorials. You can find them on YouTube via this link.

Manage data



Figure 2: The manage data stage and its graphical VFL user interfaces

The first stage in the data and AI life cycle is manage data. This stage focuses on accessing, exploring, governing, and preparing raw data so it can be trusted and used effectively for analytics and AI. By improving the quality, context, and usability of data, organizations can build greater data intelligence and make more informed decisions.

The manage data stage is typically handled by data engineers, analytics engineers, and data analysts. They are responsible for transforming raw data into trusted, analytics-ready assets by profiling, governing, and preparing data. This work makes the data suitable for reporting, analytics, AI, and decision-making.

Develop models



Figure 3: The develop models stage and its graphical VFL user interfaces

The second stage in the data and AI life cycle is develop models. This stage focuses on exploring prepared data, selecting appropriate techniques and building models that can be used to answer business questions or make predictions.

The develop models stage is most commonly handled by data scientists or advanced analysts. These personas are responsible for applying statistical and machine learning techniques to data, validating results and translating model outcomes into insights that can be operationalized or shared with the business.

Deploy insights

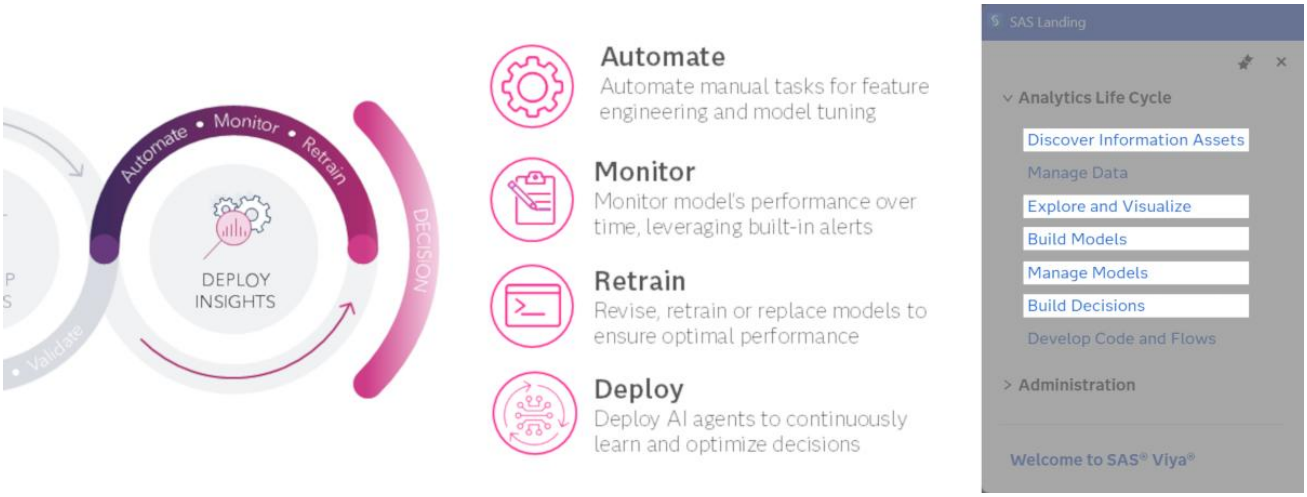


Figure 4: The deploy insights stage and its graphical VFL user interfaces

The final stage in the data and AI life cycle is deploy insights. This stage focuses on operationalizing analytics at scale, putting models and decision logic into production so insights can drive consistent, real-world actions.

The deploy insights stage typically involves two key personas: MLOps engineers, who are responsible for deploying models, managing versions, monitoring performance and maintaining reliability across environments; and business analysts or decision designers, who design, test and govern decision flows that translate analytics into clear, consistent business actions.

About SAS Viya and SAS Viya for Learners

SAS Viya is SAS's modern analytics platform for working with data, statistics, machine learning and artificial intelligence. It provides an integrated environment where you can explore data, build analytical models and turn data into useful insights.

SAS Viya for Learners gives students and educators access to SAS Viya in a learning environment. It allows you to gain hands-on experience with SAS technology and apply what you learn in courses, exercises and projects.

Business scenario

Throughout this use case, one business scenario demonstrates the complete data and AI life cycle, from accessing data to designing a decision flow.

Some background information and your role

Imagine the following story. You work as a data analyst at a large bank. The bank offers home equity loans to customers who use the value of their home as collateral. Leadership wants to improve loan approval decisions to reduce default risk and maximize profitability.

Because this use case is about lending decisions, we must also think about responsible AI. A model may help estimate risk, but it should not be used blindly. We need to consider fairness, transparency, data quality, legal requirements, and possible bias in the data.

Your objectives are to evaluate past loan performance, identify factors associated with repayment, build a model that estimates the likelihood of repayment, and use the model to support future lending decisions.

Key questions

- Which customer characteristics are most predictive of repayment?
- How can we balance approval rates with risk management?
- What metrics should be used to evaluate model performance?

The data is stored in a table named HOMEEQUITY. The table includes information about the applicant, loan details, and repayment result.

As you work through the data and AI life cycle, we will use applications and features available in a modern data and AI platform to answer these key questions and achieve our primary objectives.

The home equity dataset

During the practice, you import the HOMEEQUITY dataset from a comma-seperated value (CSV) file. The import will be done using the HTTP procedure and will import the file from this URL. The file is stored on the SAS support website. The text box shows the first three lines of the file.

```
BAD,LOAN,MORTDUE,VALUE,REASON,JOB,YOJ,DEROG,DELINQ,CLAGE,NINQ,CLNO,DEBTINC,APPDATE,CITY,STATE,DIVISION,REGION
1,1100,25860,39025,HomeImp,Other,10.5,0,0,94.366666667,1,9,,22040,oregon,Wisconsin,East North Central,Midwest
1,1300,70053,68400,HomeImp,Other,7,0,2,121.83333333,0,14,,22229,churchton,Maryland,South Atlantic,South
```

A data dictionary is a useful document or table that outlines the structure, content, and variable definitions for a dataset or collection of data. A data dictionary is a critical tool for reproducibility because it allows others to understand your data.

In legacy systems, datasets often have cryptic names for the tables and variables. In practice, especially when you are asked to work with an existing dataset from a legacy system, a data dictionary is a valuable. If it doesn't exist, you should create one. Often this involves interviewing the dataset owner or a long-term employee/colleague and can require serious time investment.

Table 1: Data dictionary for the HOMEEQUITY dataset

Name	Label	Model role	Measurement level	Explanation
BAD	Loan Status	Target	Binary	1 = applicant defaulted on loan or is seriously delinquent; 0 = applicant is current on loan payments
CLAGE	Age of oldest credit line in months	Input	Interval	Number of months since the applicant's oldest credit line was opened.
CLNO	Number of credit lines	Input	Interval	
DEBTINC	Debt-to-income ratio	Input	Interval	
DELINQ	Number of delinquent credit lines	Input	Interval	
DEROG	Number of derogatory reports	Input	Interval	
JOB	Occupational category	Input	Nominal	
LOAN	Requested loan amount	Input	Interval	
MORTDUE	Amount due on existing mortgage	Input	Interval	
NINQ	Number of recent credit inquiries	Input	Interval	
REASON	Reason for the loan request	Input	Binary	DebtCon = debt consolidation; HomeImp = home improvement
VALUE	Value of current property	Input	Interval	
YOJ	Years at present job	Input	Interval	
APPDATE	Application date	Rejected		
CITY *	City	Rejected		
STATE *	State	Rejected		
DIVISION *	Division	Rejected		
REGION *	Region	Rejected		

*) These columns were added to the original dataset and filled with sample data, to support 1) the creation of a geographical hierachy, and 2) the use of a geograhical map in a visual analytics report or dashboard.

Prerequisites

This document assumes you have gone through the steps demonstrated and explained in the video playlist **Getting started with SAS for students**. Before you continue, make sure that you have:

1. Installed and configured a browser for use with SAS' academic resources. This browser, either Google Chrome or Mozilla Firefox, must be configured with English as the primary language. Automatic or AI assisted translations should be blocked or disabled in the browser settings. The videos in the playlist explain why this is essential!
2. Created a profile for accessing SAS resources with your academic email.
3. Access to **SAS Skill Builder for Students**.
4. Access to **SAS Viya for Learners**.

NOTE: If you have not completed these four steps, complete them before you continue!

The screenshots and screen excerpts in this guide were captured while using SAS Studio as a Progressive Web App (PWA). The PWA can provide improved session persistence, faster performance, a desktop-like experience, and support for multiple instances. You can install the PWA by using the install this app option in your browser address bar.

Task 0: Customize the VFL environment

To ensure that everyone uses a similar SAS Viya for Learners environment, first change several user settings. These settings affect features used later in the exercises. If you skip them, some steps might not work as described.

Some tasks require you to enter text or open a webpage. Use the file HOMEEQUITYuseCASEhelpfile.txt if you prefer to copy and paste the required text.

Step-by-step instructions

1. Go to https://learn.sas.com/my . This page is also known as **My Training**.

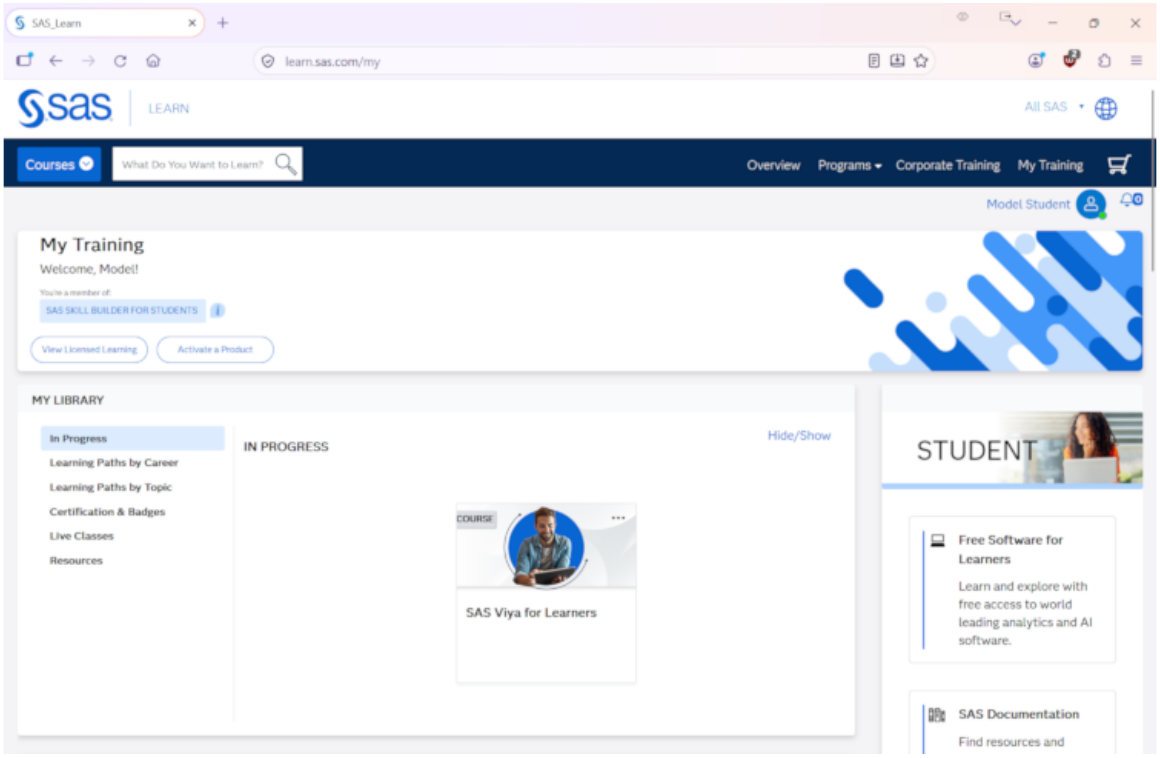
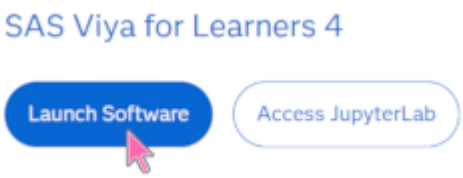


Figure 5: My Training

2. Click **SAS Viya for Learners**.
3. Click **Launch Software**.



4. In the upper-right corner of the **SASLanding** page, select your profile icon. It appears next to the Search, Notifications, and Help icons to open the options menu for the **SASLanding** page. In the illustration below this is the **M**.

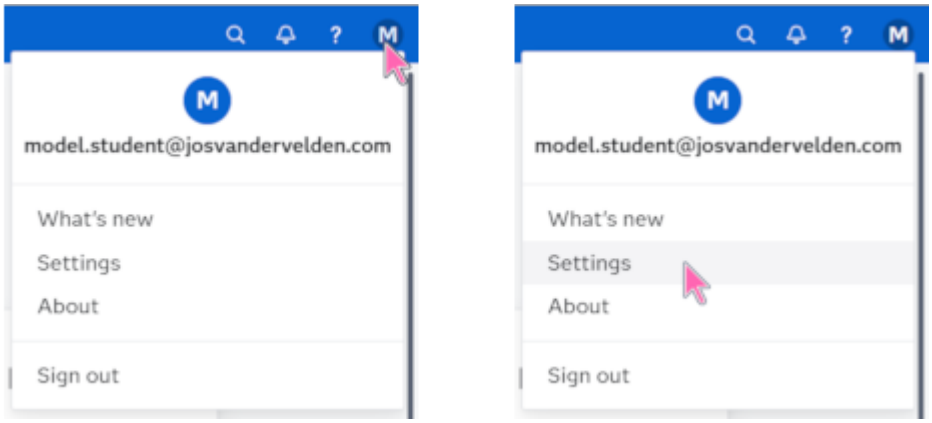


Figure 7: Open the GUI options menu and select **Settings**.

5. Select **Settings**.
6. Under **Global** select **Region and language**.

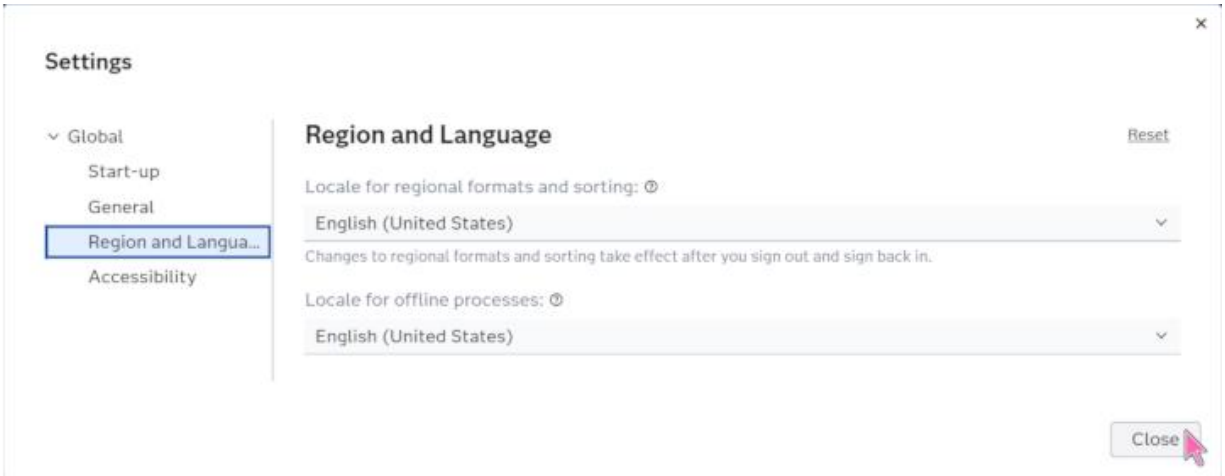


Figure 8: Customizing the language settings

7. Select **English (United States)** for both fields:
   a. **Locale for regional formats and sorting:**; and
   b. **Locale for offline processes:**.
8. Click **Close**.

9. Open **Applications menu** (nine dots).

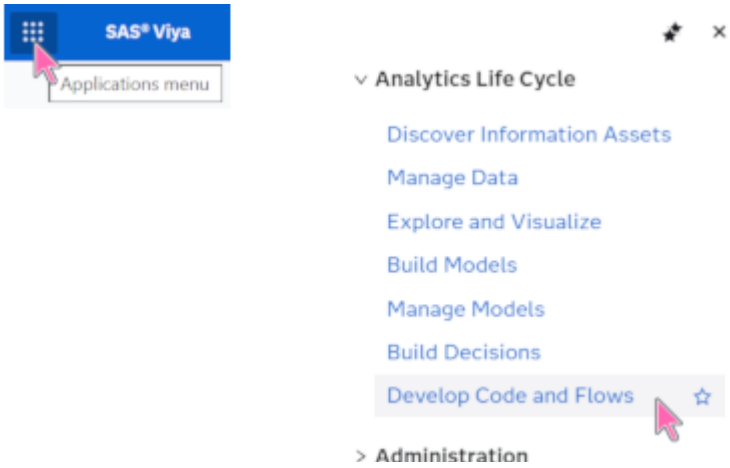


Figure 9: Open **SAS Studio** *GUI via the applications menu option* **Develop Code and Flows**.

10. Select **Develop Code and Flows**.
11. Click the letter on the top right of the page to open the options menu. This is similar to step **4**.
12. Click **Settings** to review the settings for **SAS Studio**.
13. Under **SAS Programs** select **Code and Log**.

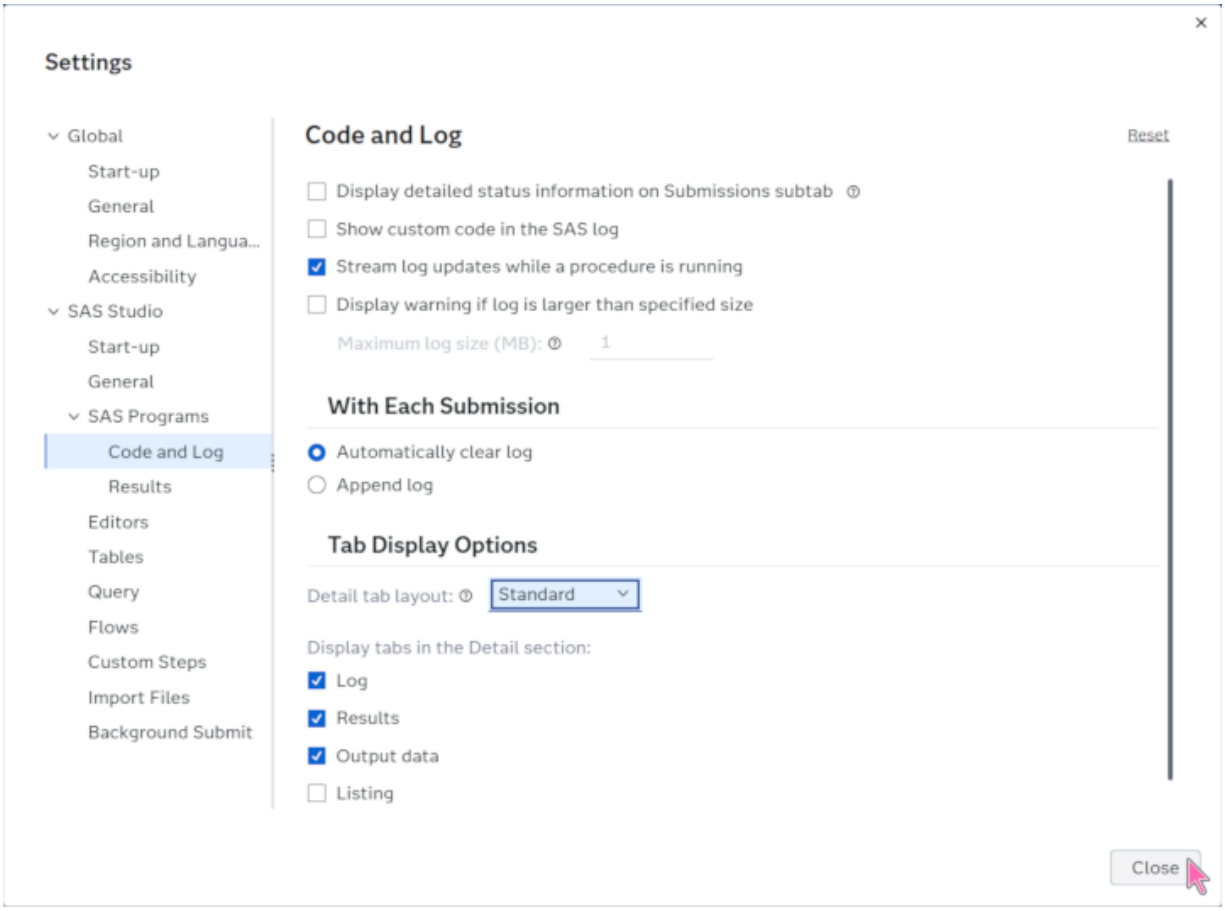


Figure 10: Change the detail tab layout to optimize workspace for SAS Studio.

14. Change **Detail tab layout**: to **Standard**.
15. Click **Close**.

16. Open **Applications menu** (nine dots).

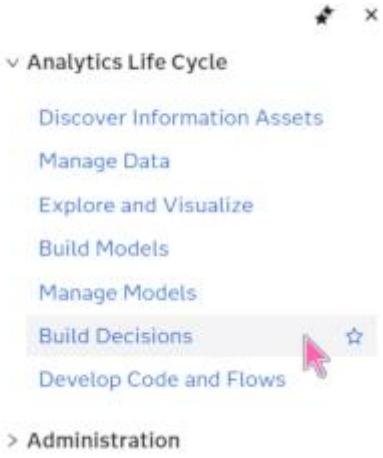


Figure 11: Open **SAS Intelligent Decisioning** *GUI via the applications menu option* **Build Decisions**.

17. Select **Build Decisions**.
18. Click the letter on the top right of the page to open the options menu. This is similar to step **4**.
19. Click **Settings** to review the settings for **SAS Intelligent Decisioning**.
20. Under **SAS Intelligent Decisioning** select **General**.

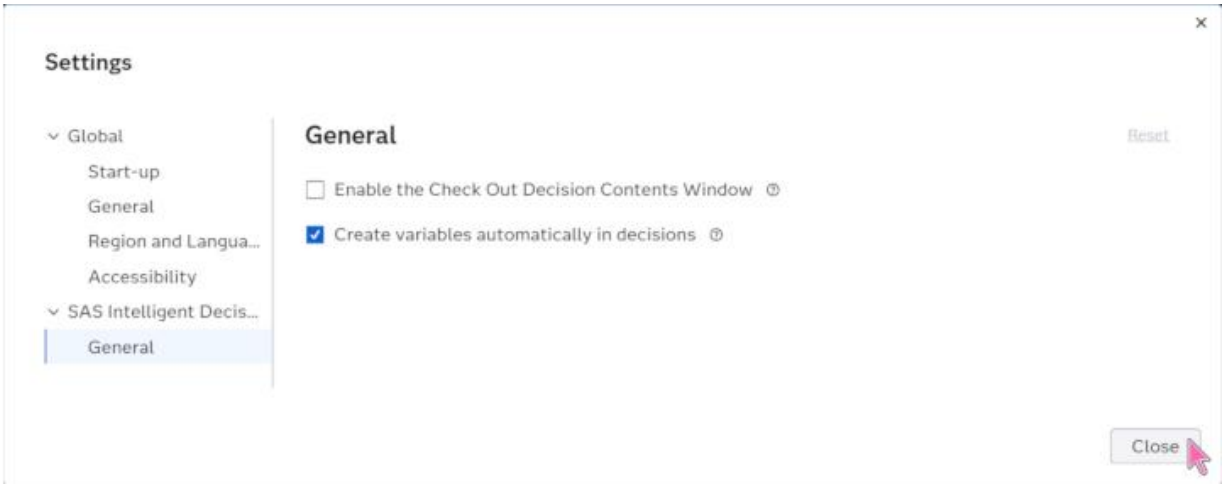


Figure 12: Enable automatic creation of variables used in decision flows.

21. Check the box for **Create variables automatically in decisions**.
22. Click **Close**.
23. Open **Applications menu** (nine dots).

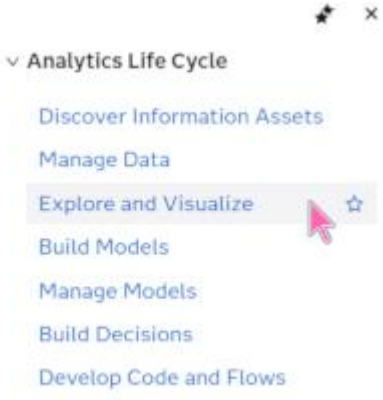


Figure 13: Open **SAS Visual Analytics** *GUI via the applications menu item* **Explore and Visualize**.

24. Select **Explore and Visualize**.
25. Click the letter on the top right of the page to open the options menu. This is similar to step **4**.
26. Click **Settings** to customize the settings for **SAS Visual Analytics**.
27. Under **SAS Visual Analytics** select **Modeling**.

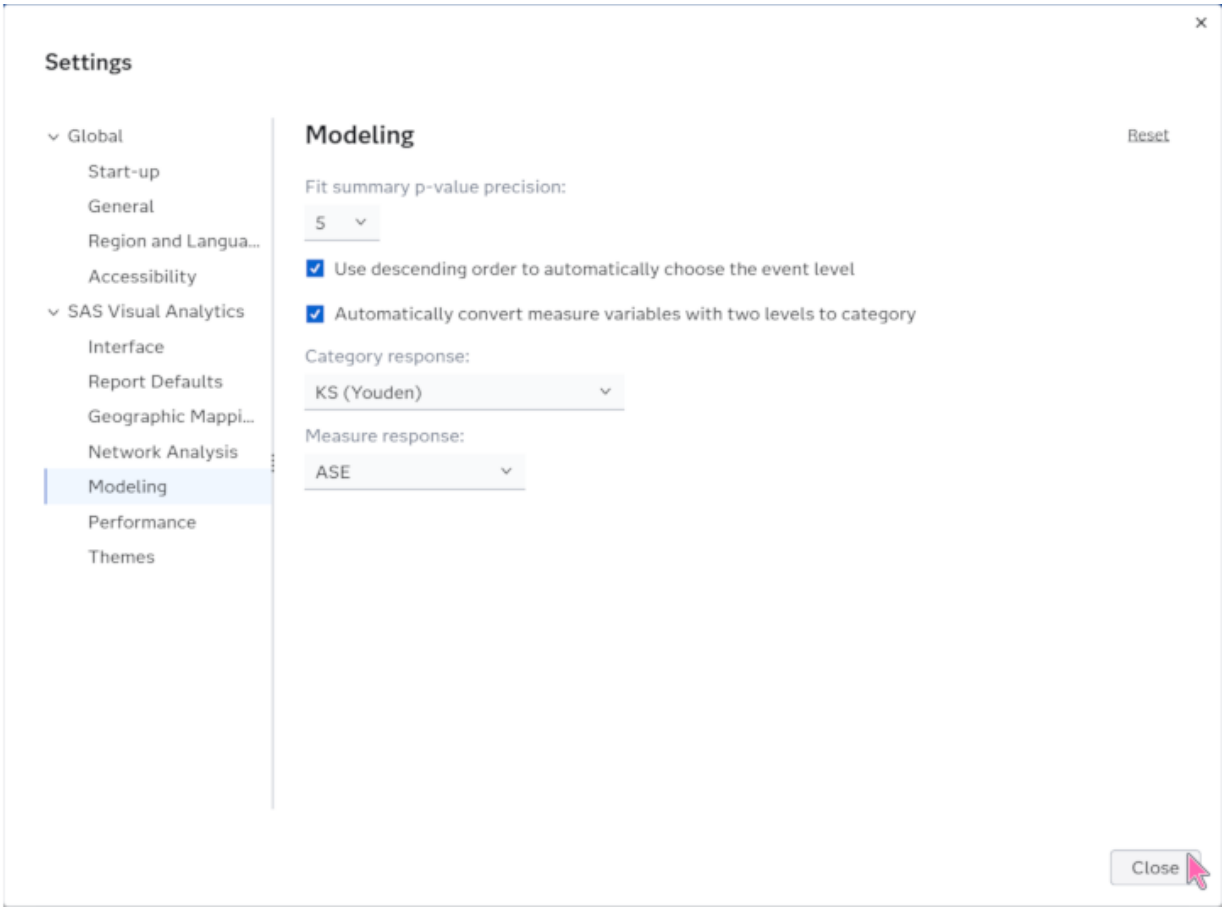


Figure 14: Enable automatic conversion of measure variables with two levels to category.

28. Check **Automatically convert measure variables with two levels to category**.
29. Click **Close**.

You have now completed the essential customization steps. The hands-on work begins with Task 1. You will run an existing SAS program to import data from a CSV file and load the resulting table into memory. You will perform these steps in the application named **SAS Studio**.

About SAS Studio

SAS Studio is a development application for SAS that you access through your web browser. With SAS Studio, you can access your data files, libraries, and existing programs, and you can write new programs. You can also organize your work in flows and use the predefined steps in SAS Studio to generate SAS code. When you run a program, query, or flow, SAS Studio connects to a SAS server to process the SAS code. The SAS server can be a hosted server in a cloud environment or a remote server in your local environment. After the code is processed, the results are returned to SAS Studio in your browser.

In short, SAS Studio is the main coding environment for SAS. You can write SAS programs or use graphical, low-code and no-code features. Recent releases also include Copilot-assisted code development. In the most recent release, the product name changed to SAS Data and AI Studio.



Figure 15: SAS Studio on the applications menu in SAS Viya.

Link to the documentation: SAS Studio User's Guide.

Coding in SAS Viya for Learners

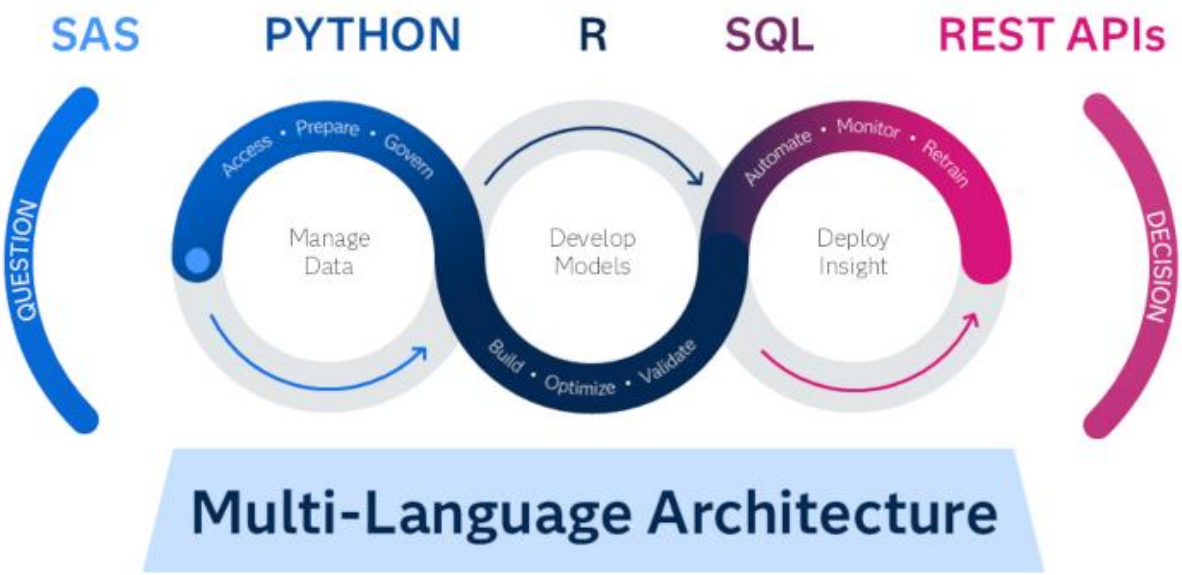


Figure 16: SAS Viya; multiple languages in a single architecture.

In SAS Viya for Learners, actions in the graphical applications are also driven by code. You can write programs yourself, or an application can generate code from your point-and-click actions. SAS Viya supports SAS, Python, R, SQL, REST APIs, and other languages in one analytics environment.

The SAS Viya platform includes two servers, the SAS Compute Server and SAS Cloud Analytics Services, or CAS. The Compute Server is equivalent to the SAS®9 workspace server, and it executes your traditional SAS code with few or modifications. The high-performance processing power of the SAS Viya platform is the CAS server. CAS provides the runtime environment for data management and analytics with SAS. CAS is fast, reliable, and fault-tolerant. In other words, it's ideal for rapid processing of very large data for both data management and analytics.

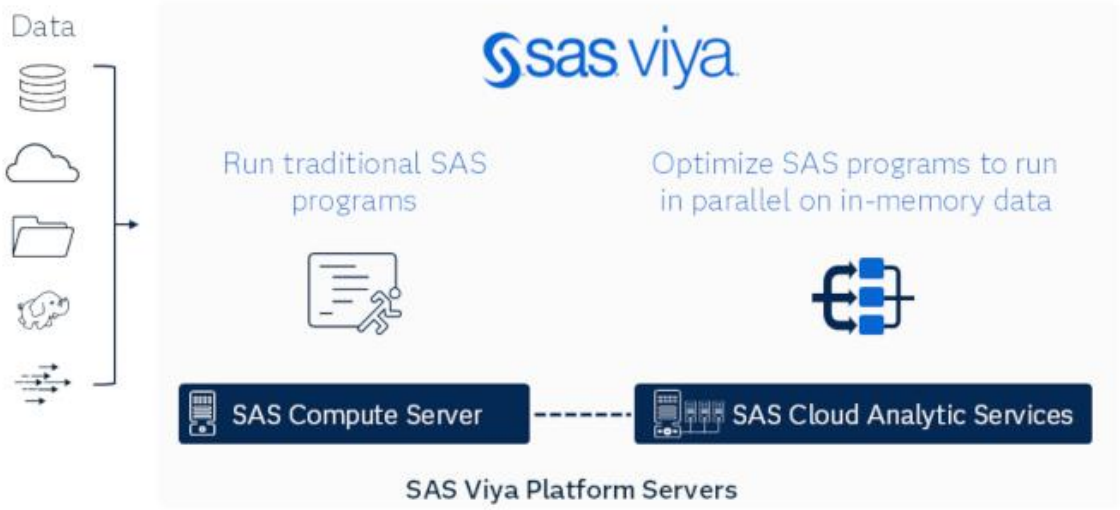


Figure 17: SAS Compute Server and SAS Cloud Analytic Services (CAS).

Which programs should you consider executing in CAS?

The answer is "it depends." As you work in your SAS Viya for Learners environment and consider writing or modifying programs to run in CAS, there are a few best practices to consider. CAS generally outperforms the Compute Server if you have programs that read data into memory multiple times, steps that take a long time to run, PROCs that are computationally demanding, or DATA steps with complex logic. You also want to use CAS to prepare data for use in other SAS Viya applications, such as SAS Visual Analytics, or if you need a specific CAS action that does not have an equivalent procedure on the Compute Server.

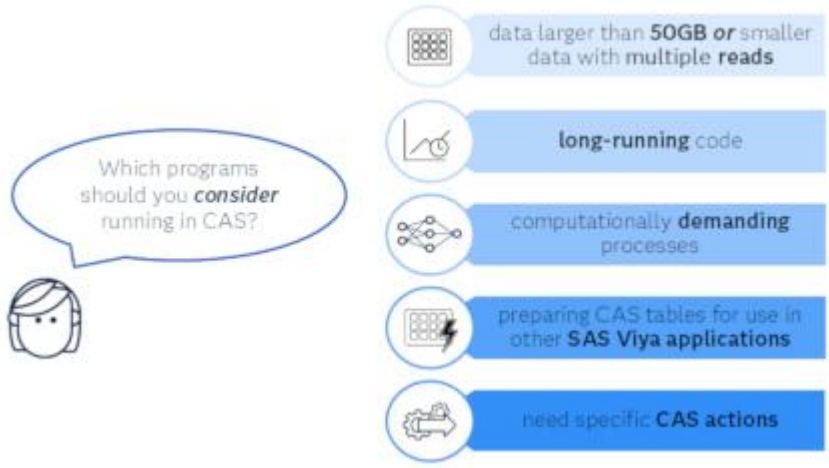


Figure 18: When should you consider to execute code in CAS.

Task 1: Import and load data

In this task, you learn how data enters SAS Viya. Reports and models can be created only after the data is available in the correct location and format. You import the HOMEEQUITY dataset from a CSV file, enrich it, and load it into SAS Cloud Analytic Services (CAS) by running an existing SAS program.

Step-by-step instructions

These instructions assume you have already entered a SAS Viya for Learners session.

1. Open **Applications menu** (nine dots).
2. Select **Develop Code and Flows**.

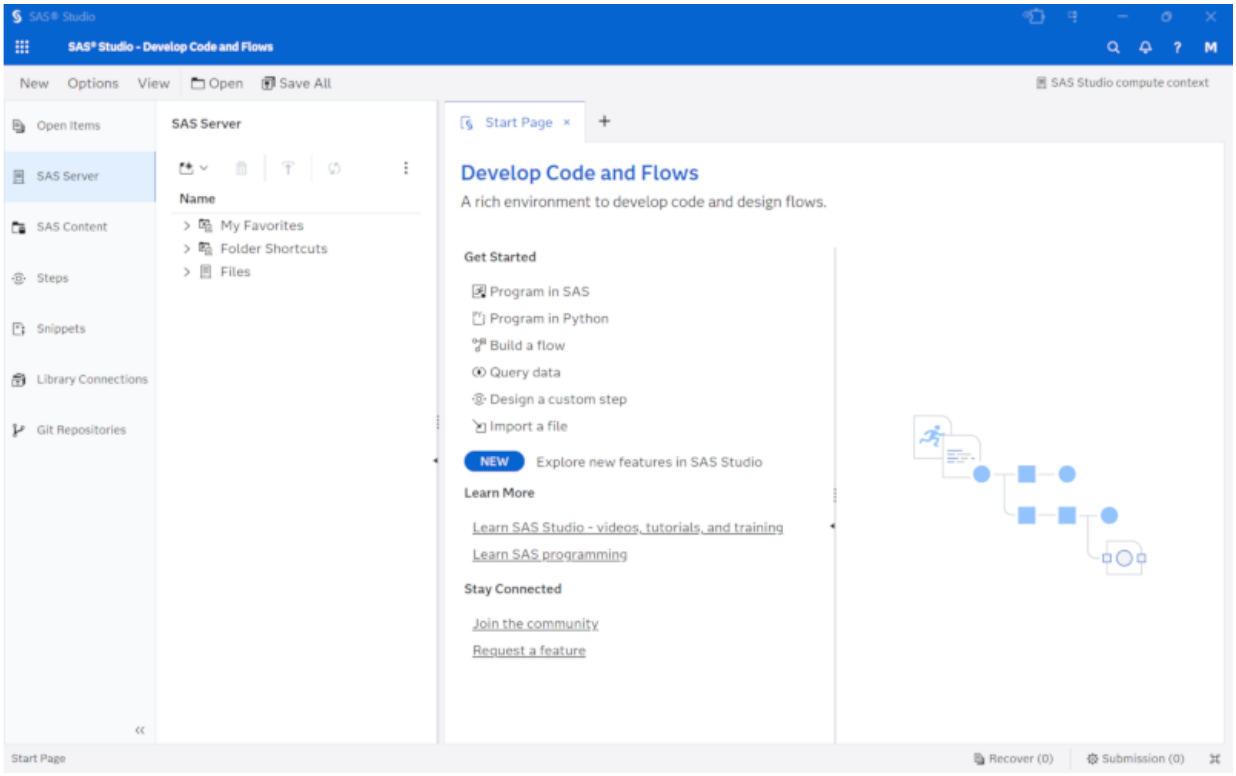


Figure 19: SAS Studio with expanded left menu and the item **SAS Server** *selected.*

3. Open the **SAS Server** navigation/explorer pane if it does not open.
   TIP1: Use the progressive webapp (PWA) for SAS Studio, a full HD or higher resolution monitor and expand the app to full screen to make efficient use of the workspace.
   TIP2: To view the menu item labels, expand the left menu with the double arrows symbol [»] on the bottom left side of the screen.
   TIP3: To close the labels use double arrows symbol [«].
4. In the **SAS Server** explorer pane, use the greater than symbol [>] to expand the folder hierarchy.
5. Expand the folders **Files** → **Home** → **Courses** → **ACADEMIC** → **workshops**.

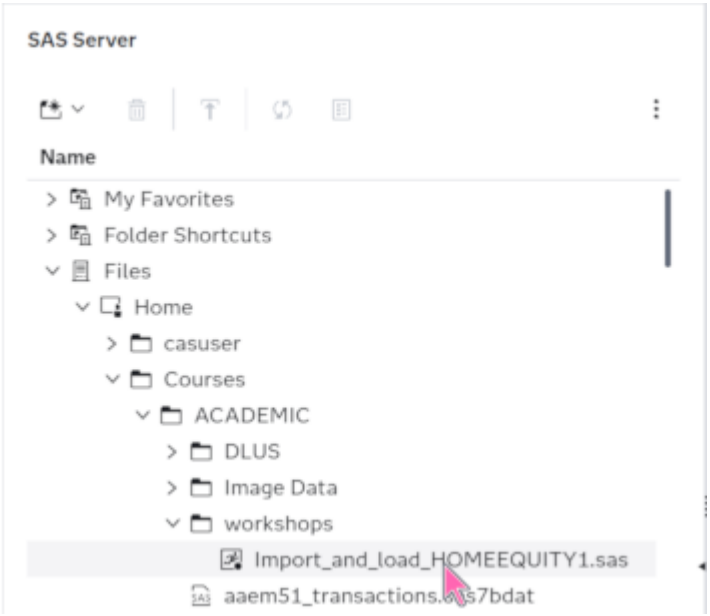


Figure 20: Navigation to the SAS program in the **SAS Server** *pane.*

6. Double-click the SAS program named **Import_and_load_HOMEEQUITY1.sas**.
7. The program is opened in the main canvas, that features a program editor.

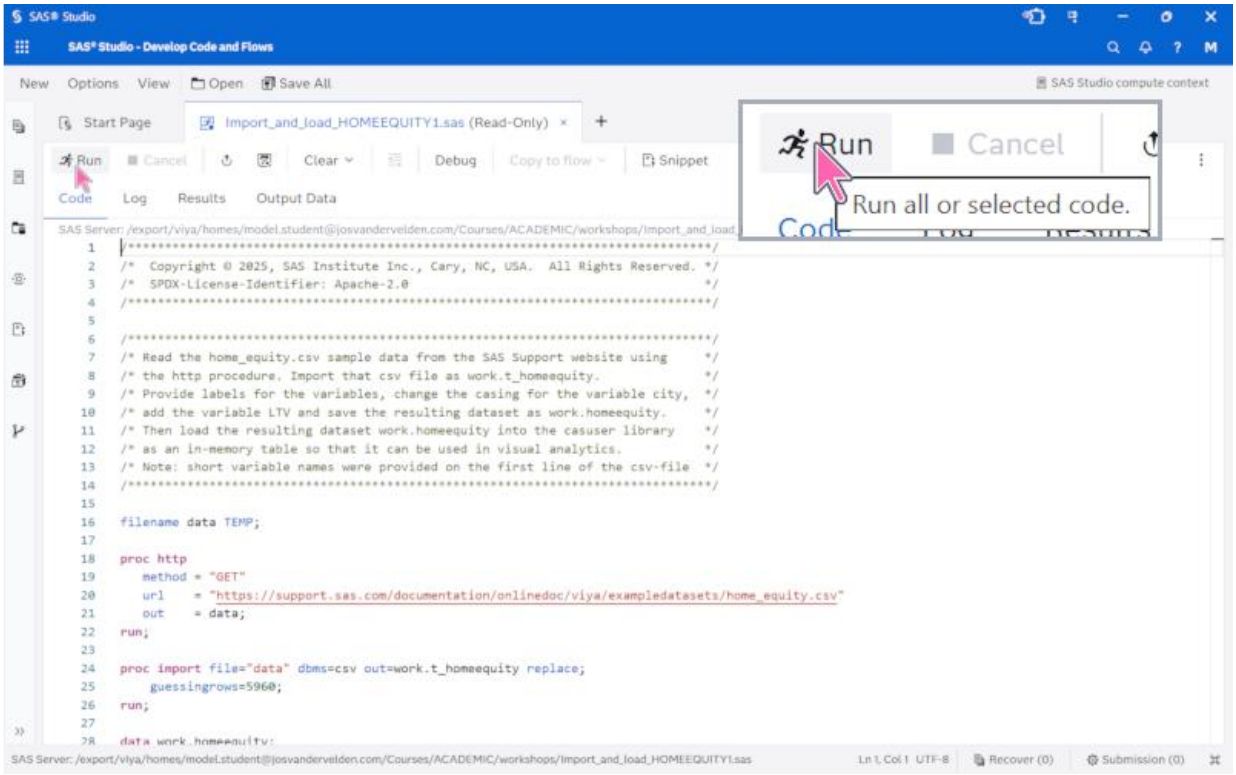


Figure 22: The opened SAS program and the running man icon.

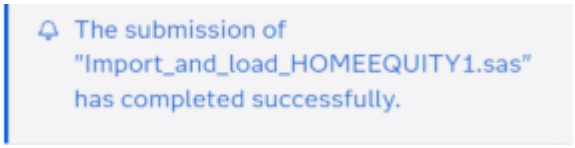


Figure 21: Message displayed after successful execution of the SAS program.

8. To run the SAS program click the **Run** button, which has a running man icon, on the top left.
9. The **Results** tab opens and allows you to scroll through the formatted output. The output was produced by the contents procedure, one of the procedure steps in the program.

10. The **Log** tab houses the SAS log, that displays information, warning, and error messages as a result of executing SAS programs or SAS global statements.
11. The dataset that was imported can be seen in the **Output Data** tab.
12. If you want to review the code you can return to the **Code** tab. It has a color-coded, syntax-checking editor for editing new or existing SAS programs.

TIP: To analyse in detail what the code, log, results, and output data tabs hold, you can review the documentation via the links provided here.

If you want to understand the SAS code in more detail, you can download a fully commented version of the program here: Import_and_load_HOMEEQUITY.sas.txt. If you want to develop strong SAS programming skills, complete courses 1 through 3 in the appendix.

The final procedure in the program loads the imported, transformed, and enriched data into memory on the CAS server. It uses your personal CAS library, or caslib, named CASUSER. In SAS Viya for Learners, this caslib is usually displayed as CASUSER(your-email). You can now use the HOMEEQUITY table in SAS Visual Analytics.

About SAS Visual Analytics

Integrated throughout the data and AI life cycle is the need to explore and visualize data. It is a critical part of all phases: manage data, develop models, and deploy insights. The SAS Viya application that makes this possible is SAS Visual Analytics. SAS Visual Analytics supports data exploration, interactive reports, dashboards, and self-service analytics. It enables you to visualize and discover relevant relationships in your data, create and share interactive reports and dashboards, and use self-service analytics to quickly assess probable outcomes.

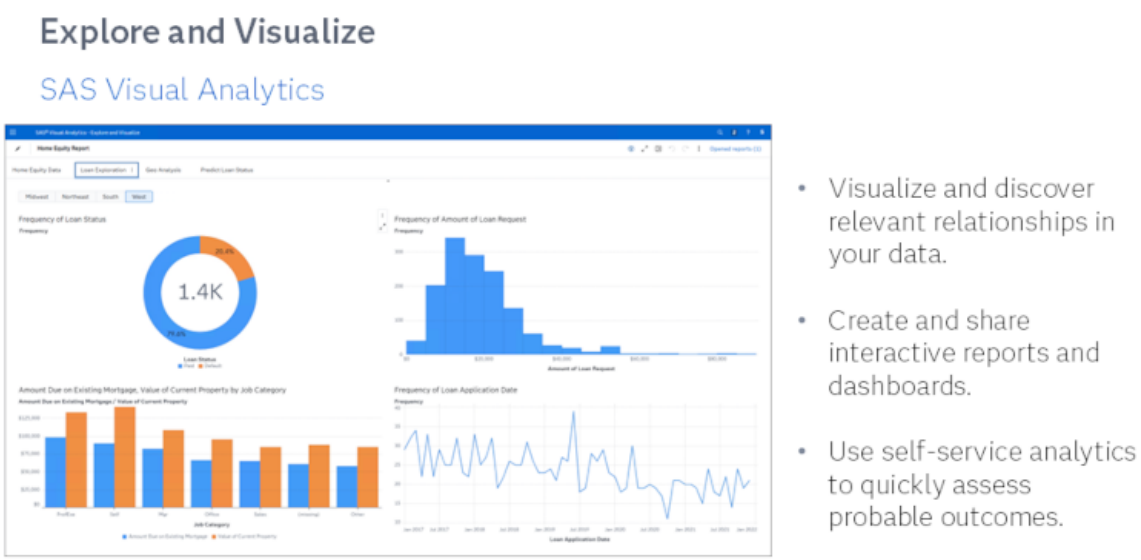


Figure 23: A dashboard created with SAS Visual Analytics.

Link to the documentation: https://documentation.sas.com/doc/en/vacdc/v_038/vawlcm/home.htm.

About automatic charts.

In Task 2, you mainly use the automatic chart feature in SAS Visual Analytics. It is useful when you start a report because SAS Visual Analytics selects a chart type based on the data items that you drag onto the canvas.

Data items	Chart type
One measure	Histogram
One category and any number of measures	Bar chart
One datetime category and any number of measures	Time series plot
One date or datetime category and one or more categories	List table
One geography and up to two measures	Geo map
One geography and three or more measures	Bar chart
One hierarchy and any number of measures	Bar chart
One hierarchy, one or more categories, and any number of other data items	Crosstab
Two or more hierarchies and any number of other data items	Crosstab
Two or more categories and any number of measures	List table
Two or three measures	Scatter plot or Heat map
Four or more measures	Scatter plot or Correlation matrix

Task 2: Design a BI dashboard

In this task, you create a dashboard. First, you add the data source and adjust metadata for selected variables. Next, you use automatic charts to create four visualizations, link the charts so that they interact, and change one visualization to a different chart type.

Step-by-step instructions

These instructions assume you have already entered a SAS Viya for Learners session.

1. Open **Applications menu** (nine dots).
2. Select **Explore and Visualize**.
3. Click **New report**.

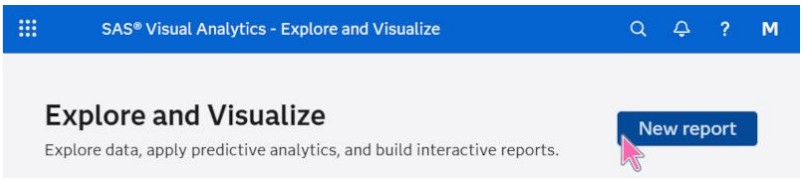


Figure 24: Create a new report.

4. Click **Add data**.

5. In the **Choose Data** dialog search for **HOMEEQUITY**.

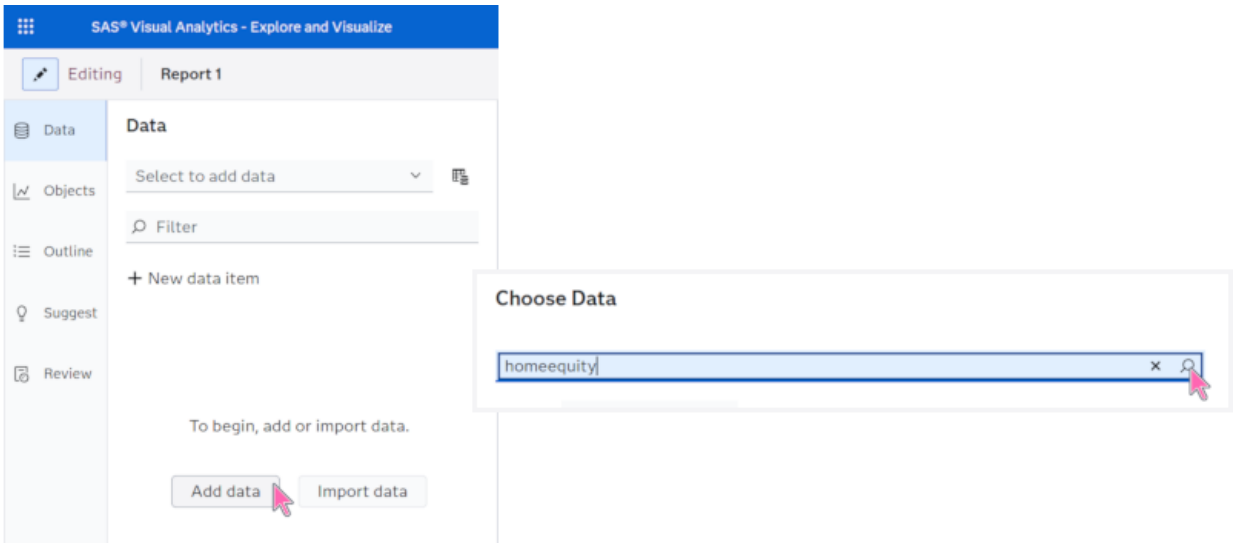


Figure 25: Add data and search for the **HOMEEQUITY** *table.*

6. Select the home equity dataset from your CAUSER library. Select the table in the CASUSER library that includes your email address in parentheses.

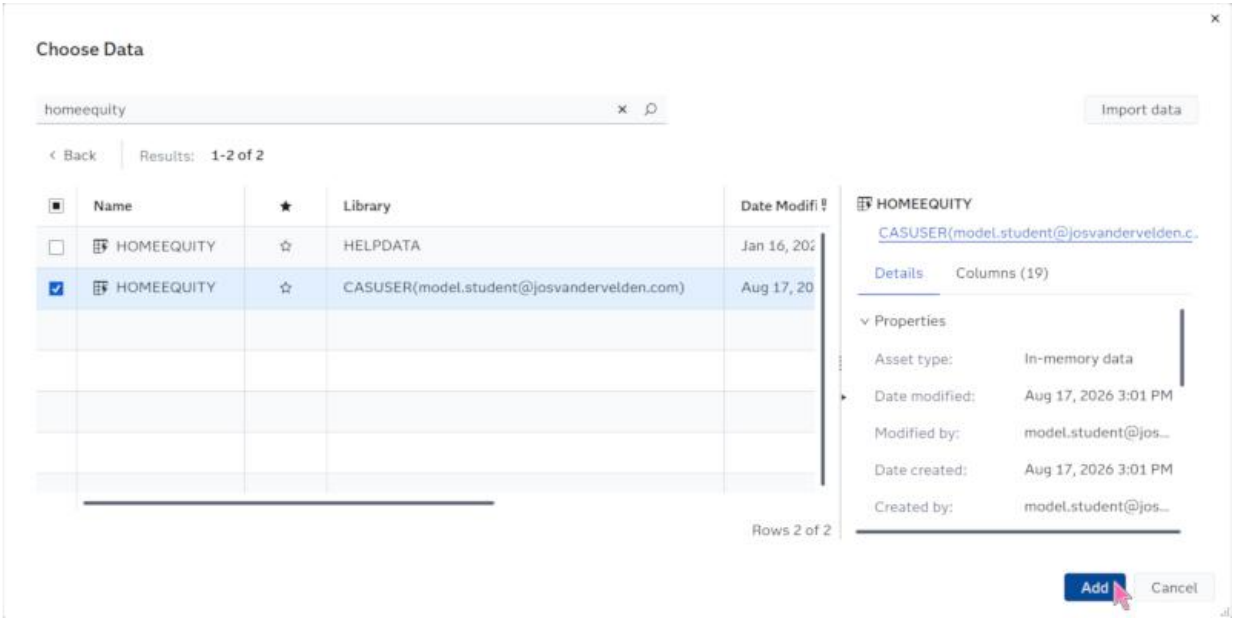


Figure 26: The **Choose Data** *dialog.*

7. Click **Add**.
8. In the **Data** pane, verify the variables under the classifications **Category** and **Measure**.

9. Hover over the variable **City** to verify the metadata for the variable.

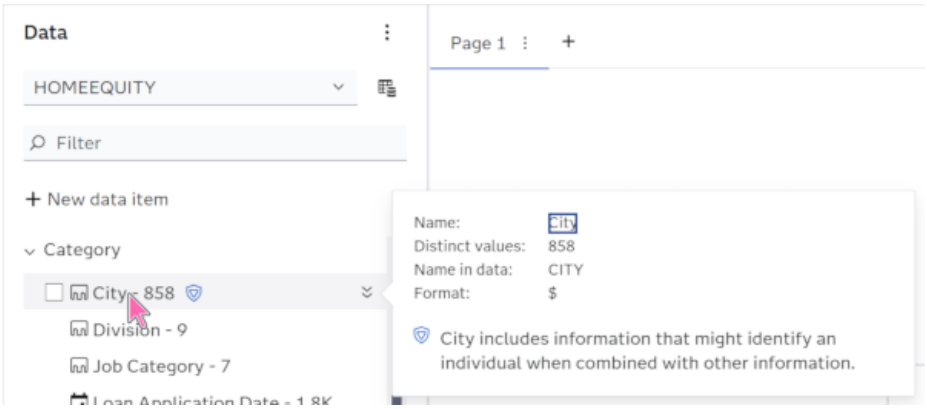


Figure 27: Metadata for the **City** *variable.*

10. Scroll down to the variables classified as **Measure**.
11. Hover over the variable **Age of Oldest Credit Line** to verify the metadata for the variable.



Figure 28: Metadata for the **Age of Oldest Credit Line** *variable.*

12. Right-click **Age of Oldest Credit Line** and click on **Measure with same aggregation** in the **Select** submenu.

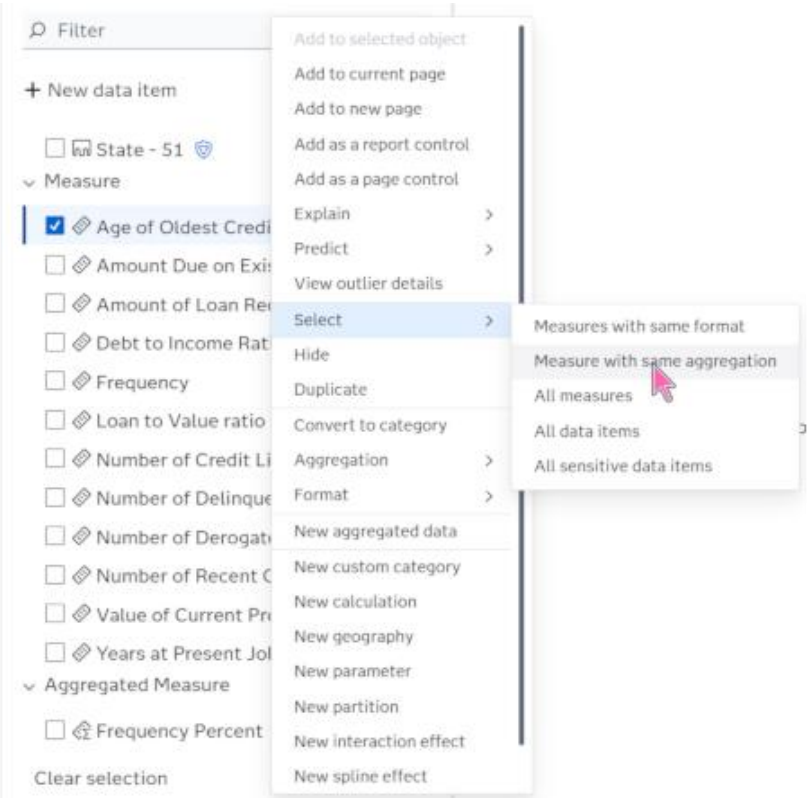


Figure 29: Select measures with same aggregation.

13. Right-click **Age of Oldest Credit Line** and click on **Average** in the **Aggregation** submenu.
14. Click **Clear Selection (11)**.

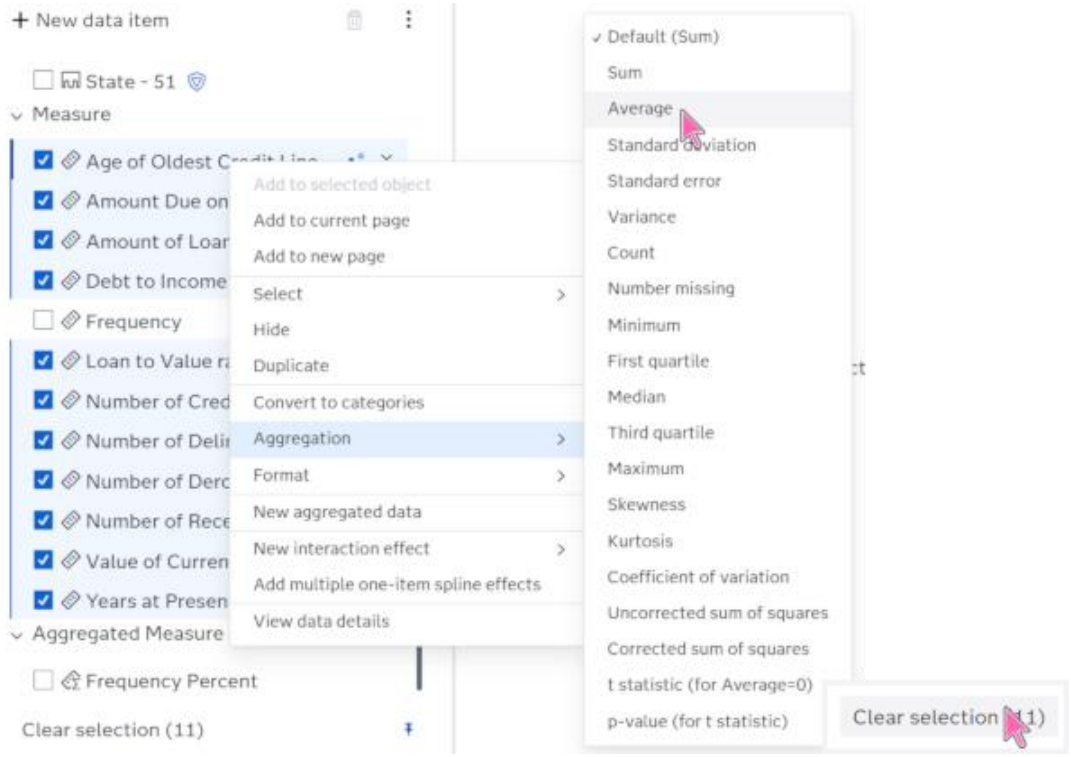


Figure 30: Change aggregation to average.

15. Right-click **Loan Application Date** and in the **Format** submenu click on **MMMYYYY (MONYY7)**.

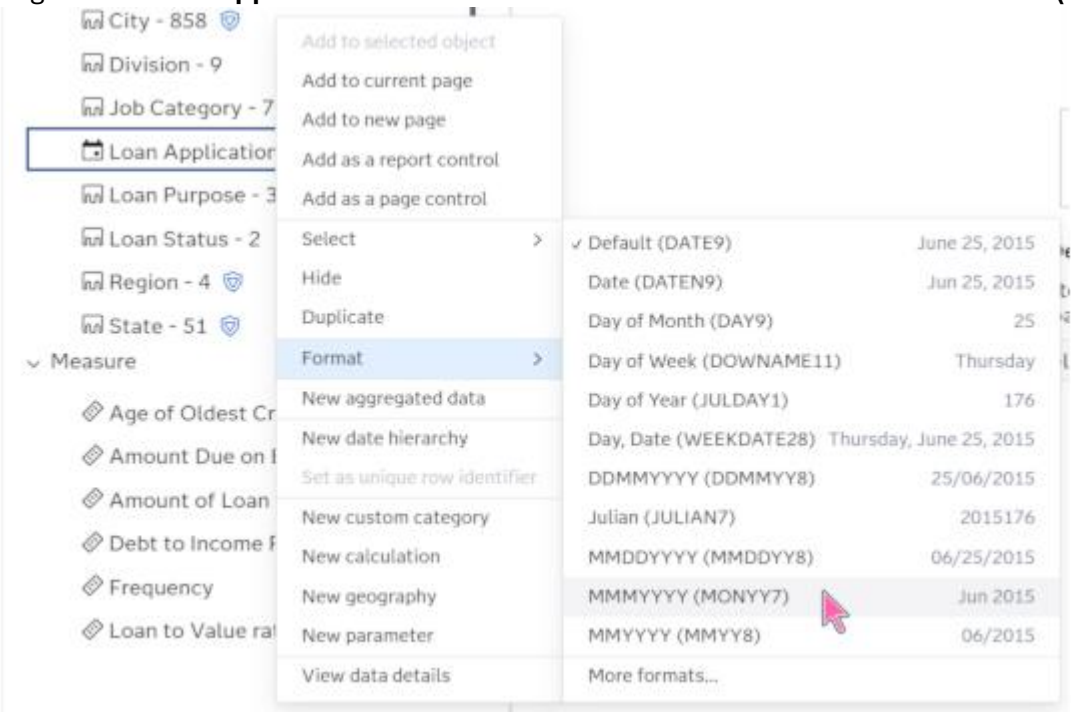


Figure 31: Change the format for the **Loan Application Date** *variable.*

16. Click **Save** and type the name **Home Equity Analysis Report** and click **Save**.

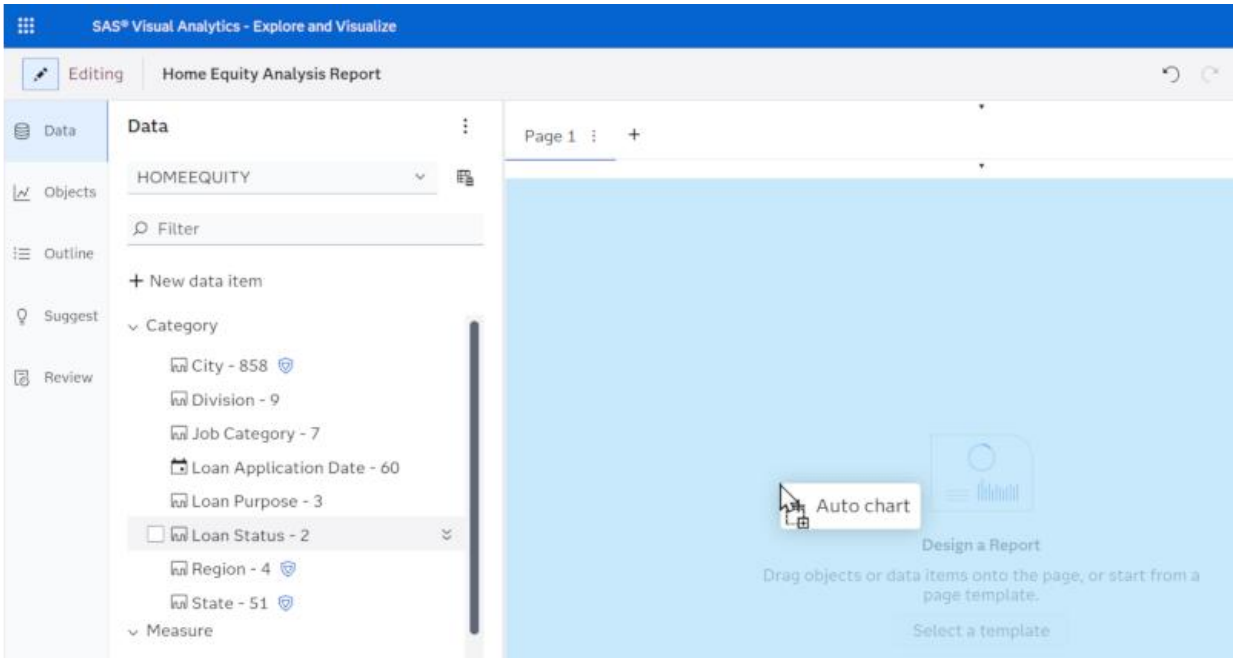


Figure 33: Add the first auto chart to the report canvas.

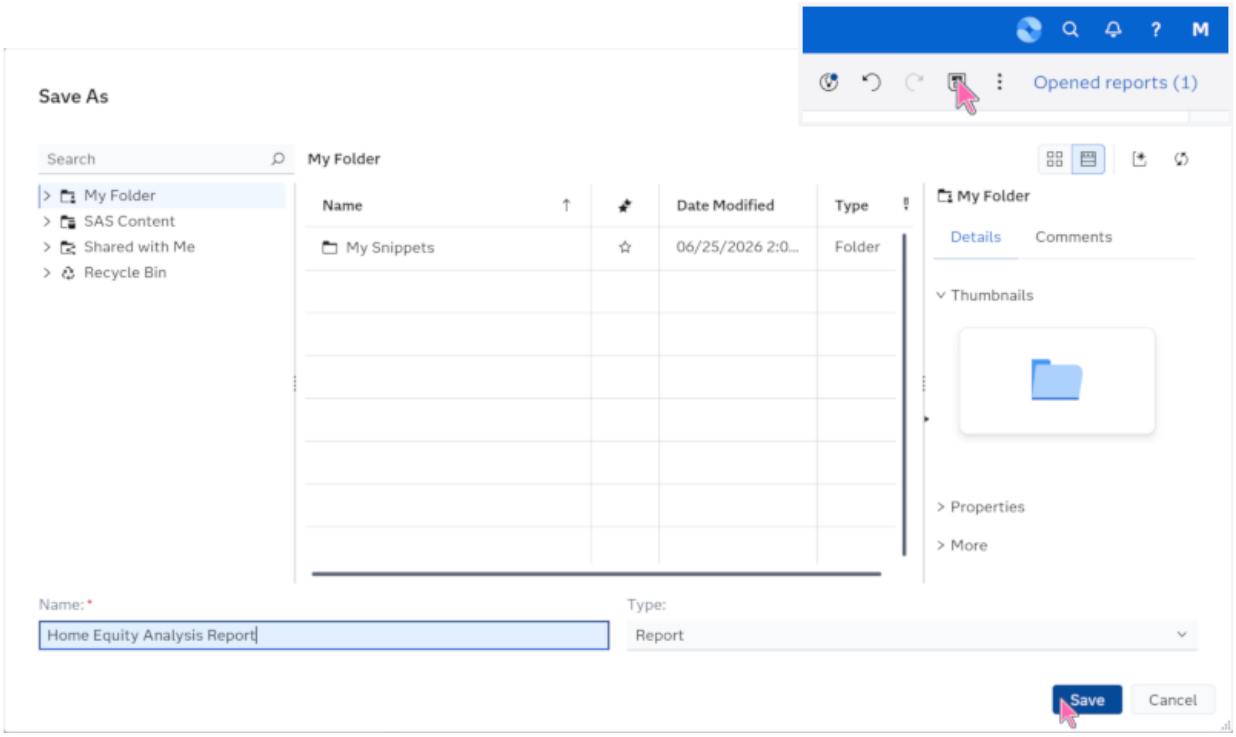


Figure 32: Name and save the report.

17. Drag-and-drop **Loan Status** to the report canvas to add the first auto chart.
18. Drag-and-drop **Amount of Loan Request** to the right side of the chart.
    NOTE: The light blue rectangle should cover only the right half of the chart when you drop it.

19. Drag-and-drop **Loan Application Date** to the bottom half of the **Amount of Loan Request** histogram.

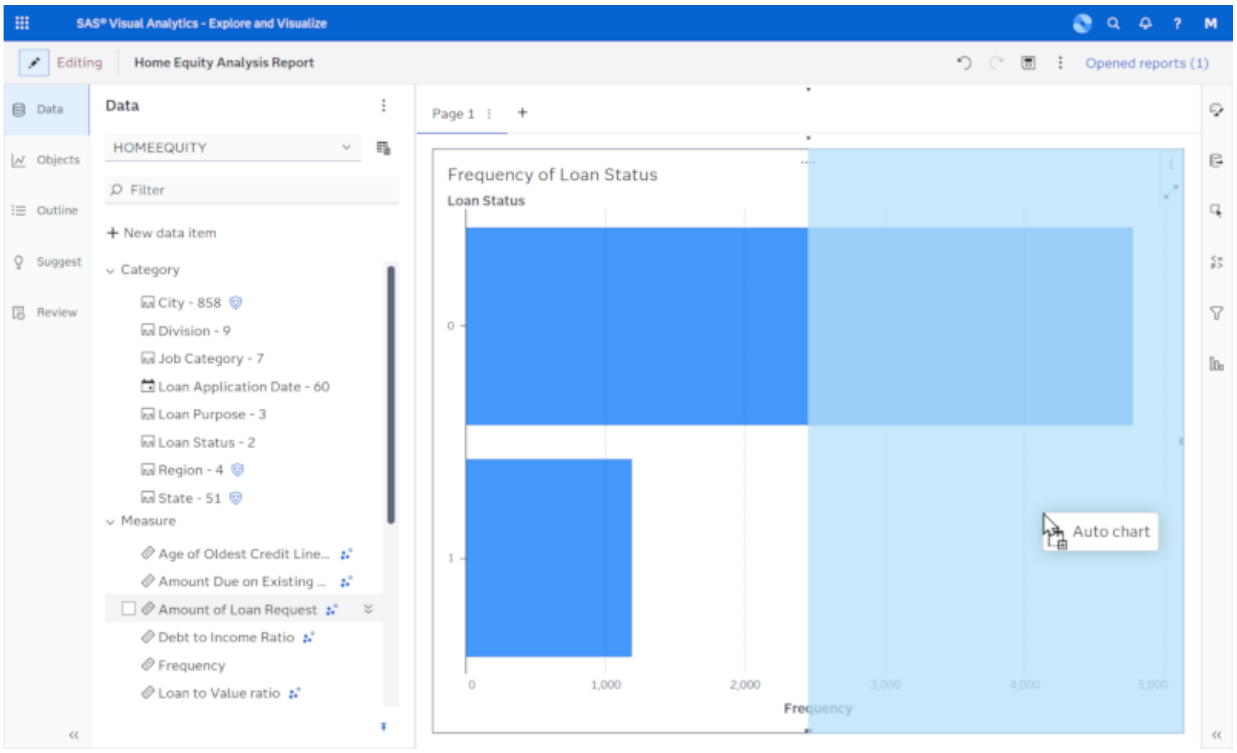


Figure 34: Drag-and-drop the next variable to the report canvas.

20. Right-click **Age of Oldest Credit Line** and in the **Select** submenu click on **Measure with same aggregation**.
21. Drag-and-drop them on the bottom half of the **Loan Status** chart.

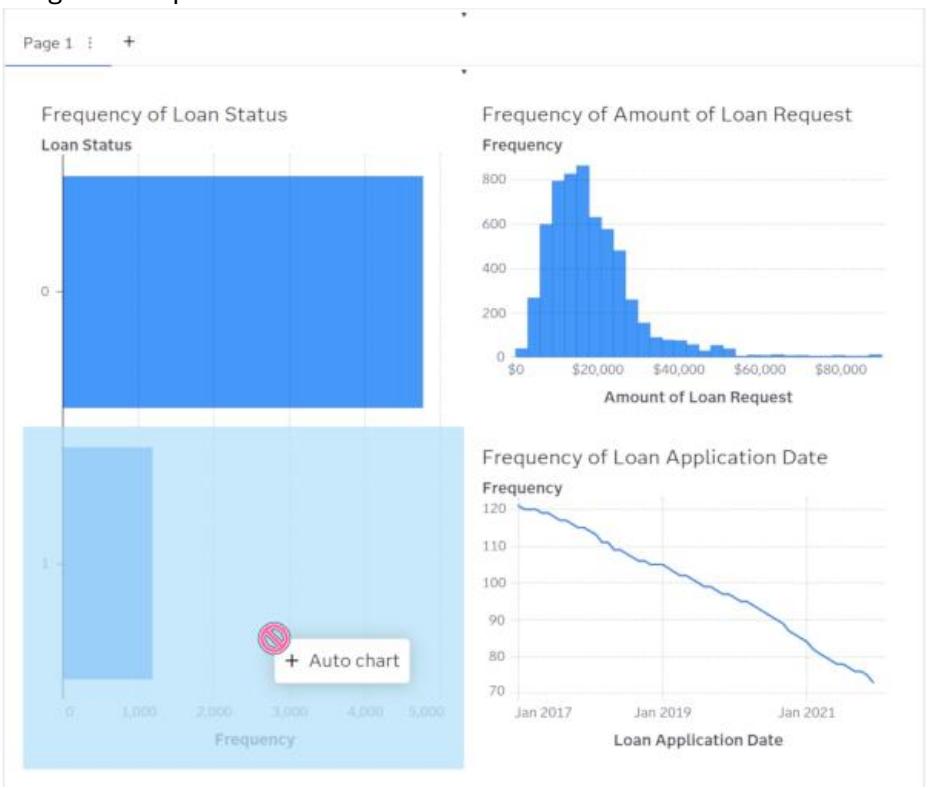


Figure 35: Add the fourth auto chart to the report canvas.

22. Select the **Loan Status** chart.



Figure 36: Page 1 with the selected Loan Status chart.

23. Click the options menu icon [⋮] and in the **Change Bar chart to** submenu select **Pie chart**.

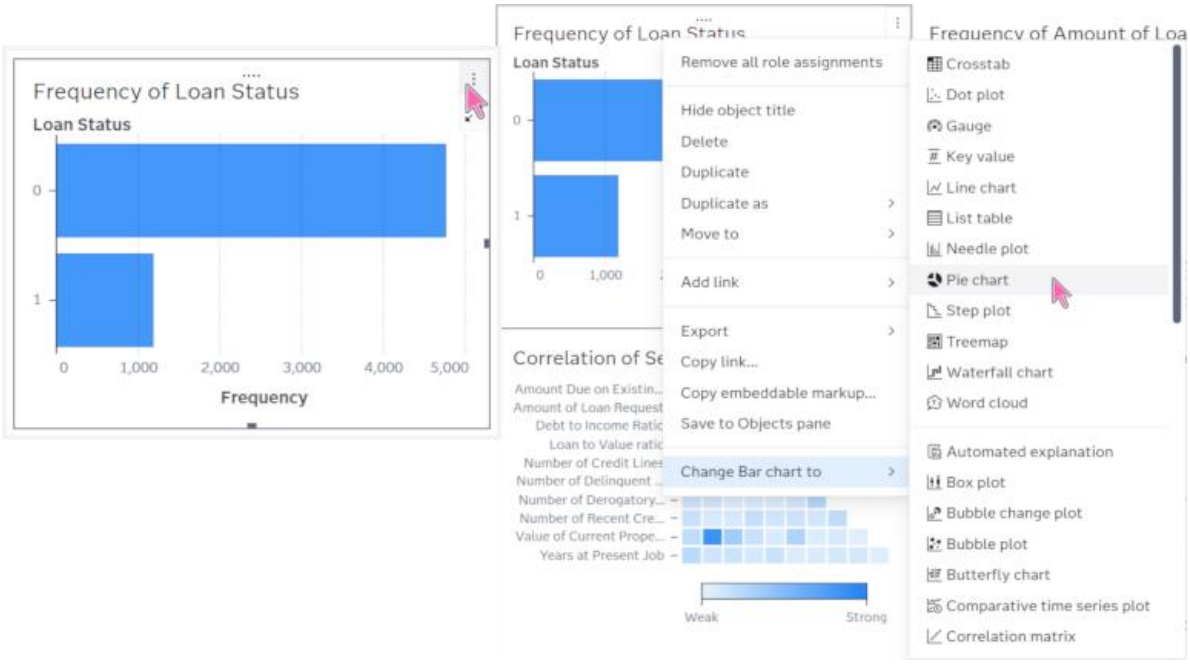


Figure 37: Change a bar chart to a pie chart.

24. With the **Pie chart** still selected open the **Options** item in the right side menu.
25. Scroll down to the **Data labels** section and check the field **Percent of total**.
26. For the **Data label text style** section change the size to **16**, the style to Bold [**B**], and change the color to **Deep purple**.

27. For the **Data label location** select **Outside**.

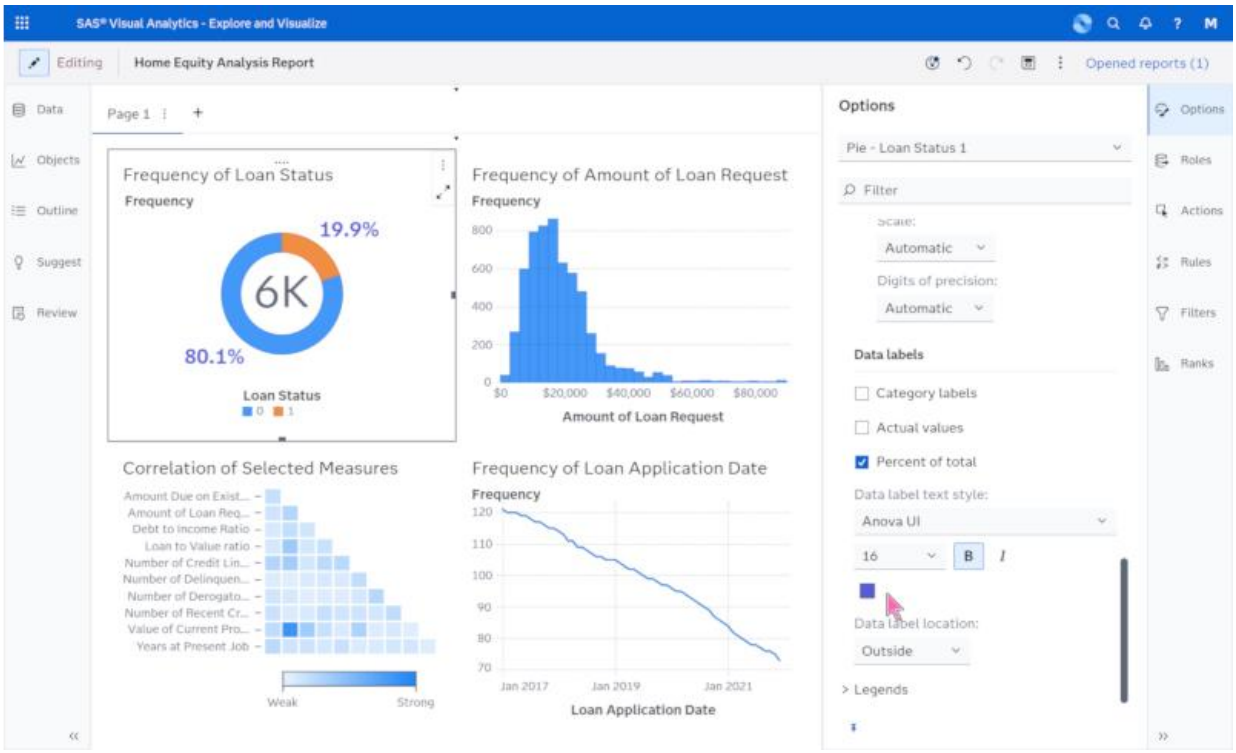


Figure 38: Customize the labels in the pie chart.

28. With the **Pie chart** still selected open the **Actions** item in the right side menu.
29. Check the box for the field **Automatic actions on all objects**.

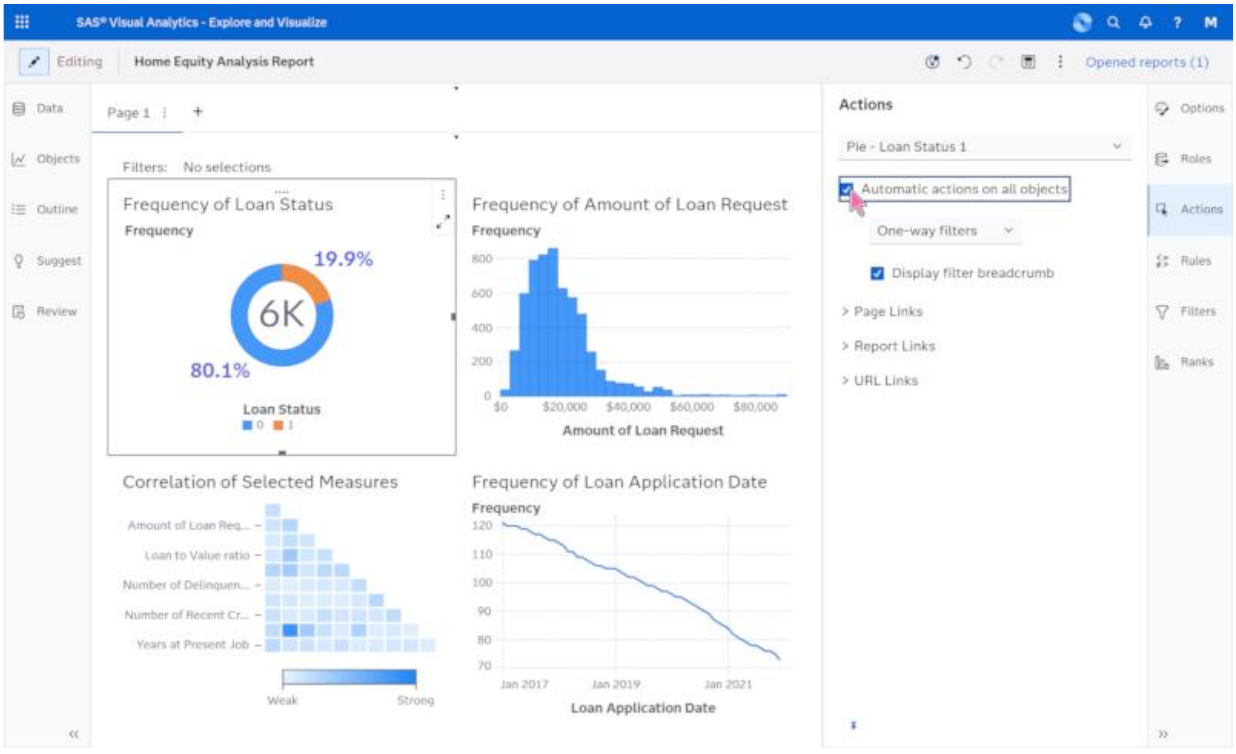


Figure 39: Define automatic actions on all object with the pie chart selected.

30. Click the Page 1 menu icon [⋮] and rename the page to **Interactive charts**.
31. Click **Save** to complete the task.
32. Click the pencil icon on the top left to switch from editing mode to viewing mode.
    NOTE: To return to editing mode just click the pencil icon.

33. Explore the interactive report. Select values in different charts and observe how the other charts respond.

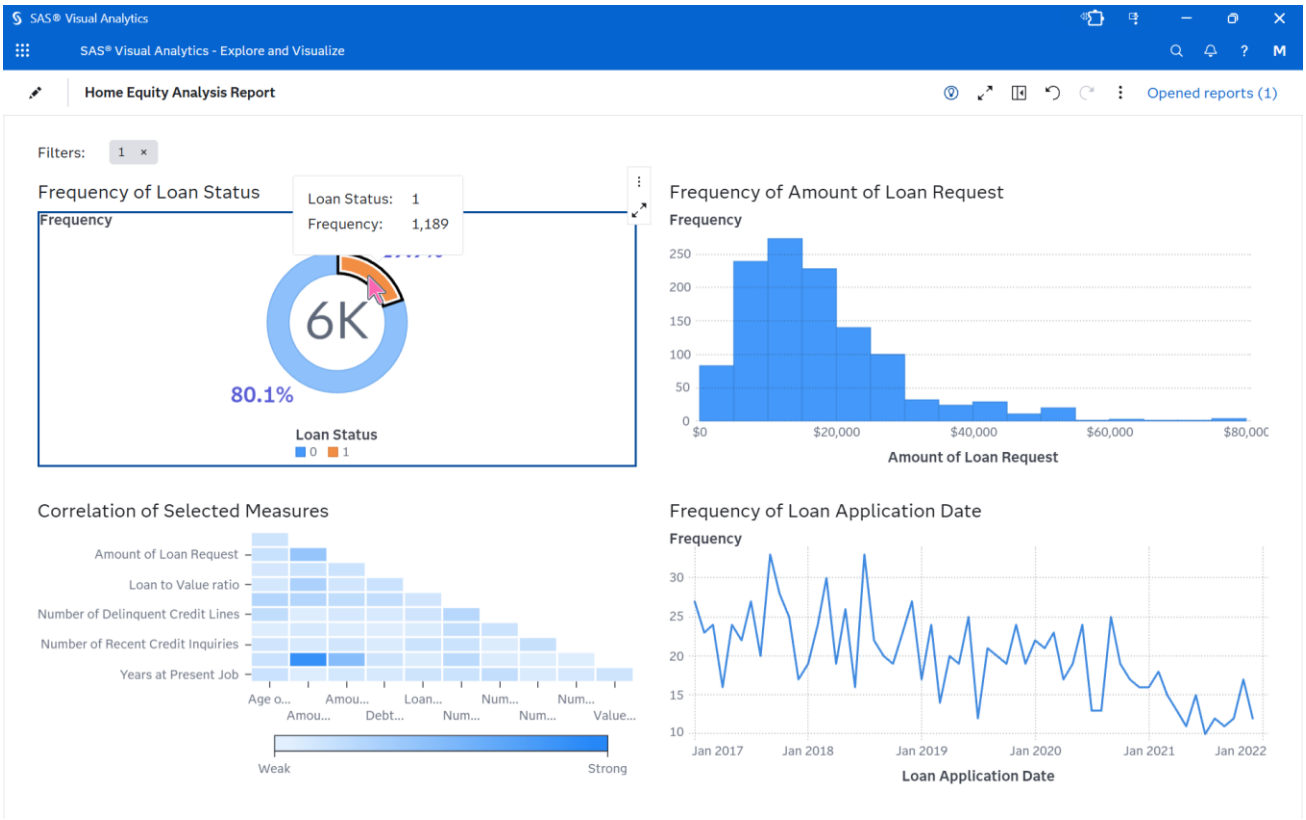


Figure 40: The saved report with a focus on applicants that defaulted on the loan.

Reflection question: Which chart helped you understand the loan data most quickly? Why?

For those interested in understanding the essential features in SAS Visual Analytics we recommend completing courses 4 and 5 from the appendix.

The automated explanation object

An automated explanation helps you explore possible reasons behind a result. In this workshop, the target or response variable is **Loan Status**. The automated explanation helps us identify which variables may be related to the target variable. This does not prove cause and effect, but it gives us a useful starting point for further analysis.

An automated explanation object determines the most important underlying factors for a specific response variable. The results include relative importance, relationship plots, and the highest and lowest groups.

The automated explanation object displays the following information:

- Summary information about the response factor. If you assign a data item to the time role, then forecast information is displayed.
- A relative importance score for each underlying factor appears in a bar chart. The most important underlying factor is assigned a score of 1, and all other scores are proportional to that value.
- If you select show high and low groups, then groups are displayed in the high tab and the low tab. Groups are calculated based on the selected underlying factor.

- Several decision trees are run on the specified response variable, and each group describes a leaf from one of those decision trees. For measure responses, the three groups that result in the highest average values of the response are displayed on the high tab. The three groups that result in the lowest average values of the response are displayed on the low tab. For category responses, the three groups that contain the highest percentage of the selected event level of the response are displayed on the high tab. The three groups that contain the lowest percentage of the selected event level of the response are displayed on the low tab. The high tab and the low tab display only groups that include the selected factor in the calculation of the group.
- A relationship plot for the selected underlying factor. The contents of this plot depend on the variable type of both the response variable and the selected underlying factor. The following table indicates which plot is used for each combination of response variable and underlying factor.

Link to the documentation:
https://documentation.sas.com/doc/en/vacdc/v_038/vaobj/n0ewwfd6udhv7qn1nropd18lof9j.htm.

Task 3: Investigate using an automated explanation

In this task, you add a page that contains an automated explanation object.

Step-by-step instructions

These instructions assume you have already entered a SAS Viya for Learners session.

1. Return to the report and if necessary, activate editing mode by clicking the pencil icon.
2. Open the **Data** pane and right-click on **Loan Status**.
3. In the **Explain** submenu select **Explain on new page**.

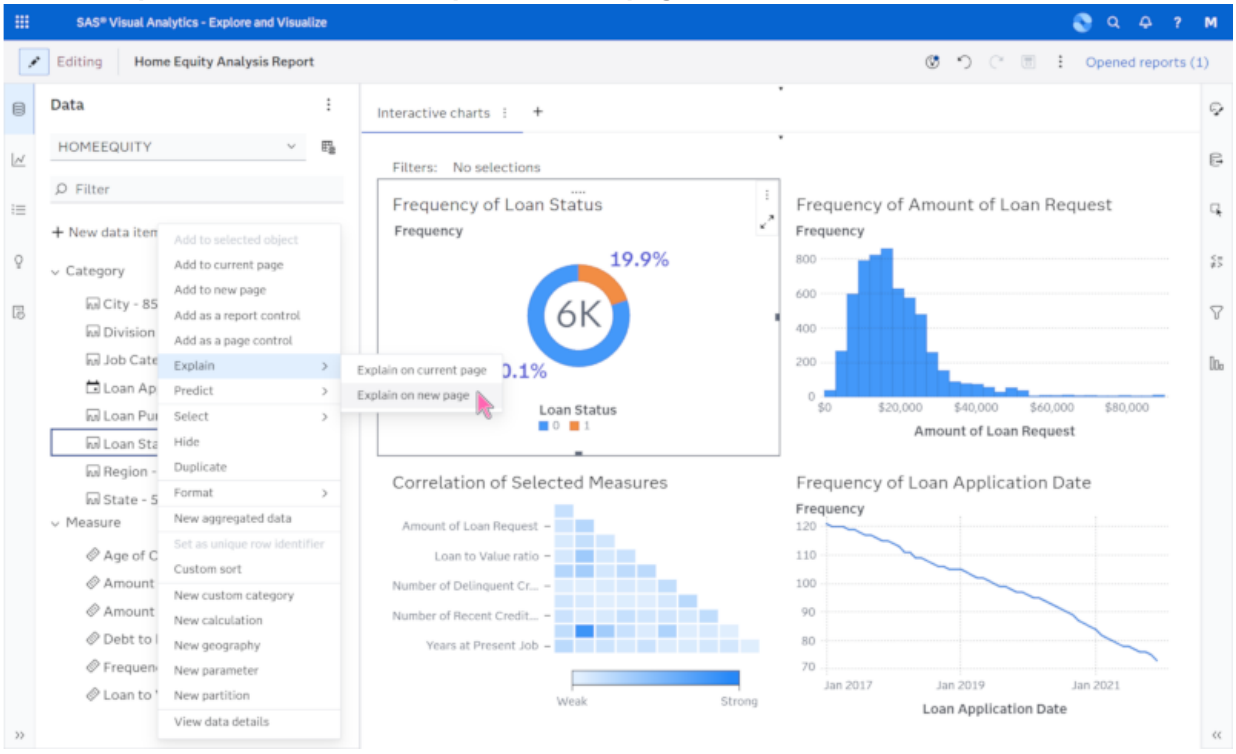


Figure 41: Create a new page with an automated explanation.

4. Wait while the new page loads and then rename the page to **Explain Loan Status**.

5. On the top right select **1**.
   NOTE: By selecting 1 we focus on the participants who defaulted on the loan.

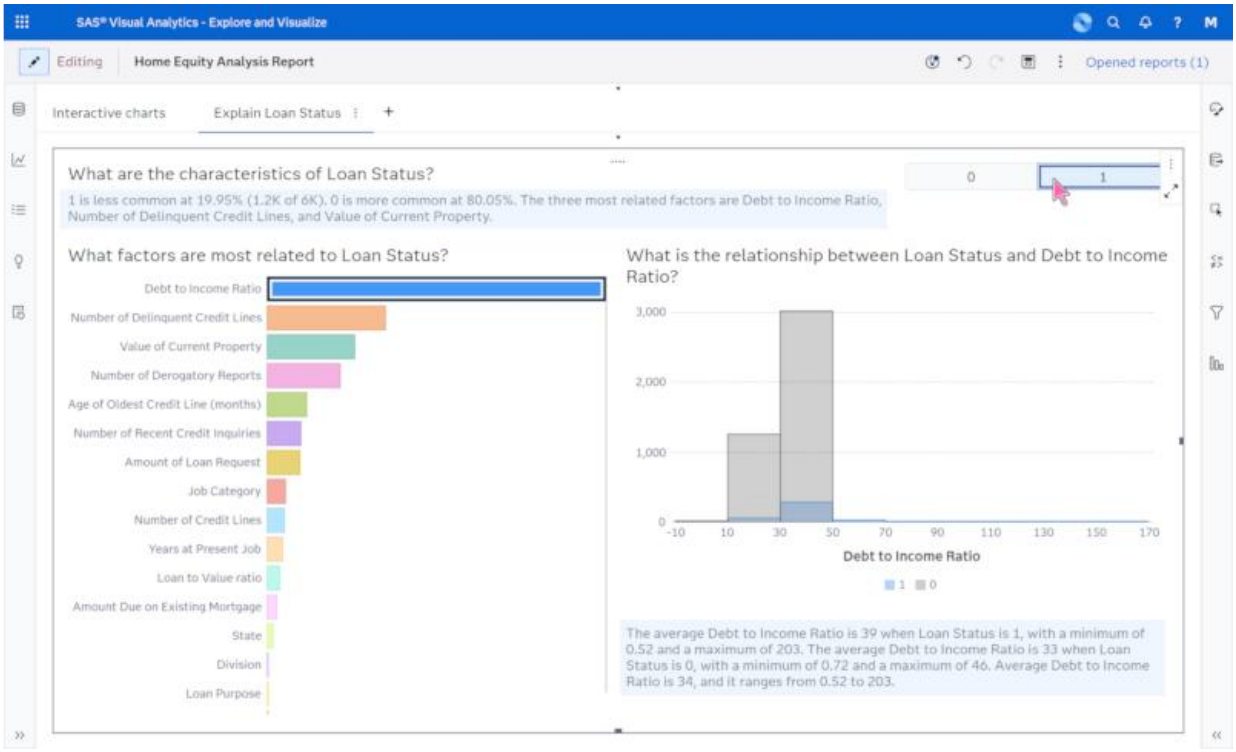


Figure 42: Focus on characteristics of participants that defaulted on the loan.

6. Analyse the page and focus on:
   a. The text provided via Natural Language Generation (NLG) on the top and bottom right.
      NOTE: The texts are highlighted with light blue in the illustration below. The generated text summarizes the results of the algorithm.
   b. The changes when selecting other variables in the bar chart with factors on the left.
7. Click **Maximize** (the icon with diagonal arrows under the three vertical dots).



Figure 43: Maximize the view for the object.

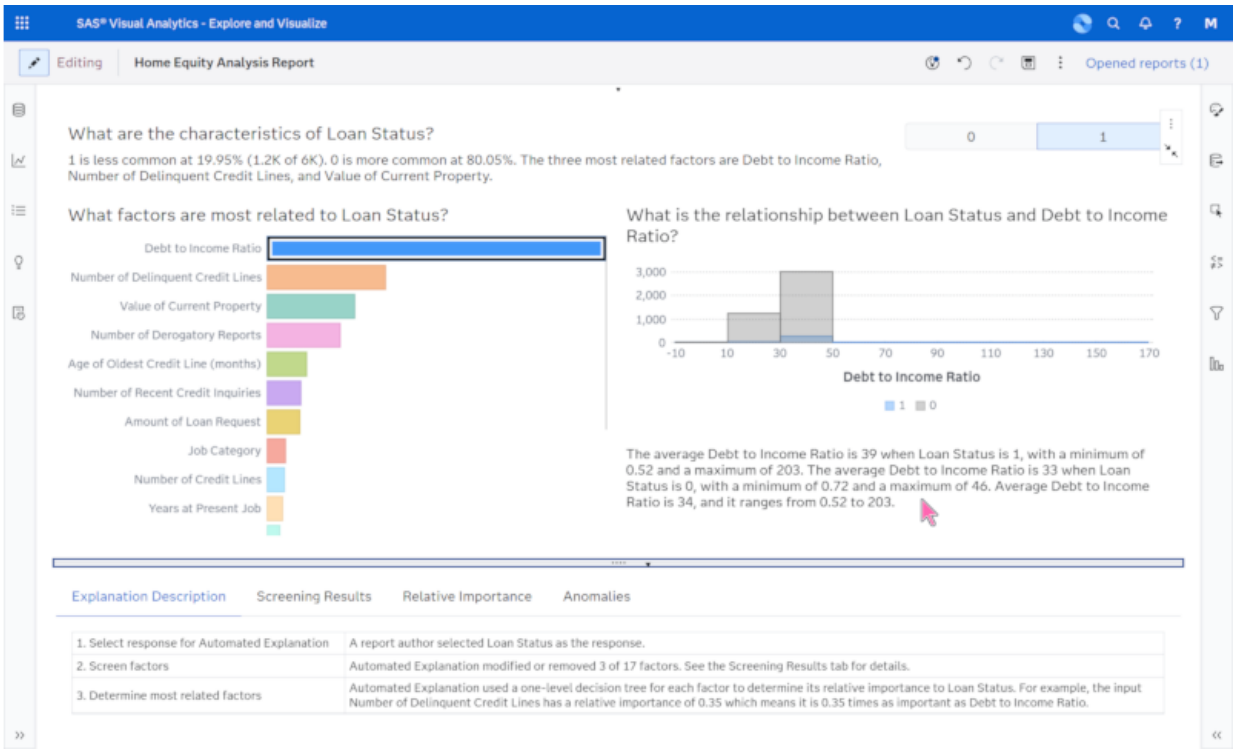


Figure 44: The maximized view for the automated explanation object.

8. Review the information provided in the four tabs named:
    a. **Explanation Description**
    b. **Screening results**
    c. **Relative Importance**
    d. **Anomalies**
9. Click **Minimize** and **Save** the report to complete this task.

Course 7, listed in the appendix, provides statistical background information that is essential for working with the automated explanation and the other features that are shown in the next task.

Task 4: Explore a data mining and machine learning pipeline

In this task, you use Copilot as an assistant to create data mining and machine learning pipelines. You also compare several models and pipelines, including one that contains an open-source model. During preparation steps 1 through 7, you use **SAS Information Catalog** to analyze the data. This step is essential to provide copilot with metadata about the dataset. Copilot uses this to guide the advice creation.

In this section whenever you encounter ***CC*_*UNI*_*MMMYY*_*yourinitials*** you must replace:

- ***CC*** with your country code (in this document **PT**);
- ***UNI*** with two or three characters that inidicate your institution or study program (in this document **SAS**);
- ***MMMYY*** with the a three character month abbreviation, followed by the year with two digits (in this document **NOV26**);
- ***yourinitials*** with two or three characters that represent your initials (in this document **JDV**).

Step-by-step instructions

These instructions assume you have already entered a SAS Viya for Learners session.

1. Open **Applications menu**.
2. Click **Discover Information Assets**.
3. Search **HOMEEQUITY**.
4. Select the home equity table from **CASUSER(your_email)**.
5. In **Actions** select **Analyze data**.

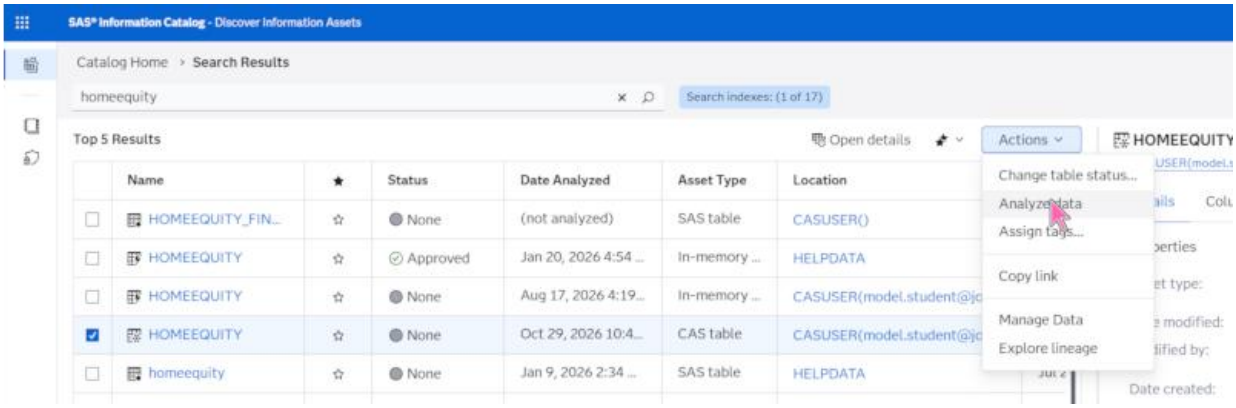


Figure 45: Selecting the action Analyze data for the CASUSER(your_email) table.

6. Click **Submit request**.
7. Verify the process completed successfully.

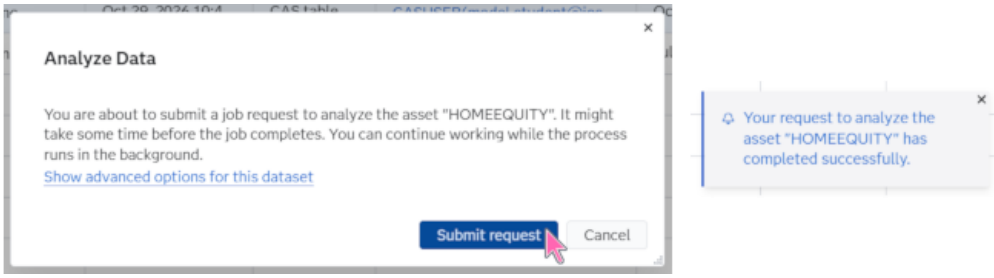


Figure 46: After you've submitted the analysis request you should wait till the message box appears.

8. Open **Applications menu**.
9. Select **Build Models**.
10. Click **New project**.
11. As **Name** use ***CC*_*UNI*_*MMMYY*_*yourinitials*_HOMEEQUITYuseCASE**.

12. Next to the **Data** fileld click **Browse**.

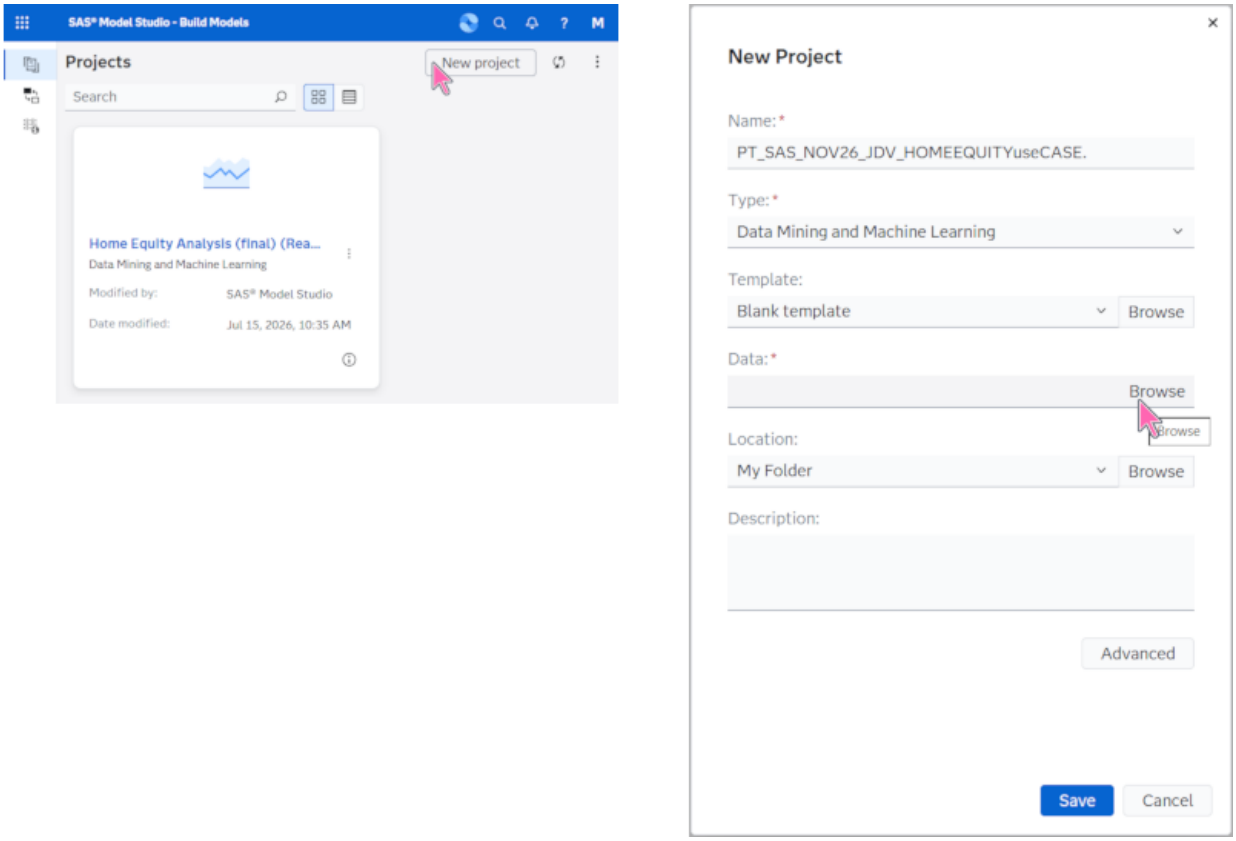


Figure 47: In the New Project dialog box click Browse for the Data field.

13. Search and select **HOMEEQUITY** in CASUSER(your_email).
14. Click **OK**.

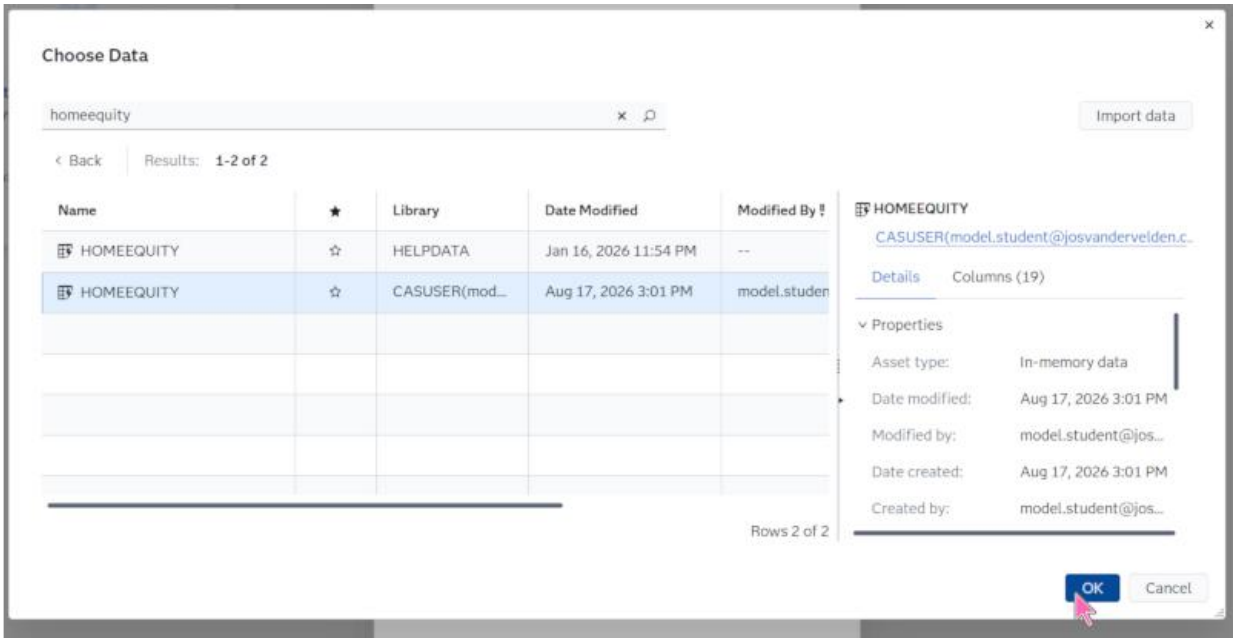


Figure 48: Select the HOMEEQUITY table in CASUSER(your_email) and click OK.

15. Click **Save**.

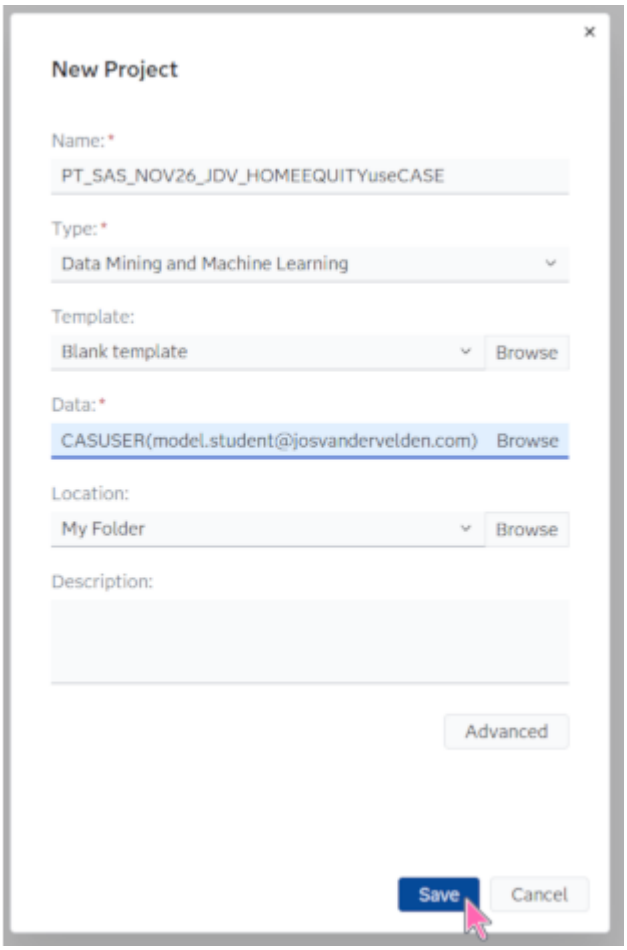


Figure 50: Save the new project.

16. Select **BAD** and change the **Role** to **Target**.

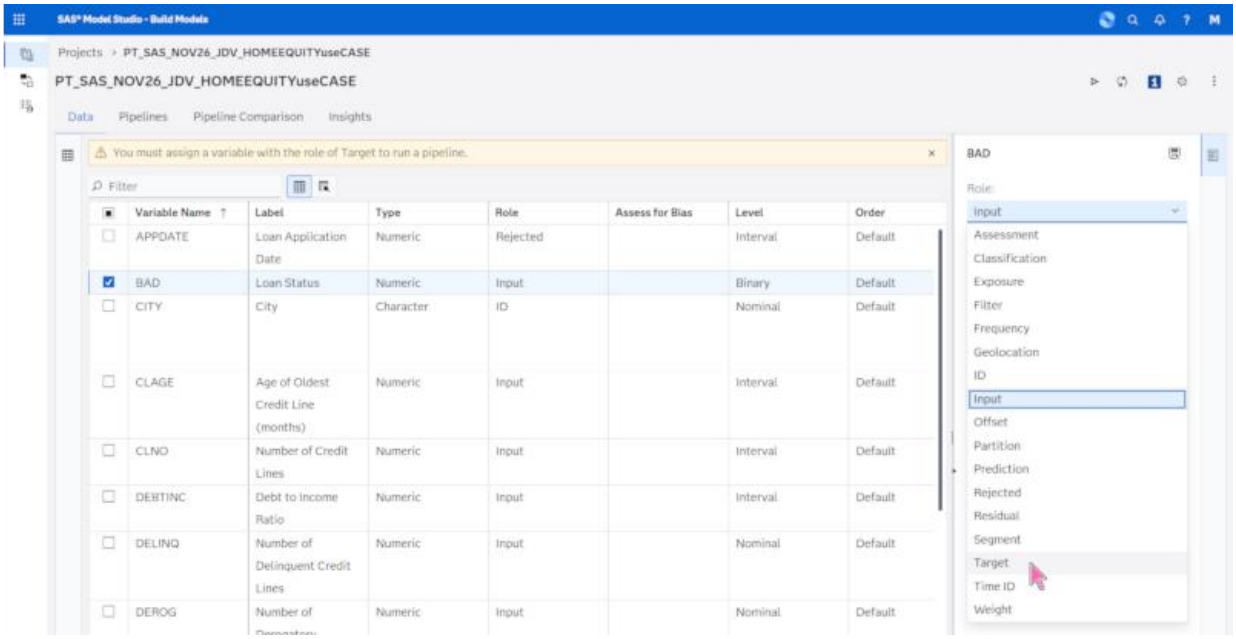


Figure 49: Change the role for the BAD variable.

17. Make sure **BAD** is not selected and then select **JOB** to change its **Role** to **Rejected**.

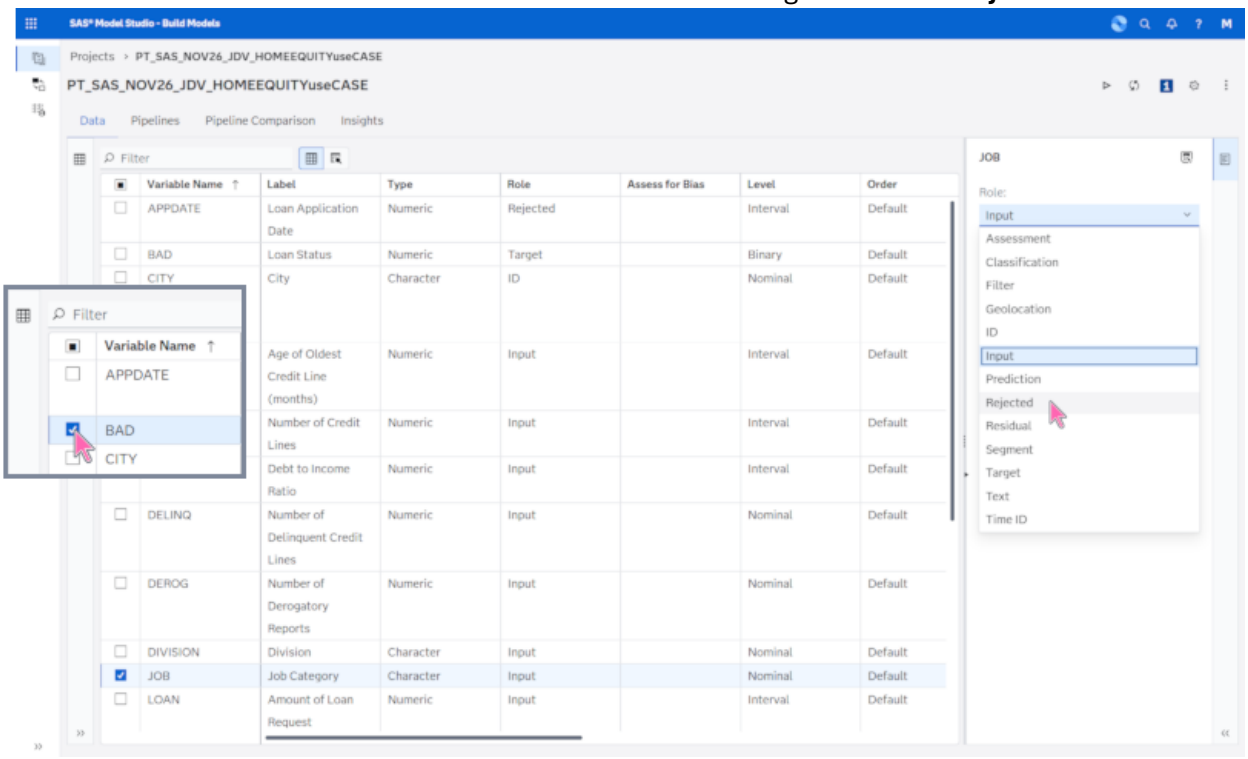


Figure 51: Unselect BAD before changing the role of JOB.

18. Check the box for **Assess this variable for Bias**.

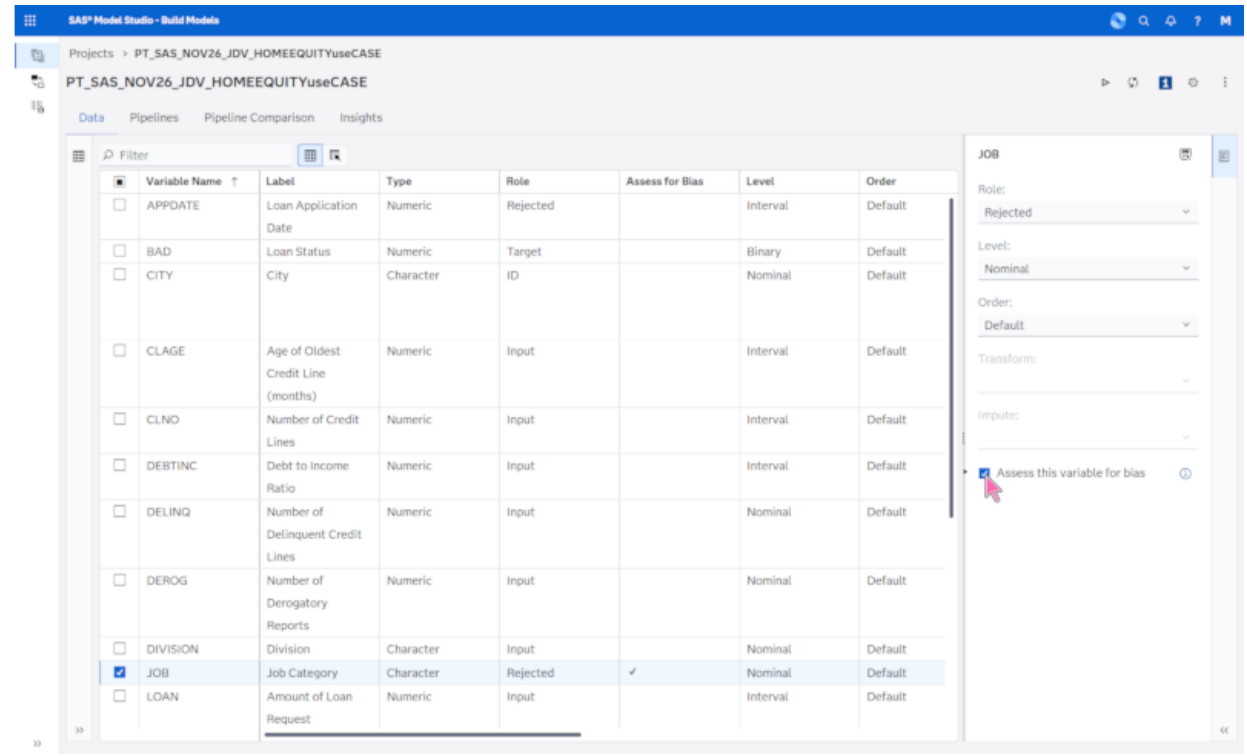


Figure 52: Check the box to assess for bias for the JOB variable.

19. Clear variable selection by double-clicking the top checkbox.

20. Go to the **Pipelines** tab.

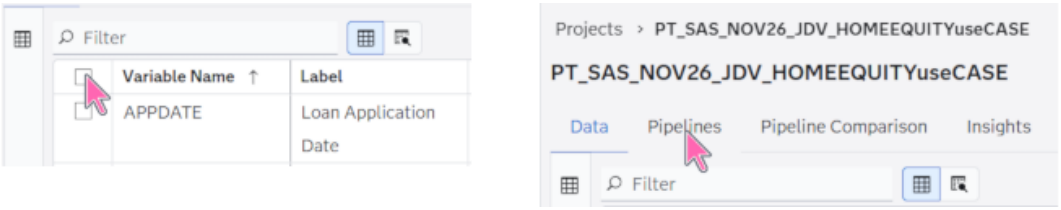


Figure 53: Deselect all variables before selecting the Pipelines tab.

21. Open the **Copilot** side pane.

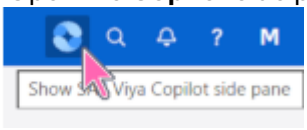


Figure 54: Click the copilot icon to open the copilot side pane.

22. Click **Run this node**.

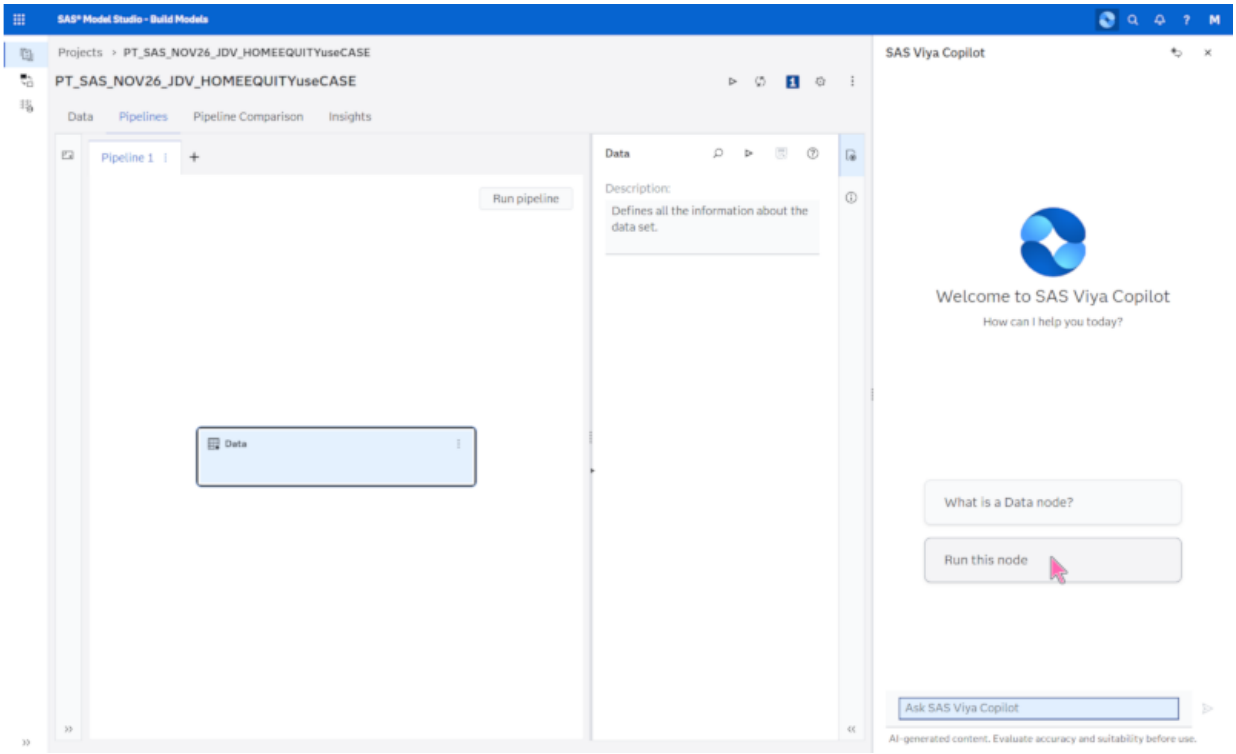


Figure 55: Run the selected Data node.

23. Wait for green check mark.
24. Type **Suggest next node** and click Send or press [enter].



Figure 56: After the checkmark appears ask copilot to suggest the next node.

25. Click **Start exploration**.
26. Click **Add Feature Machine and Gradient Boosting Nodes** (or the node(s) that has the highest KS in your session).
27. Click **Run node(s)**.
28. Click 3 dots and select **Rename**.
29. Type ***CC*_*UNI*_*MMMYY*_*yourinitials*_Copilot_Pipeline** and click **OK**.

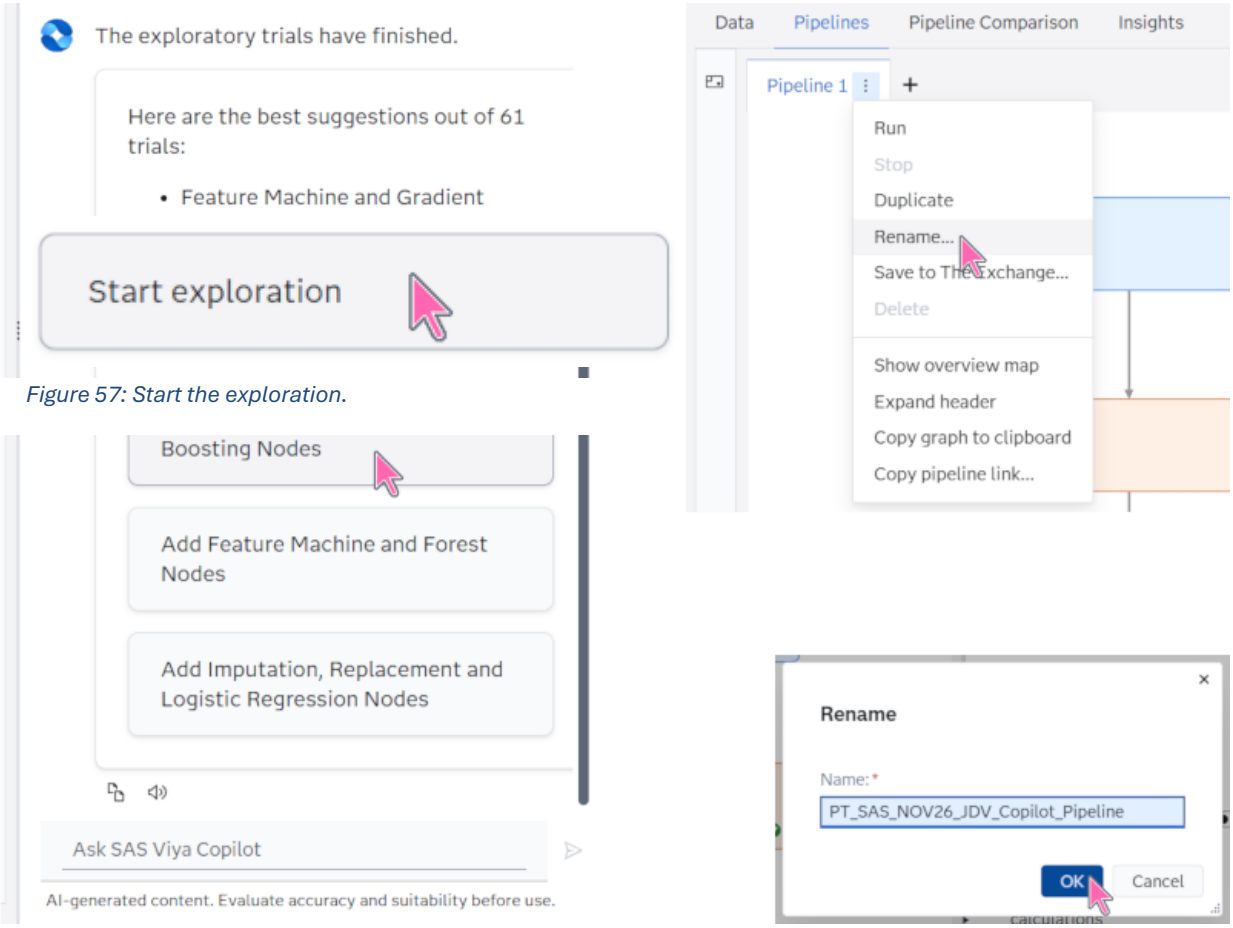


Figure 57: Start the exploration.

Figure 58: Add the node(s) and rename the pipeline.

30. In the Copilot side pane type **Add a logistic regression node** and click **Run node**.
31. Type **Improve the model** and press [enter].
    NOTE: This step takes a while to complete, so we can add another pipeline while this step completes.

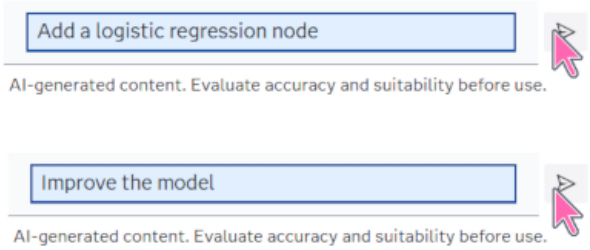


Figure 59: Add a logistic regression node, run it, and improve the model.

32. If you have a small monitor hide the **Copilot** side pane.
33. Click **+** to add a new pipeline.
34. Click **Browse**, search for **advanced** and select **Advanced template for class target**.
35. Click **OK**.

36. Name it **Advanced template for class target** and click **Save**.

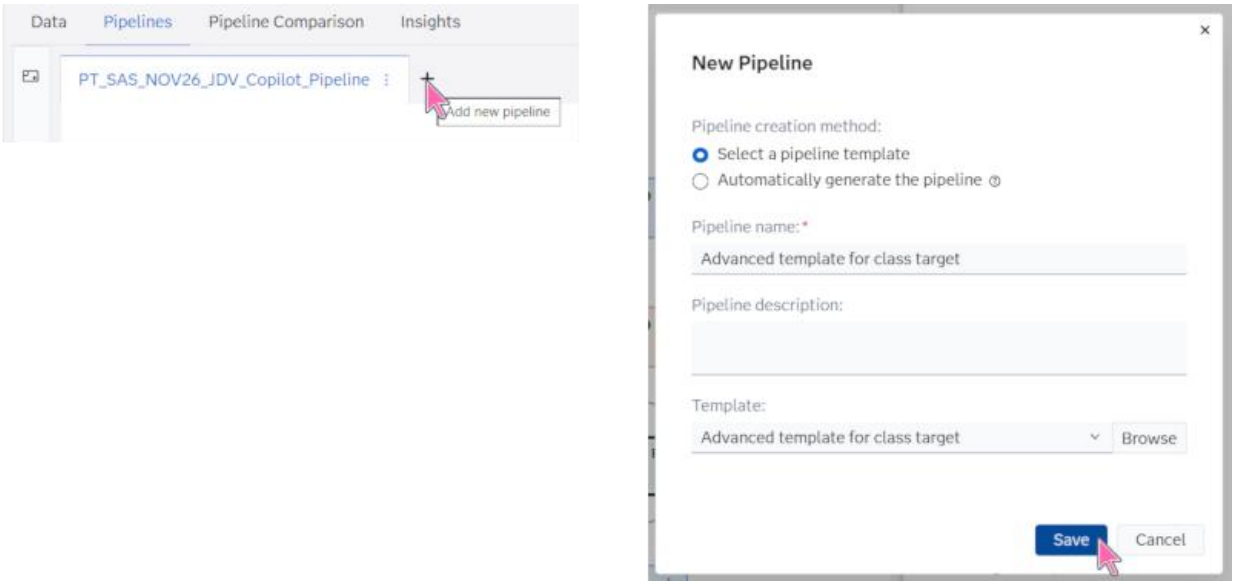


Figure 60: Add a new pipeline.

37. Click **Run pipeline**.

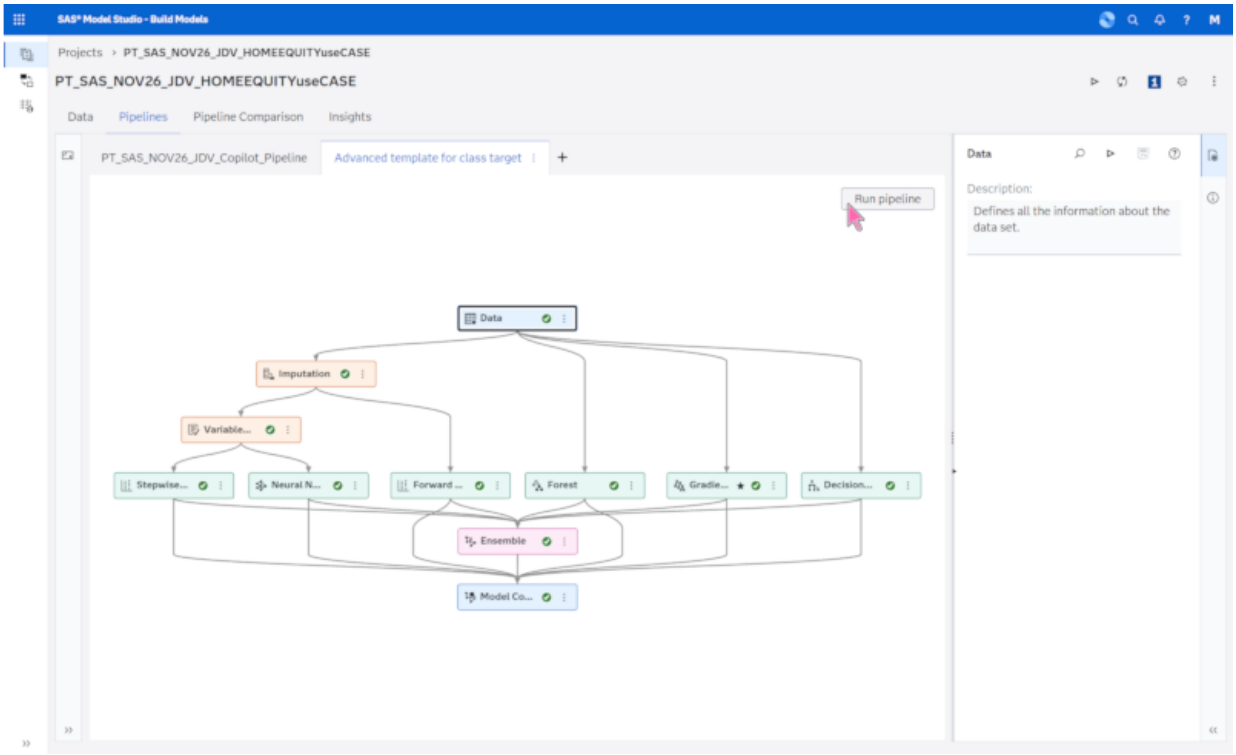


Figure 61: Run the pipeline created based on a template.

38. Return to the original pipeline, open the copilot side pane if you closed it, and click **Apply Improvements**.
39. In Copilot type **What is the difference between Logistic Regression and Gradient Boosting** and press [enter].
    NOTE: Copilot provides a lot of information and can provide more on your request!
    TIP: If you have a small monitor hide the **Copilot** side pane.
40. Open the **Nodes** menu on the left.

41. Expand **Miscellaneous** and drag **Open Source Code** to drop it on the **Data** node.

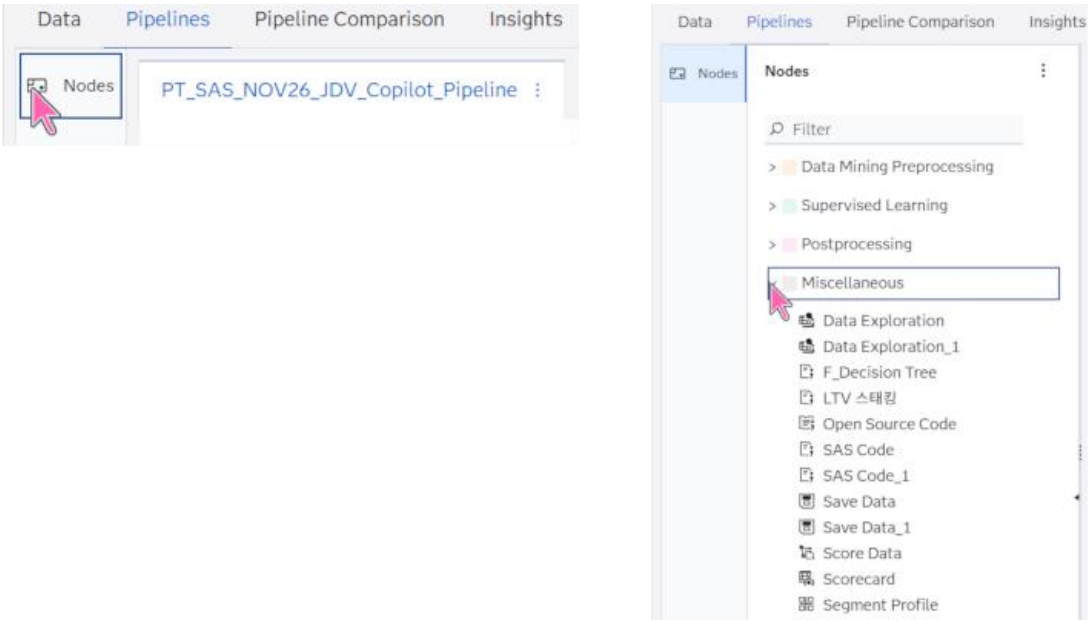


Figure 62:Open the Nodes pane and expand Miscellaneous.

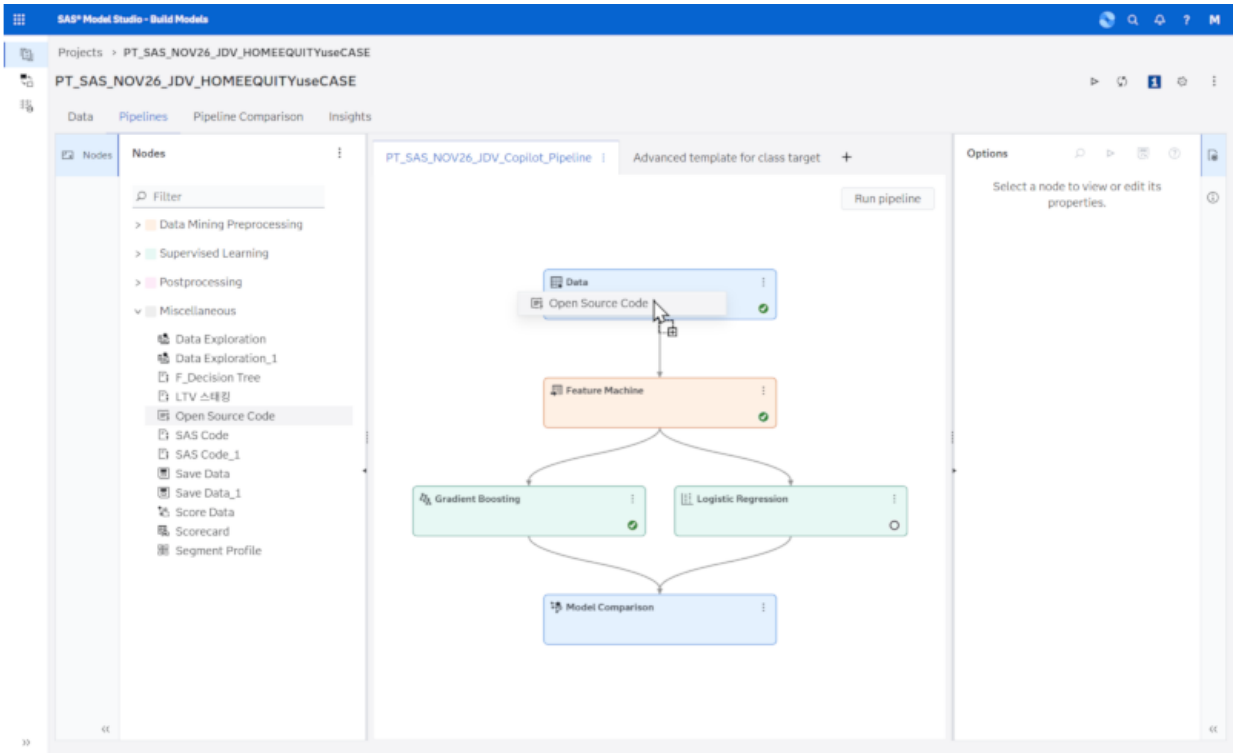


Figure 63: Drag the **Open Source Code** *node and drop it on top of the* **Data** *node.*

42. Click **Open code editor**.

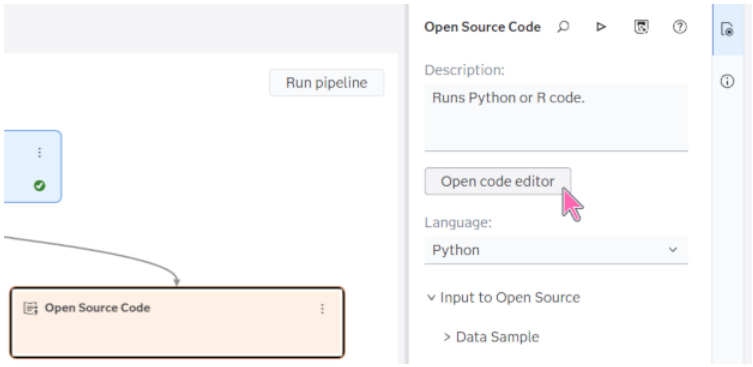


Figure 64: With the open source code node selected open the code editor.

43. Go to GitHub using the link **https://github.com/sassoftware/sas-viya-quick-start**.
44. Open **Train_HOMEEQUITY.txt**.
45. Click **Copy raw file**.

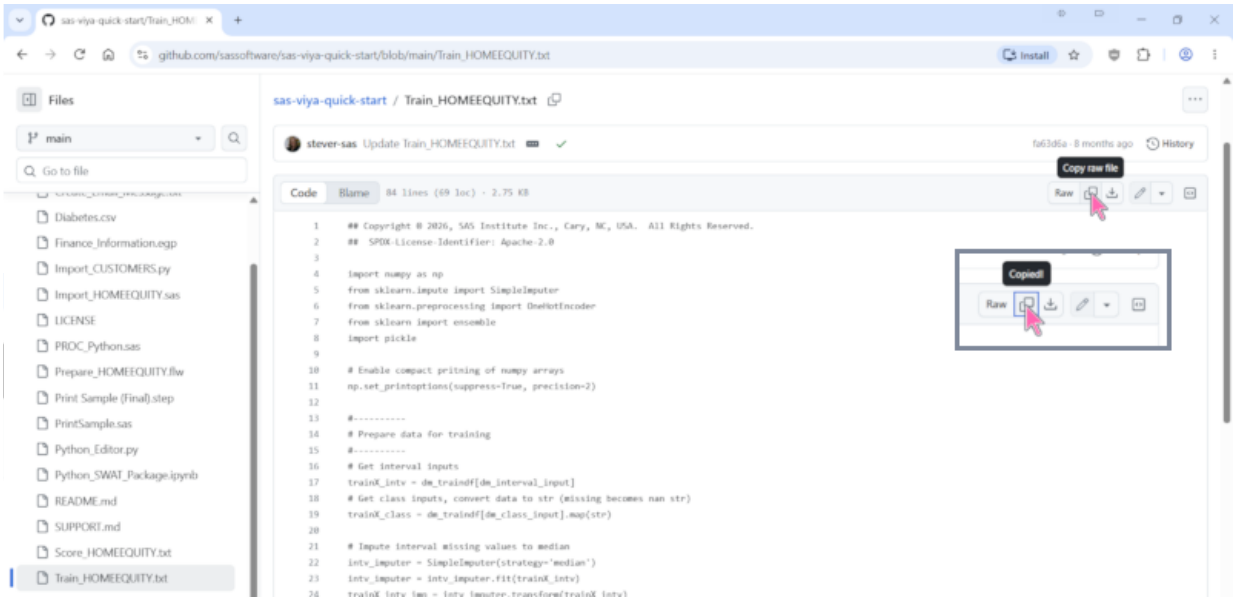


Figure 65: Copy the raw file from github.

46. And in the Training Code section use **Ctrl + V** to paste.

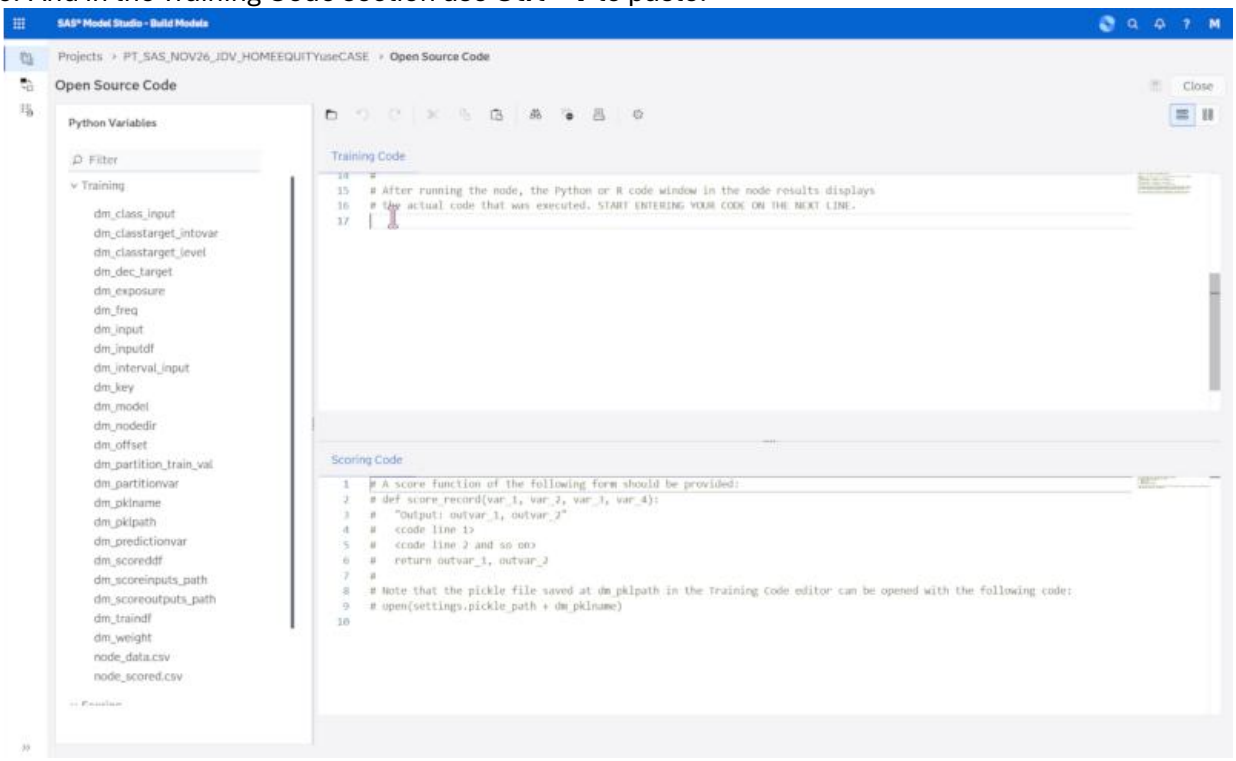


Figure 66: Add the copied text in the **Training Code** *section using* **Ctrl+V**.

47. Return to github.
48. Open **Score_HOMEEQUITY.txt**.
49. Click **Copy raw file**.
50. Paste in the **Scoring Code** section using **Ctrl + V**.
51. Click **Save**.
52. Click **Close**.

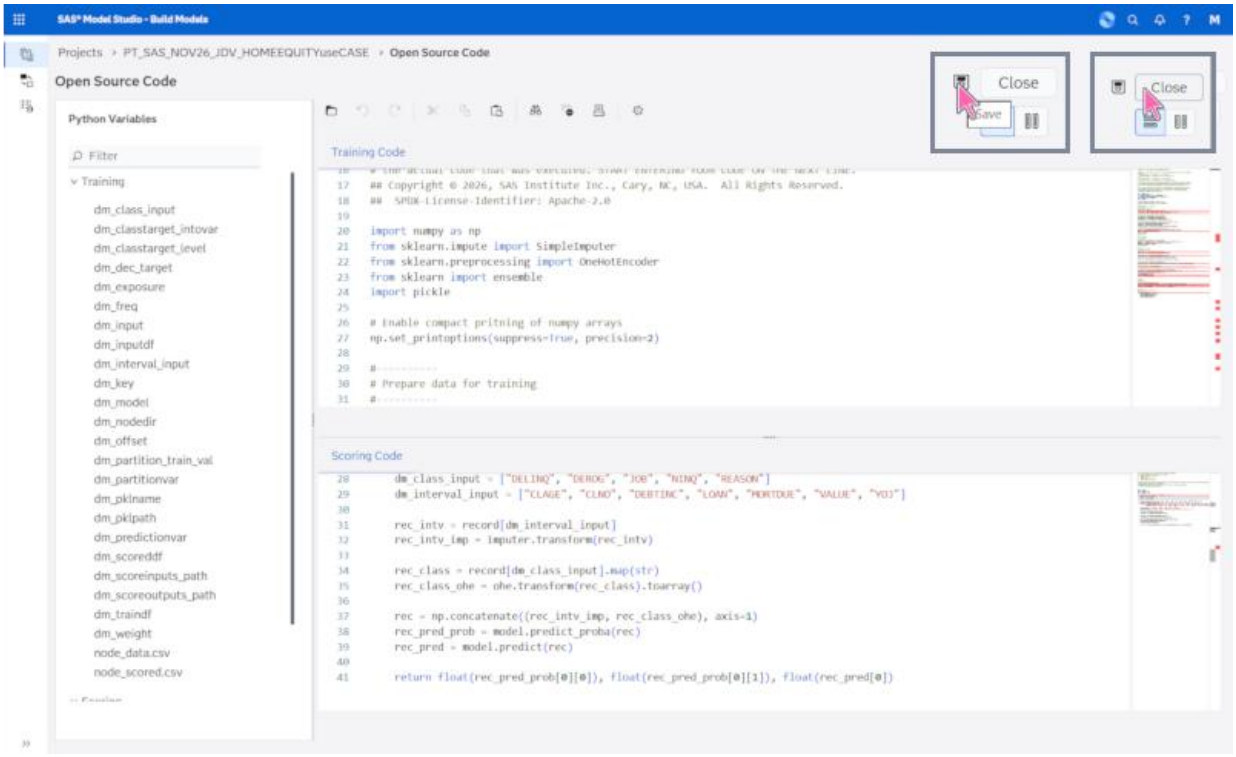


Figure 67: Paste the scoring code in the **Scoring Tab** *section and click* **Save** *and* **Close**.

53. In the Options pane (with the Open source code node selected) expand **Data Sample** and remove the checkmark for **Include SAS formats**.
54. Right-click **Open Source Code**.
55. In the Move menu select **Supervised Learning**.

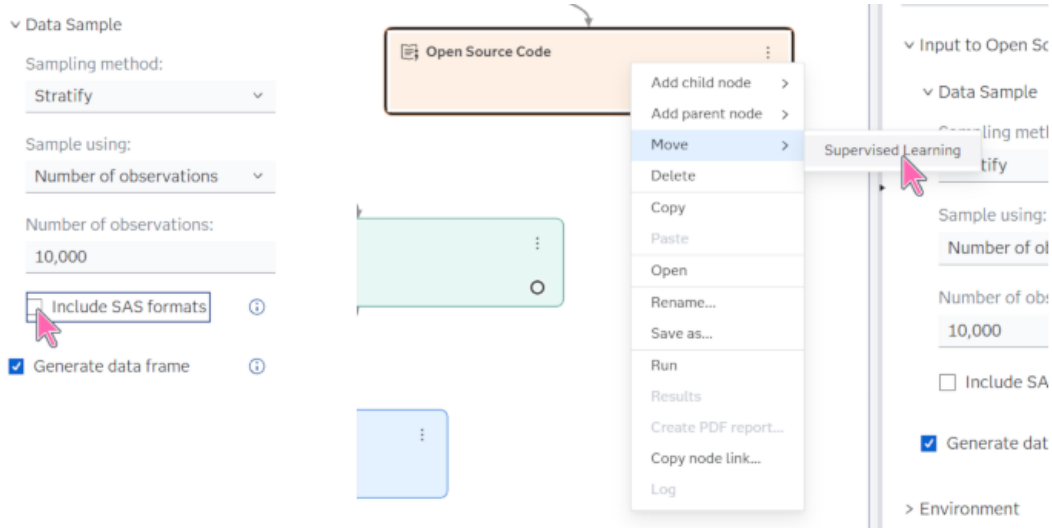


Figure 68: Move the open source code node to supervised learning.

56. Click **Run pipeline**.

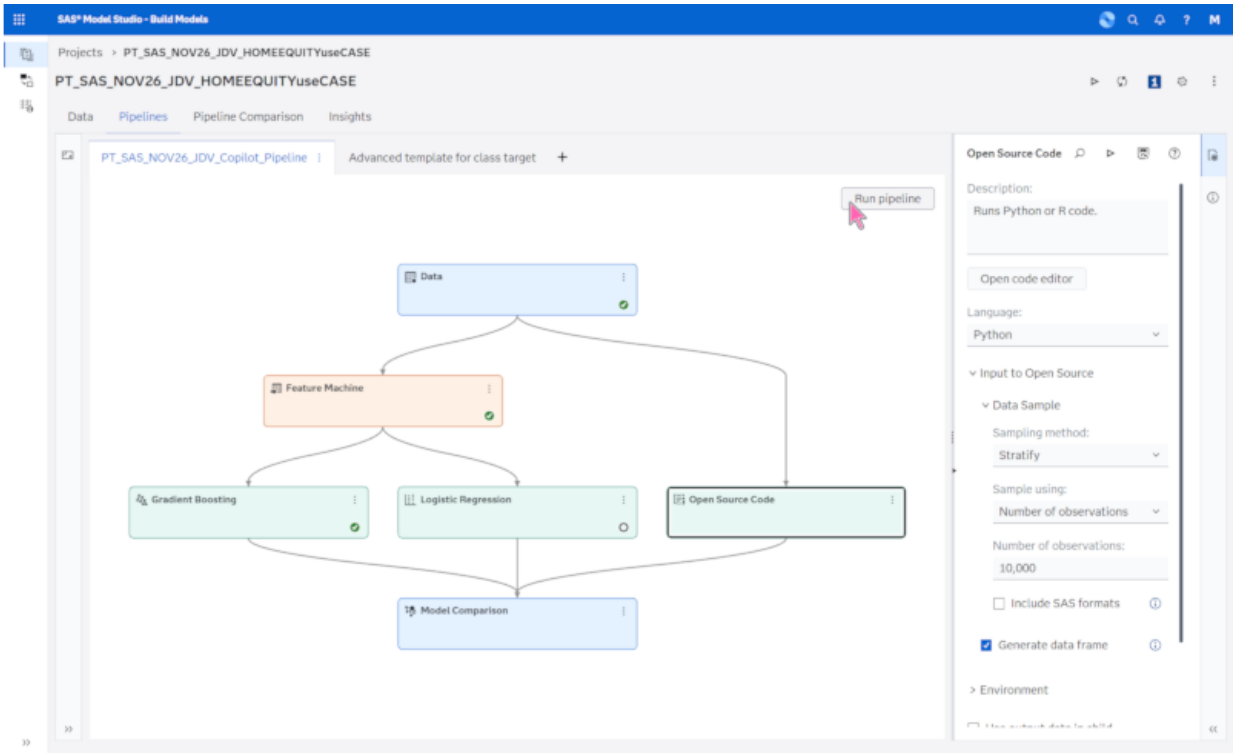


Figure 69: Run the pipeline.

57. Right-click the **Model Comparison** node.
58. Click **Results** and analyze the provided information.

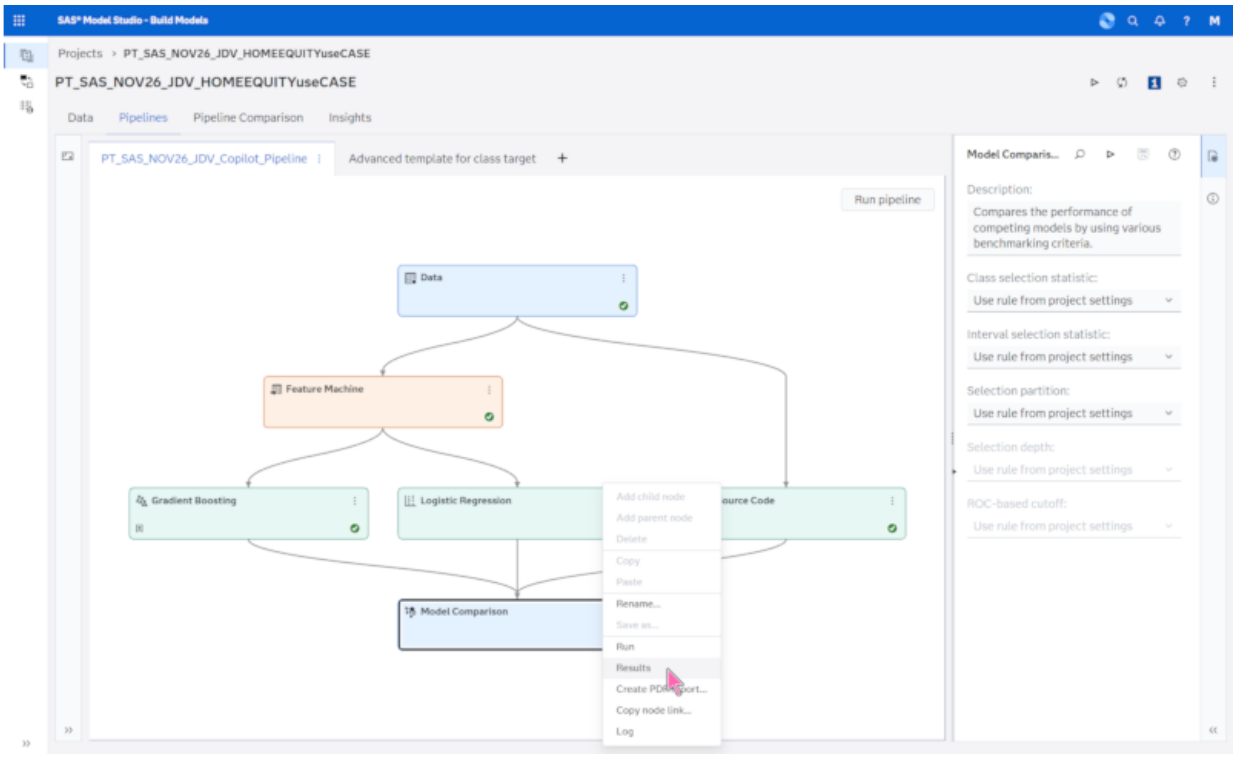


Figure 70: Open the results for the ***Model Comparison****.*

59. Click **Maximize view** and verify the available details for the **Model comparison** object. Take special of those for the **Champion** model.

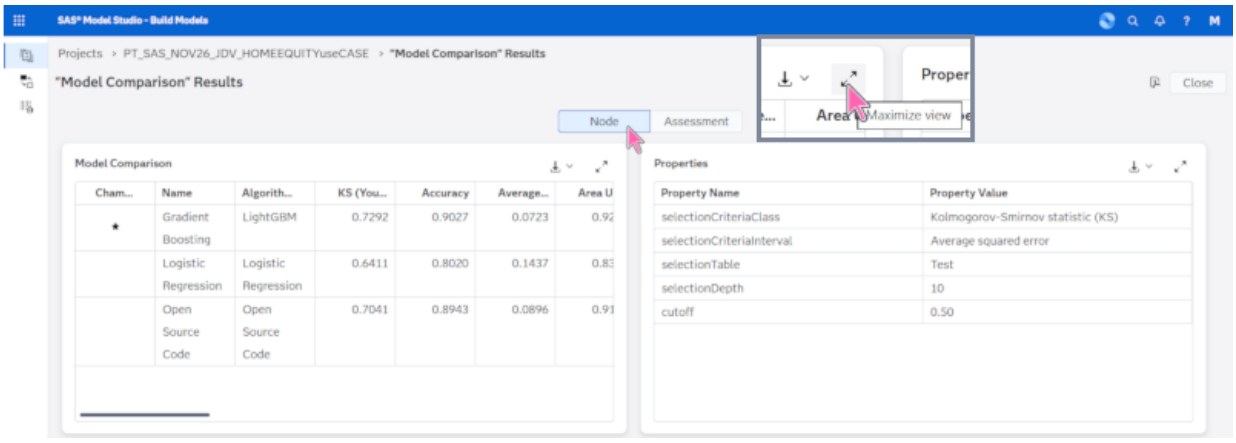


Figure 71: In the Node tab and most other tab you can maximize the objects.

60. Click **Close**.



Figure 72: To leave the maximized object box click **Close**.

61. Open the **Assessment** pane to analyze the available options for the various objects. Change the graph types to analyze all aspects.
62. Click **Export to PDF** and verify the available options.

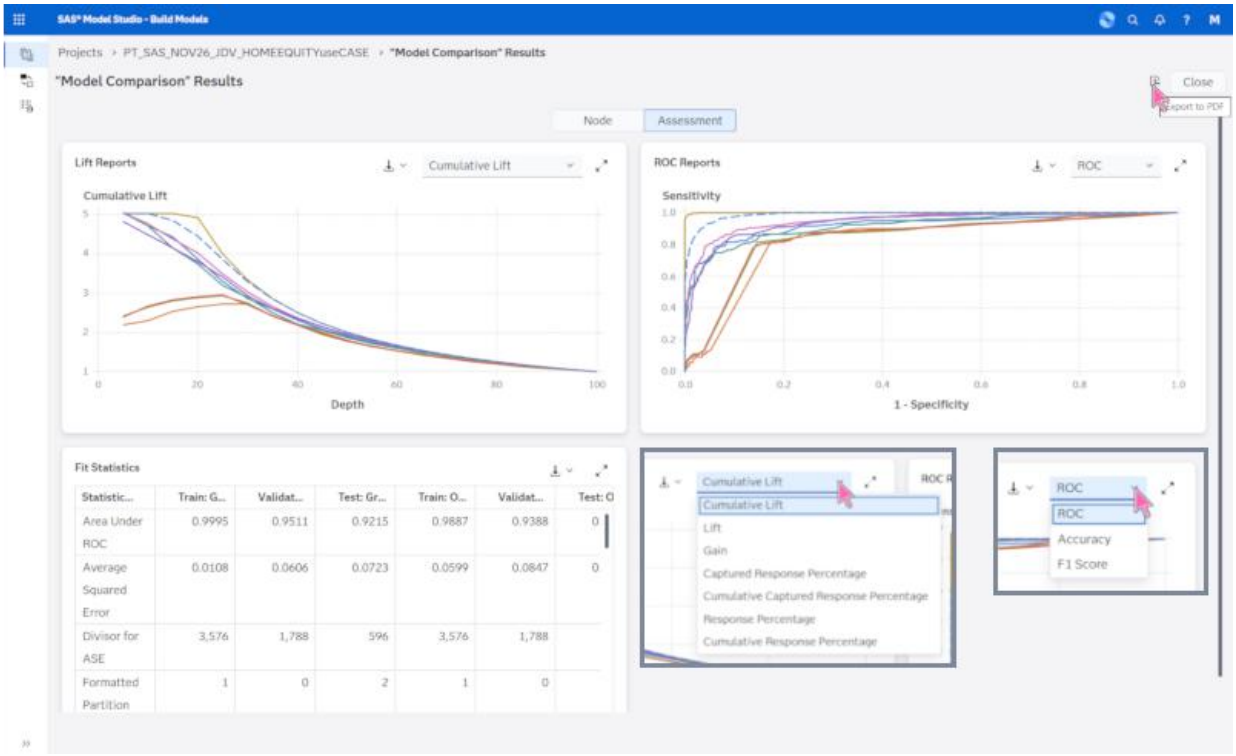


Figure 73: The objects in the assessment tab. Graph types can be changes easily.

63. Click **Cancel**.
64. Click **Close**.

65. Right-click the **Gradient Boosting** node. It has the champion icon in the bottom left corner.
    NOTE: If you have a different champion node select that node.

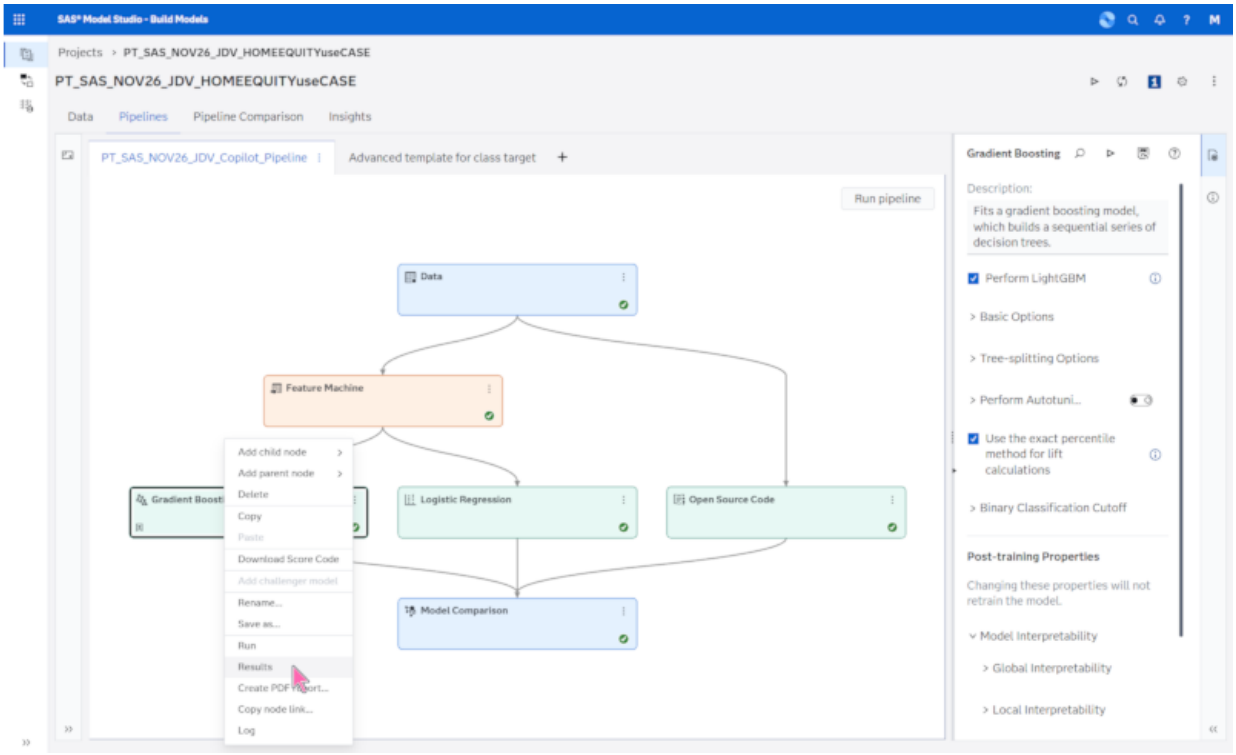


Figure 74: Open the **Results**.

66. Click **Results** and analyze the provided information in the **Node** tab. Use the **Maximize view** option for the objects.
    NOTE: The text produced with NLG that provides explanatory information.

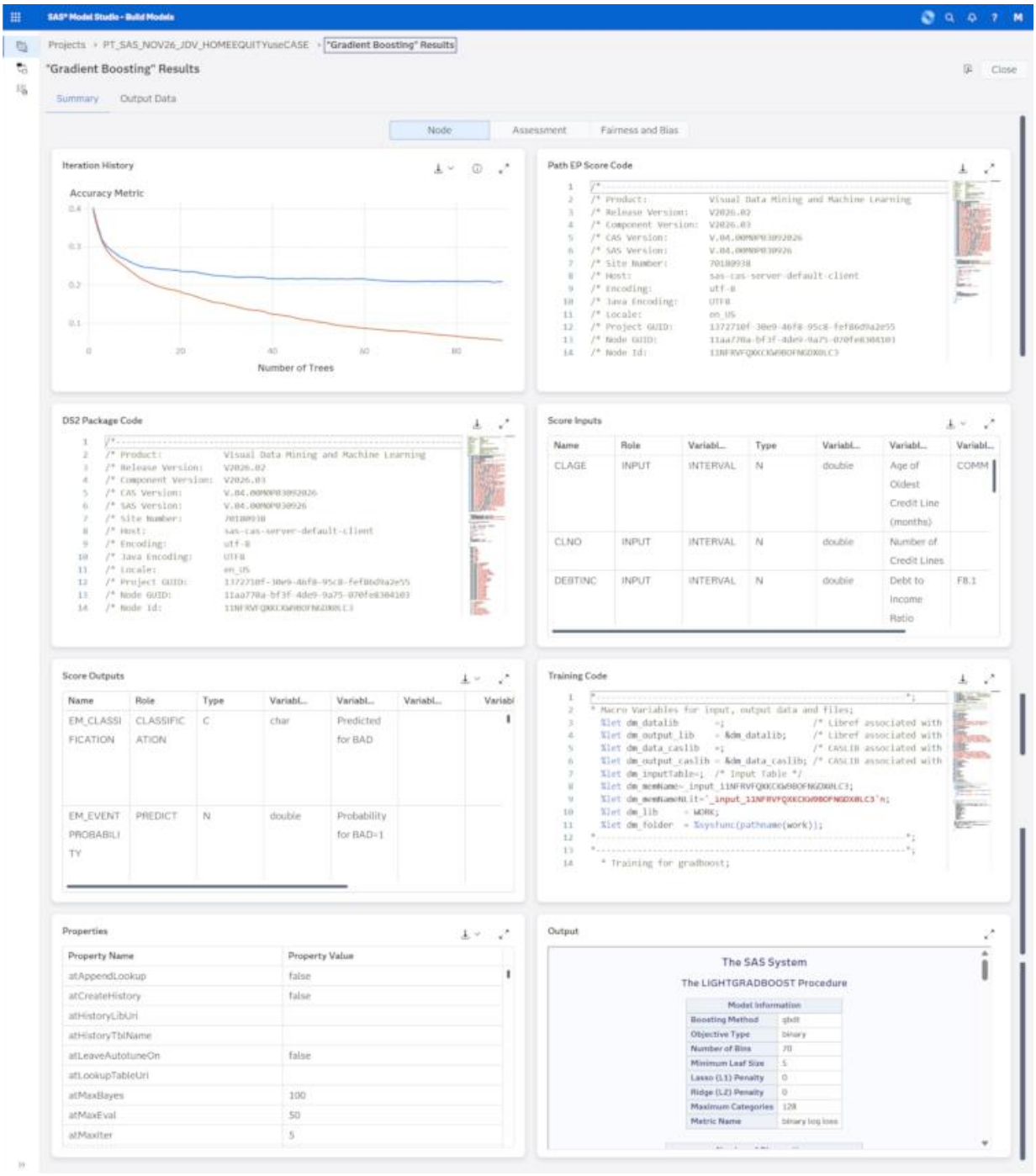


Figure 75: The results for the supervised learning nodes have detailed information.
Don't forget to maximize the objects in the results view for access to NLG explanations.

67. Open the **Assessment** tab and anlyze the information provided in the many objects.

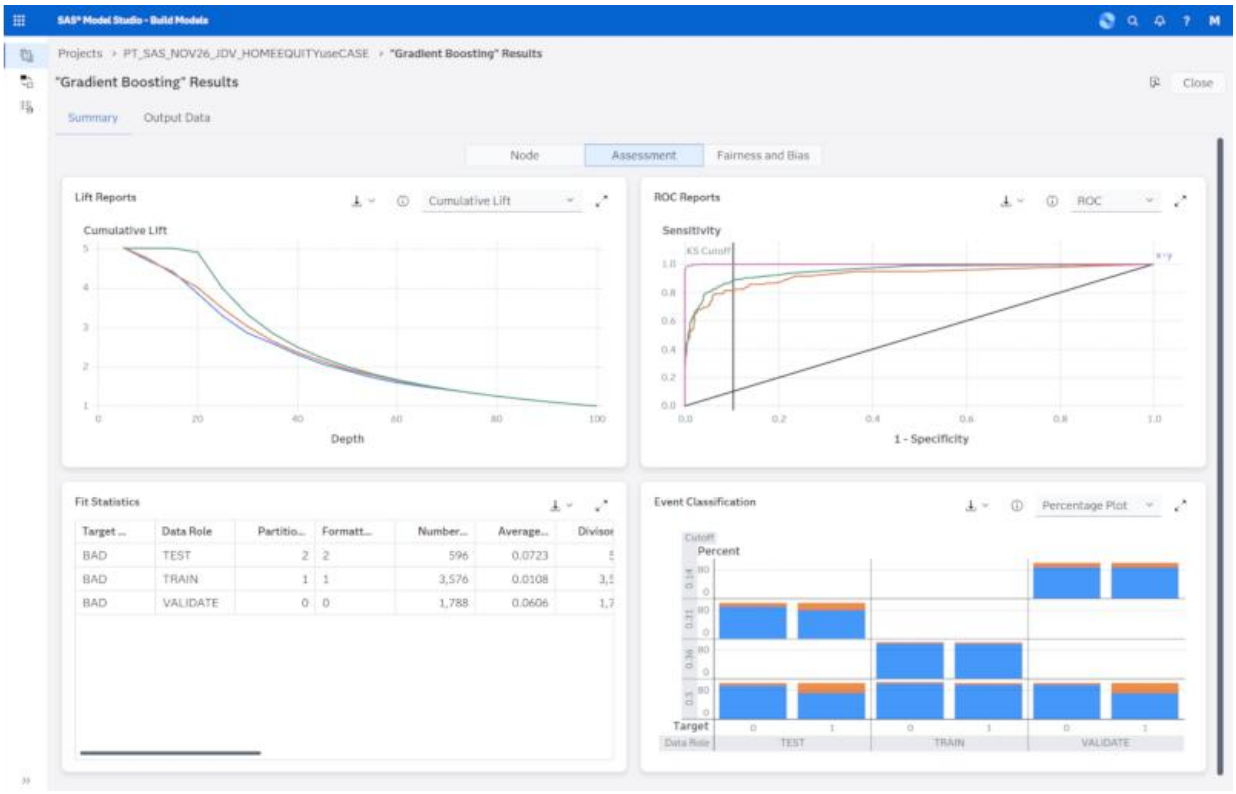


Figure 76: The objects in the assessment tab.

68. Open the **Fairness and Bias** tab that has the information related to fairness and bias for the **JOB** variable.
    NOTE: Take special notice of the differences for **Performance Bias**.
    QUESTION: What can explain these differences in your opinion?

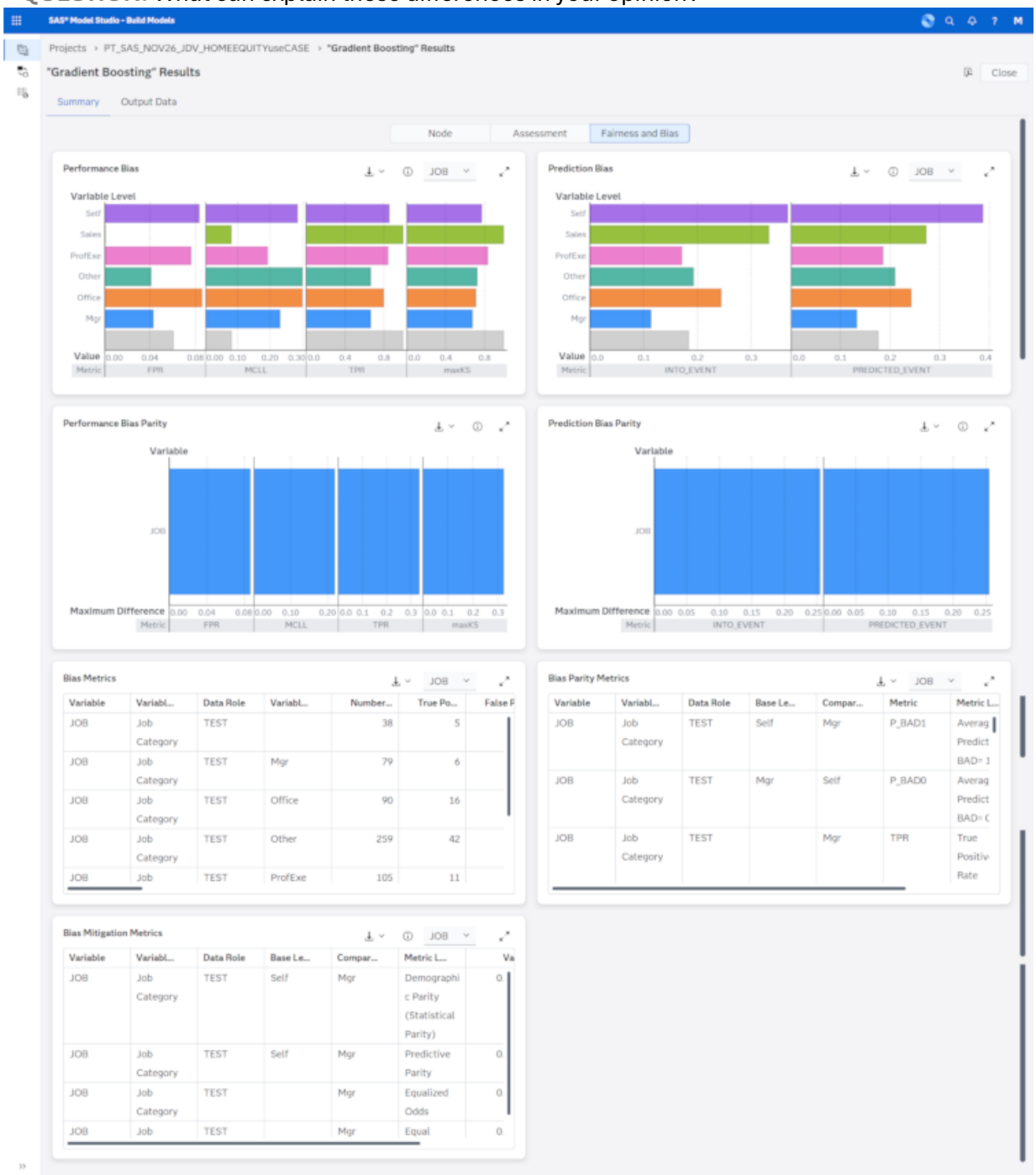


Figure 77: The objects in the **Fairness and Bias** *tab.*

69. Click **Close**.
70. Right-click **Gradient Boosting**.
71. Select **Rename**.
72. Type ***CC*_*UNI*_*MMMYY*_*yourinitials*_Champion**.

73. Click **OK**.

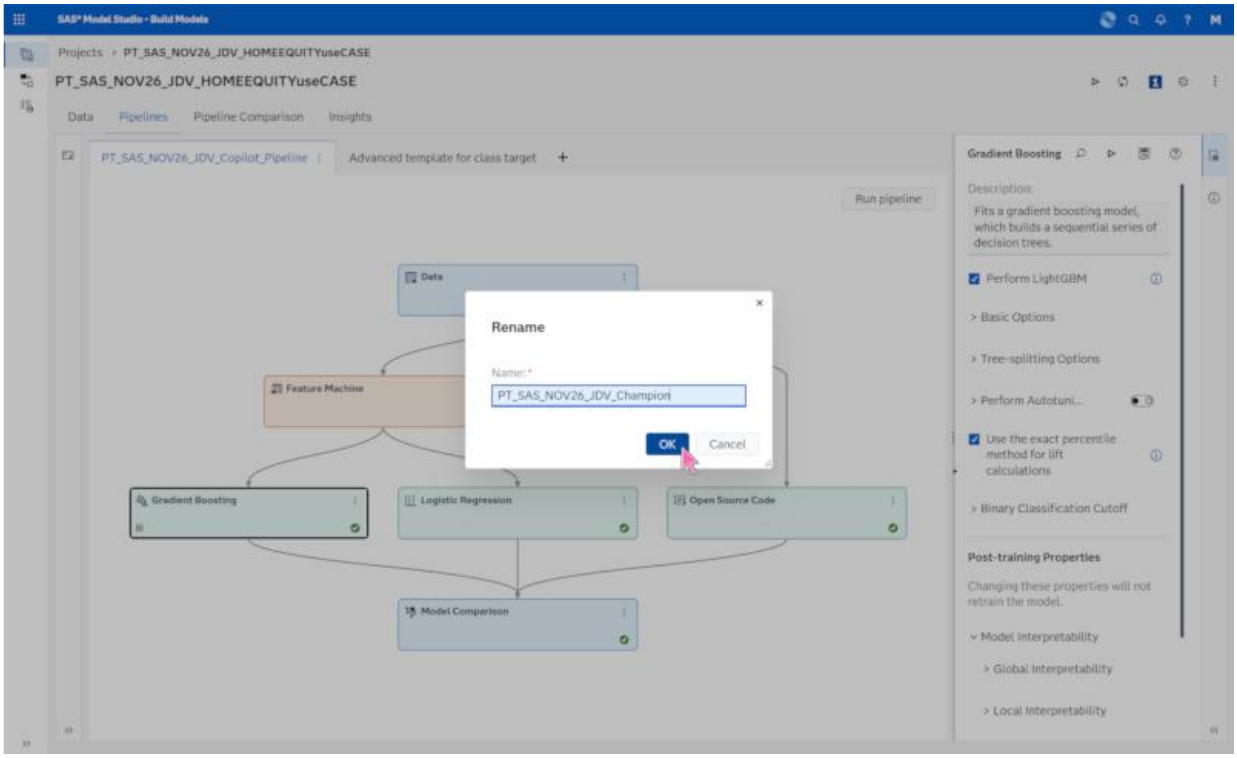


Figure 78: Rename the champion node in all pipelines. **TIP:***The champion node has a small icon bottom left.*

74. Also rename the champion model in the pipeline named Advanced template for class target.
    NOTE: Renaming these champion models will make it easier to find them when the overall champion model, registered in the next step, is needed for the decision flow in part 6.
75. Click **Pipeline Comparison** to review the opened tab and analyze the items. The overall champion model has a star in table at the top.

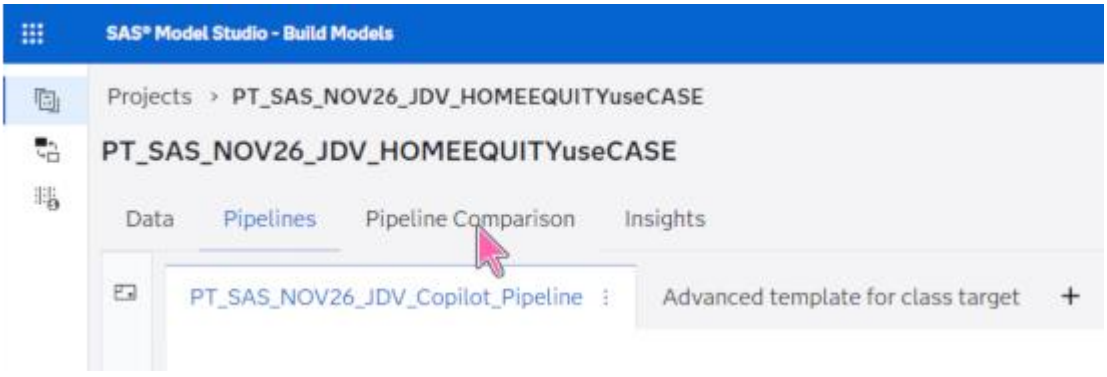


Figure 79: Open the **Pipeline Comparison** *tab.*

NOTE: The full pipeline comparison tab is displayed on the next page.

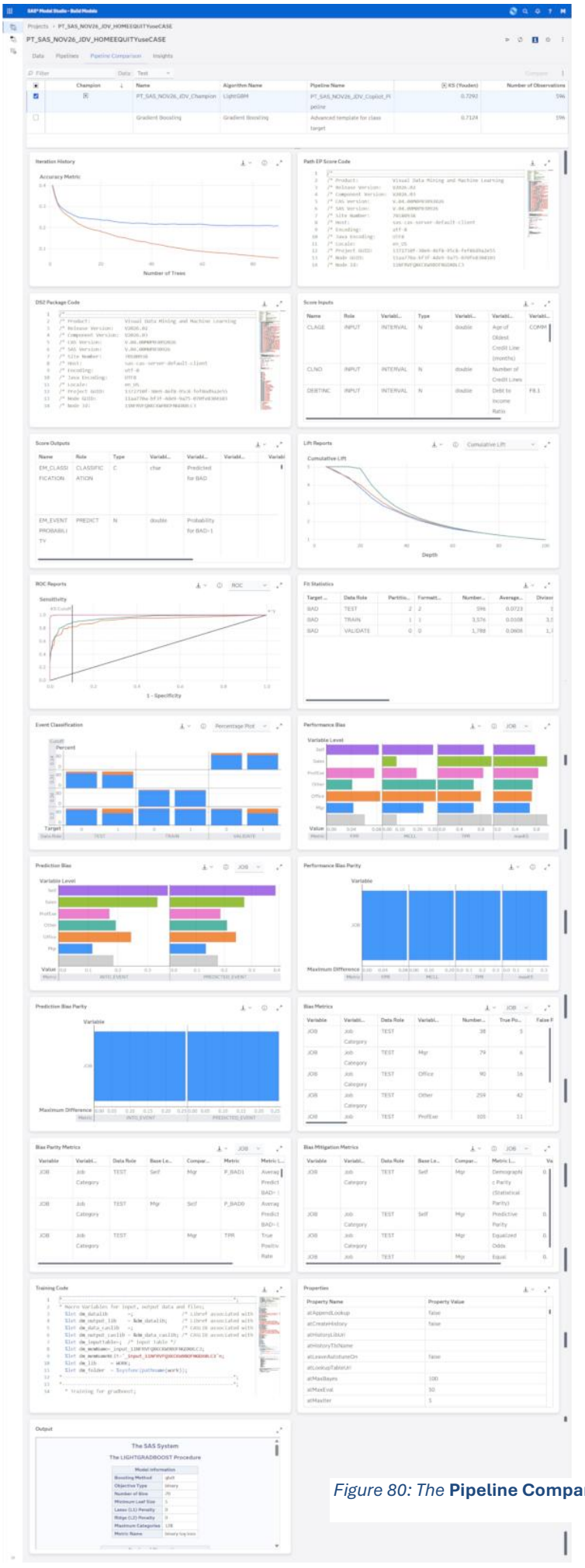


Figure 80: The **Pipeline Comparison** *page and its objects.*

76. Right-click the **Champion** row.
77. Click **Register models**.

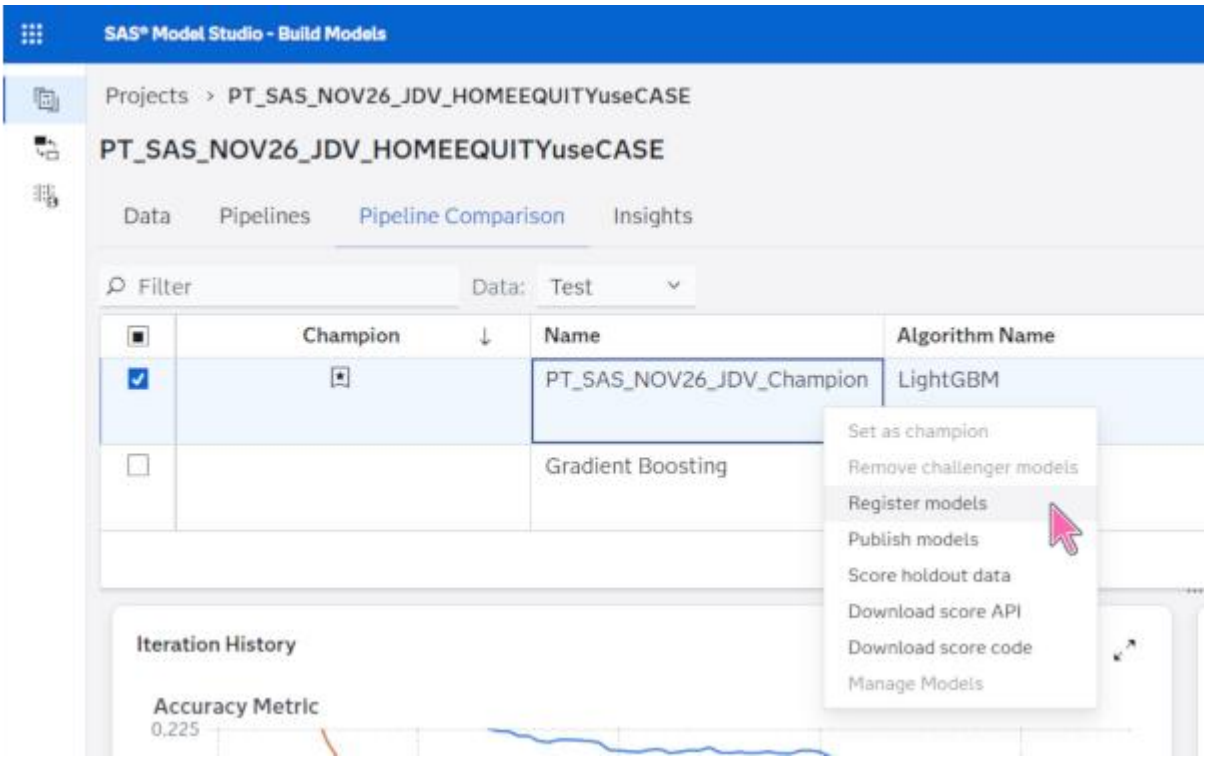


Figure 81: Register the overall champion model.

78. Click **OK**.
79. Click **Close**.

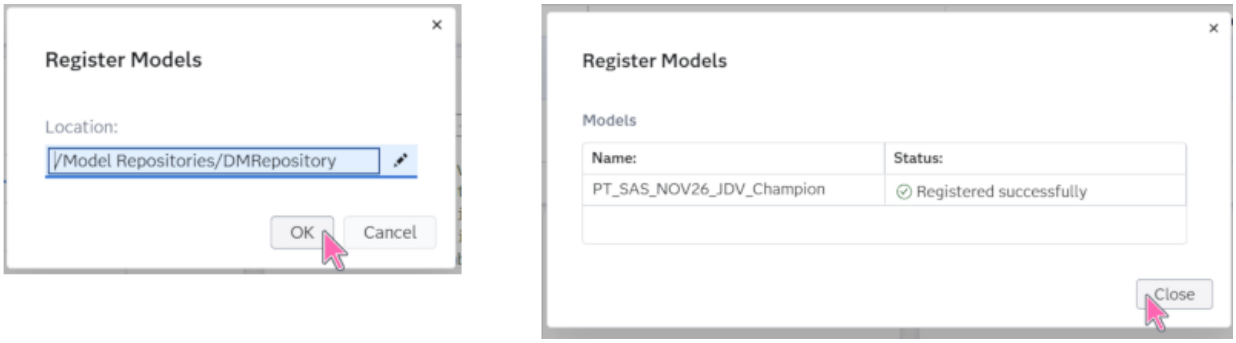


Figure 82: Complete the registration process.

80. Select **Insights**.

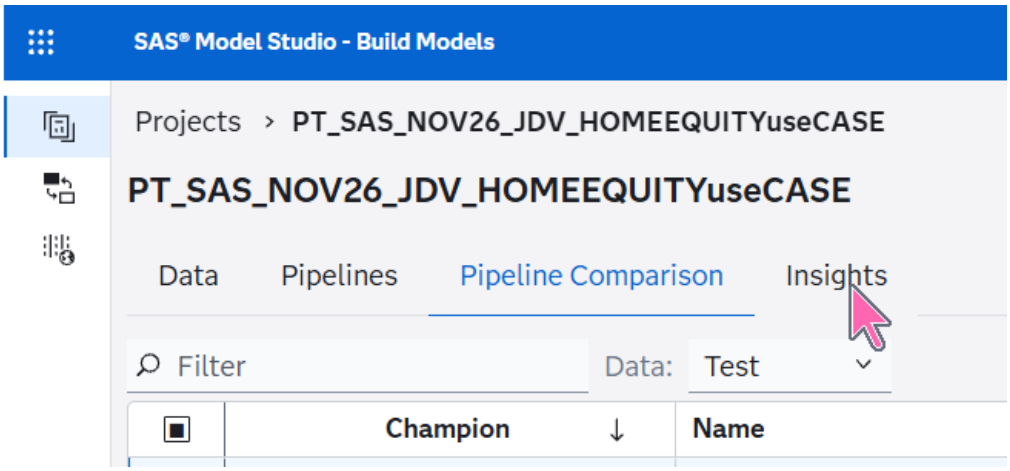


Figure 83: Open the Insights tab.

81. The **Project Summary** is provided through NLG.
82. Click **Maximize view** to analyze details for objects.

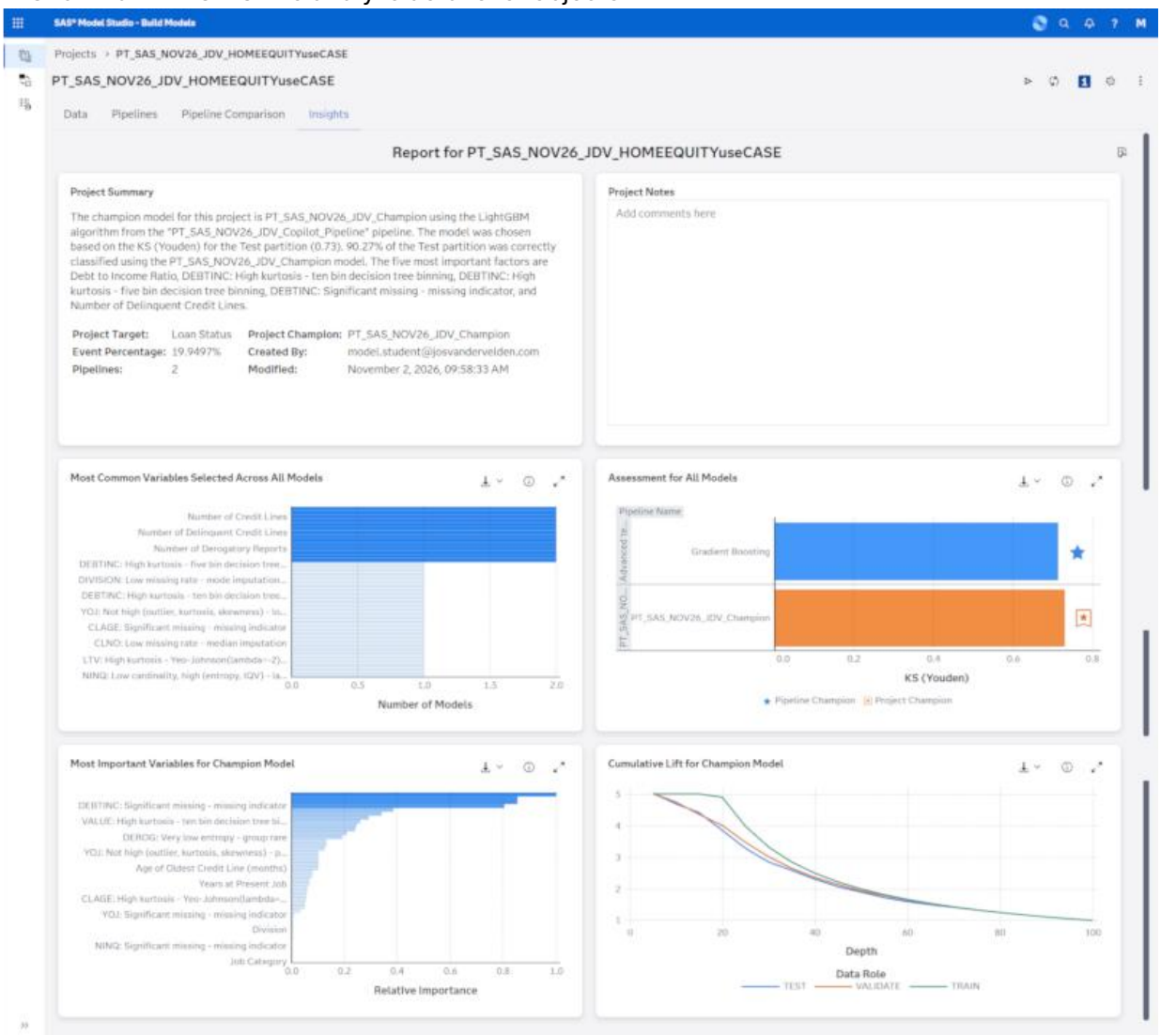


Figure 84: The Insights page and its objects.

Course 8, listed in the appendix, provides detailed information and practices to learn more about some of the available supervised machine learning algorithms in SAS Viya for Learners.

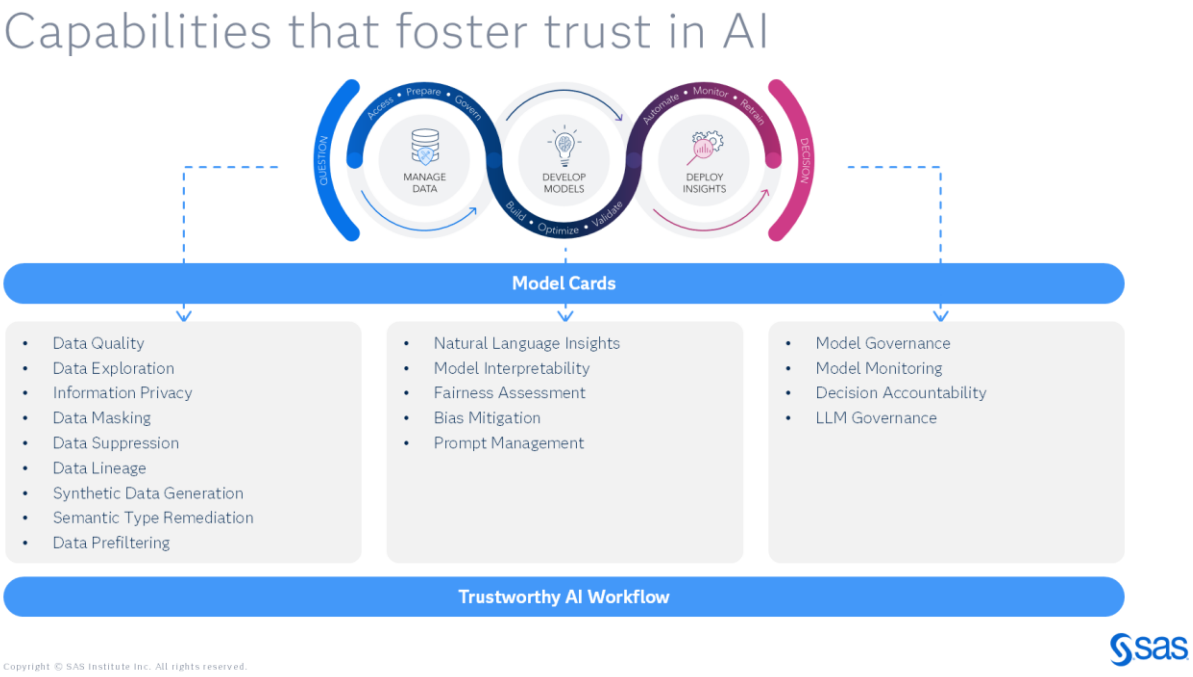
Trustworthy AI across the data and AI life cycle

Advances in AI have reshaped business, societies and individual lives, and has given us a deeper understanding of our planet and the cosmos. We've only just begun to see how it could unlock human potential. But without the right safeguards, it could also produce harm. That's why it's essential to create AI systems that are ethical, equitable and sustainable.

The SAS Trustworthy AI Life Cycle outlines steps for evaluating and deploying a more trustworthy AI system. It aims to make the U.S. National Institute of Standards and Technology's (NIST) recommendations, standards, and best practices for AI risk management easier to adopt and follow.

The life cycle helps organizational stakeholders specify individual roles and expectations, gather required documentation, and outline factors for consideration. As a result, teams can produce documentation to support the assertion that the organization has done its due diligence to provide evidence that the model is fair and its processes do not cause harm.

The SAS Trustworthy AI Life Cycle is a stepwise process organized into sections. Many tasks in the life cycle reflect guidelines put forth by the NIST AI Risk Management Framework Playbook as it appeared in October 2023. You can find more information about the trustworthy AI life cycle sample workflow via this GitHub link.



A Trustworthy AI approach injects fairness and oversight at every step of the AI and analytics journey, so you can tackle complex data and analytics challenges in a responsible manner.

SAS Viya is an end-to-end data and analytics platform that enables trustworthy practices at every stage of the AI and analytics lifecycle, from development to deployment. With intuitive visual interfaces, a centralized platform that enhances and encourages collaboration, AI-driven automation, and more, anyone can generate trusted outcomes from their data.

The SAS Model Manager model card provides an easy-to-understand summary of a model, its training data, performance, intended use, limitations and governance. It helps stakeholders assess whether the model is appropriate, trustworthy and being used responsibly. When a model moves into production, the model card also provides an excellent starting point for documenting the AI project and the information needed to manage it responsibly throughout its life cycle.

The model card does not declare that a model is trustworthy. It provides evidence that people can use to assess its trustworthiness.

Task 5: View a model card

In this task, you open and review the model card for the registered champion model. The model card is a good starting point for documentation. It should be completed with relevant information.

Step-by-step instructions

These instructions assume you have already entered a SAS Viya for Learners session.

1. Open **Applications menu** (nine dots).
2. Select **Manage Models**.
3. In the left menu click **Models**.
4. In the search field type ***CC*_*UNI*** (for me that is **PT_SAS**) and click the search icon.

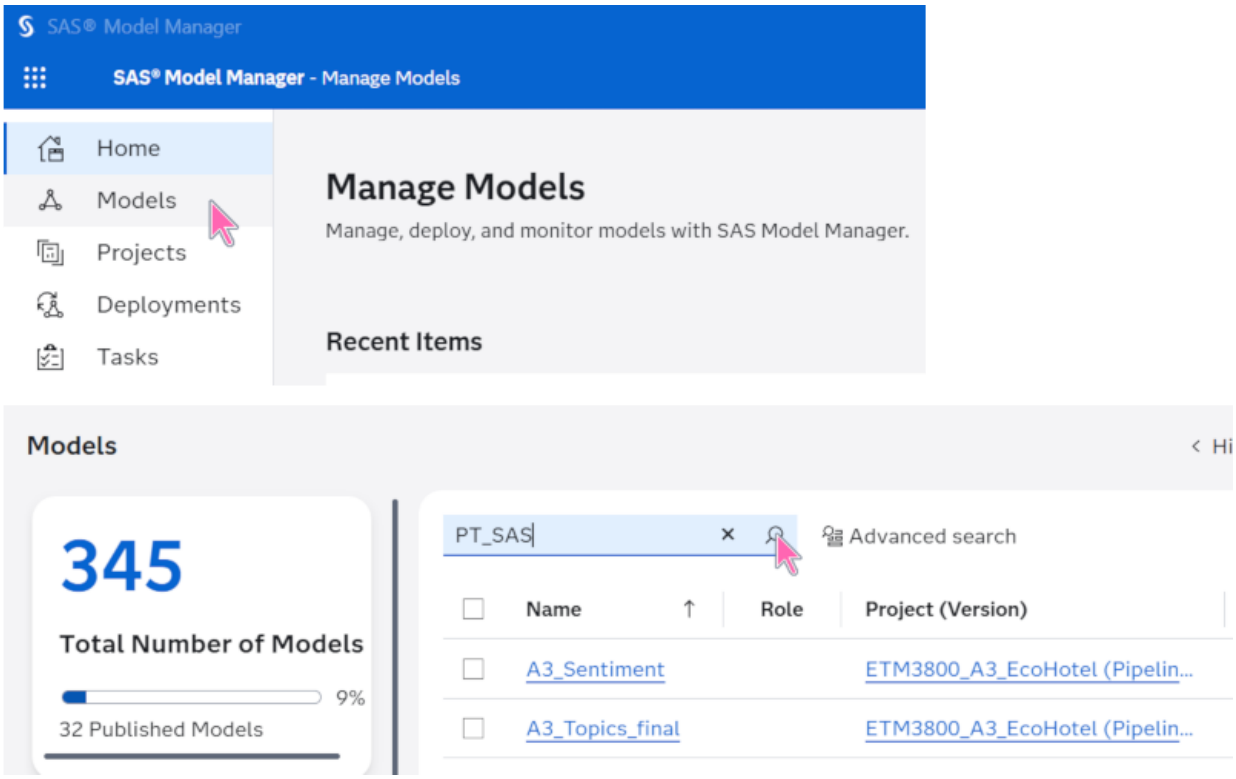


*Figure 85: Open the models menu item and search for **CC*_*UNI***.*

5. Click the model with the name you provided.

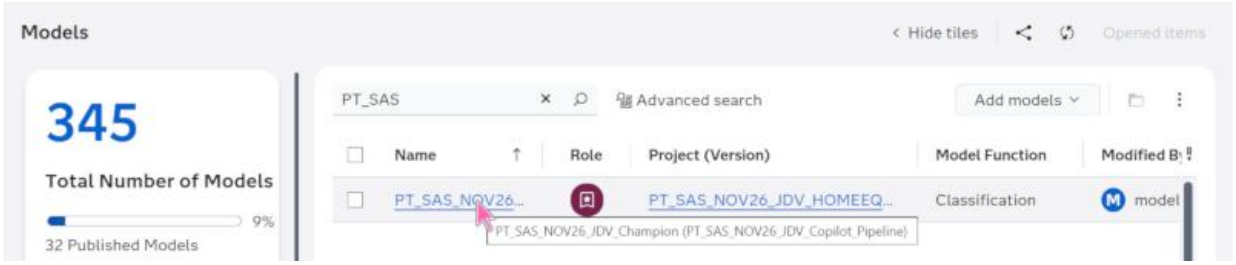


Figure 86: Open the model you have registered.

6. Review the model card.

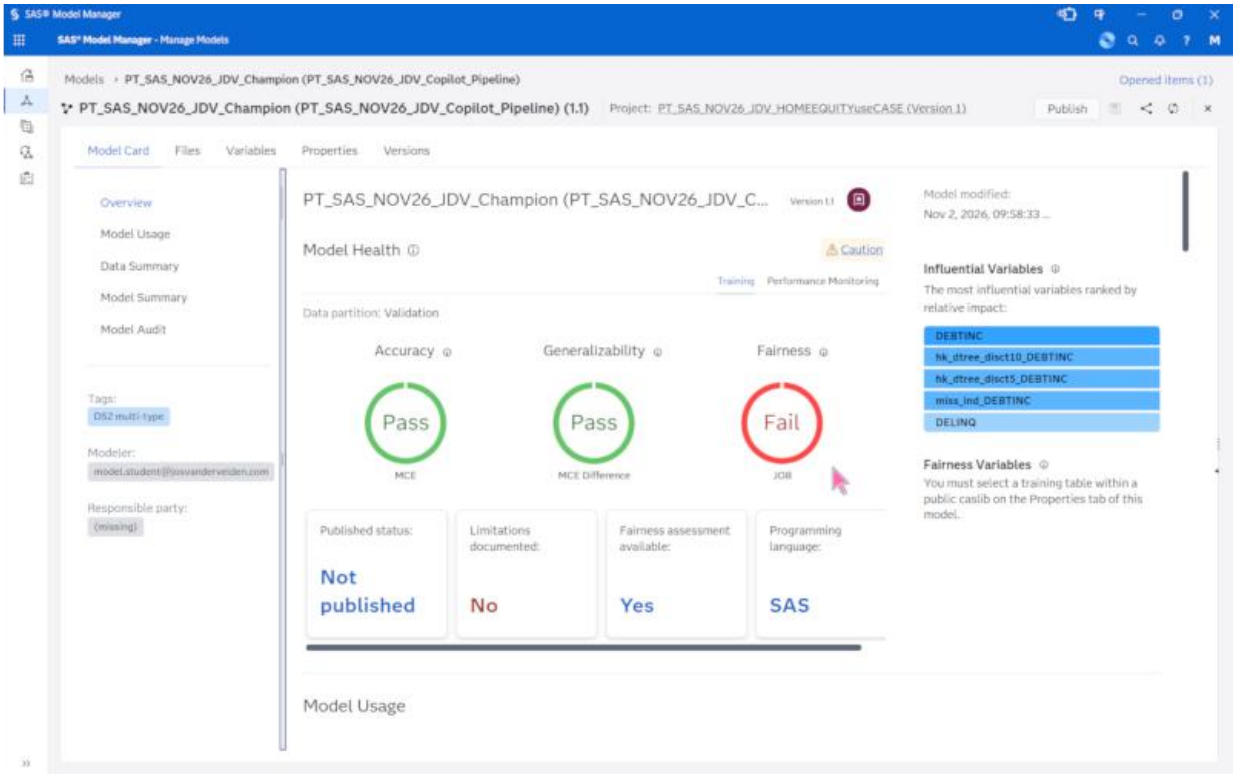


Figure 87: The initial view of the model card.

A standard hands-on session does not provide enough time to explore and customize the model card in detail. If you are interested please send an email to emeagapsupport@sas.com with a request for more details on model cards referring to this document.

Task 6: Create a decision flow

You learn how to integrate the champion model into a decision flow that can automatically send emails to loan applicants to inform them that their loan request has been accepted or rejected. The email provides a reason for the decision.

Step-by-step instructions

These instructions assume you have already entered a SAS Viya for Learners session.

1. Open **Applications menu** (nine dots).
2. Select **Build Decisions**.
3. In the left menu click **Decisions**.

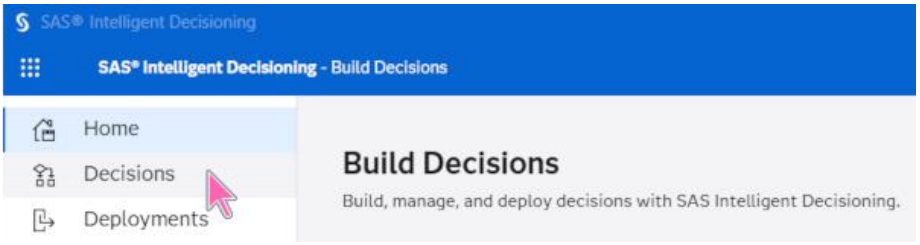


Figure 88: Open **Decisions**.

4. Open the **Loan Decision (Final)**.

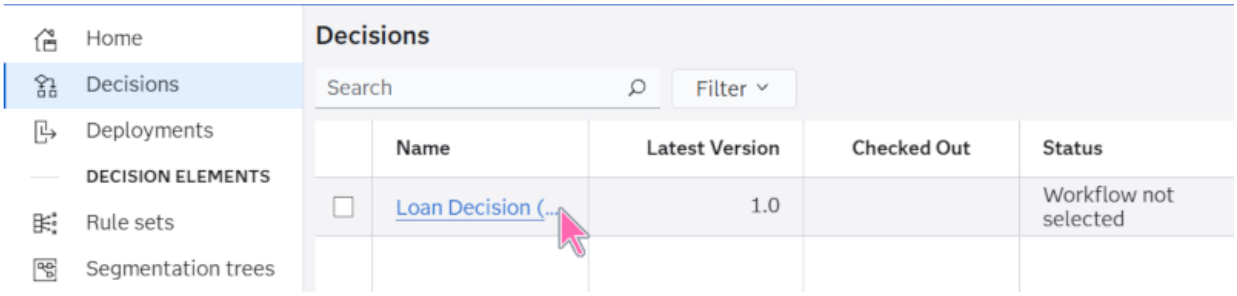


Figure 89: Open **Loan Decision (Final)**.

5. The **Decision Flow** tab has a visual representation of the decision flow.

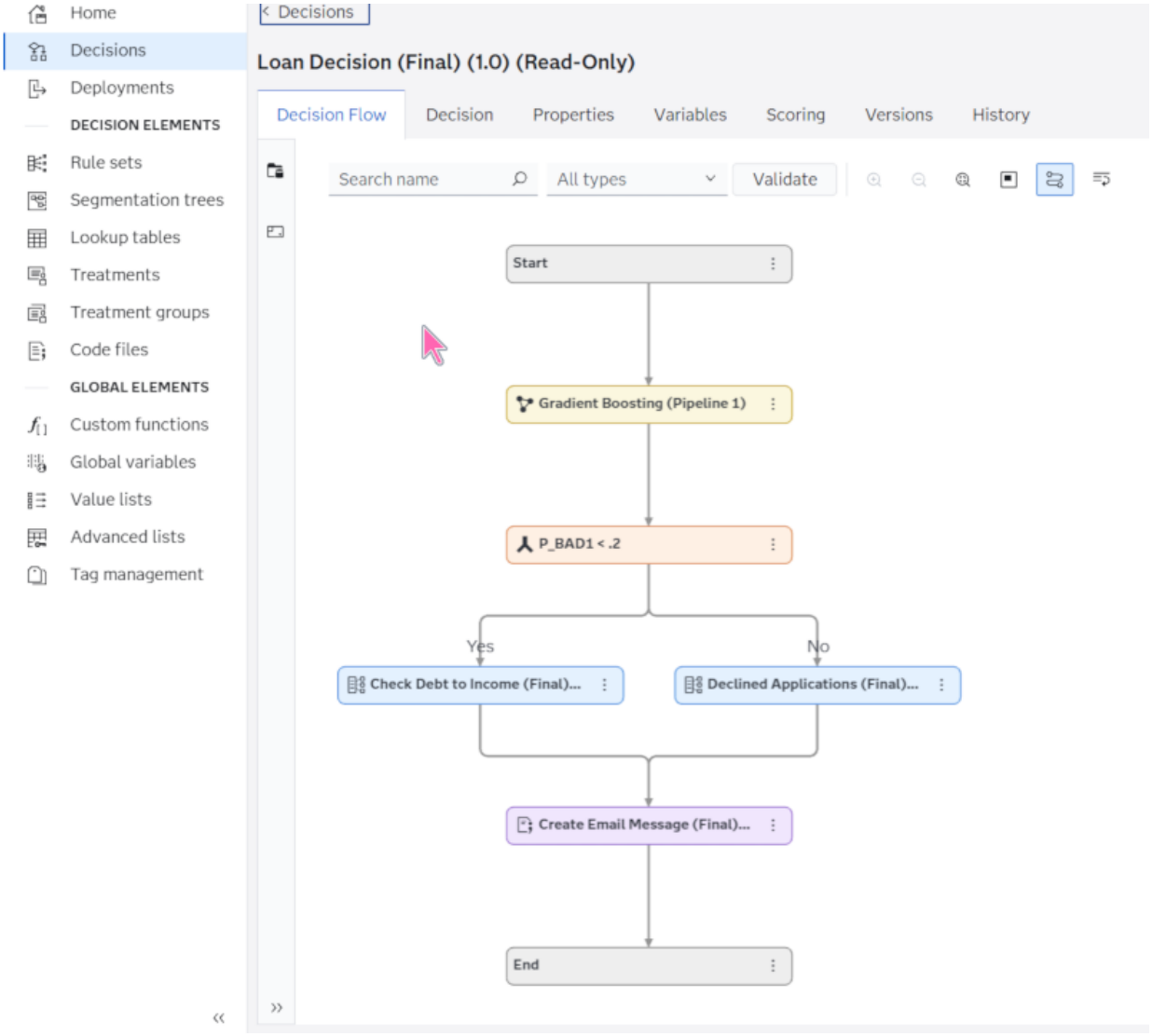


Figure 90: The decision flow.

6. The second node in the pipeline, called a model node, refers to a model that is registered with the name **Gradient Boosting (Pipeline 1)**.

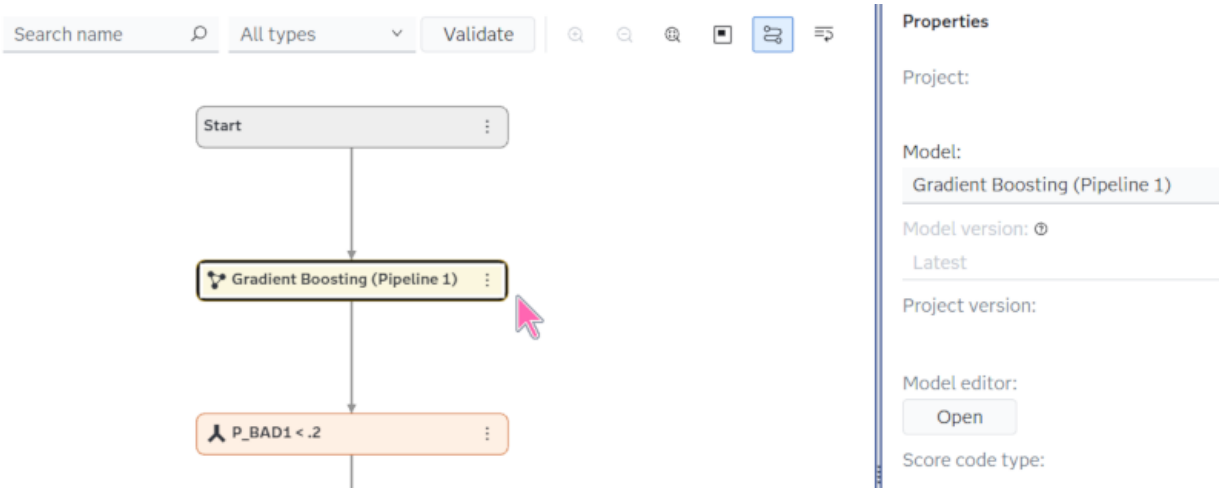


Figure 91: The model node.

7. The third node, named **P_BAD1<.2**, is a **Yes/No** branch node.
   The flow follows the **Yes** path, if the probability of default is less than 0.2.
   The **No** path is followed for the other cases.
8. The blue rule set node in the **No** branch has the name **Declined Application (Final)**.

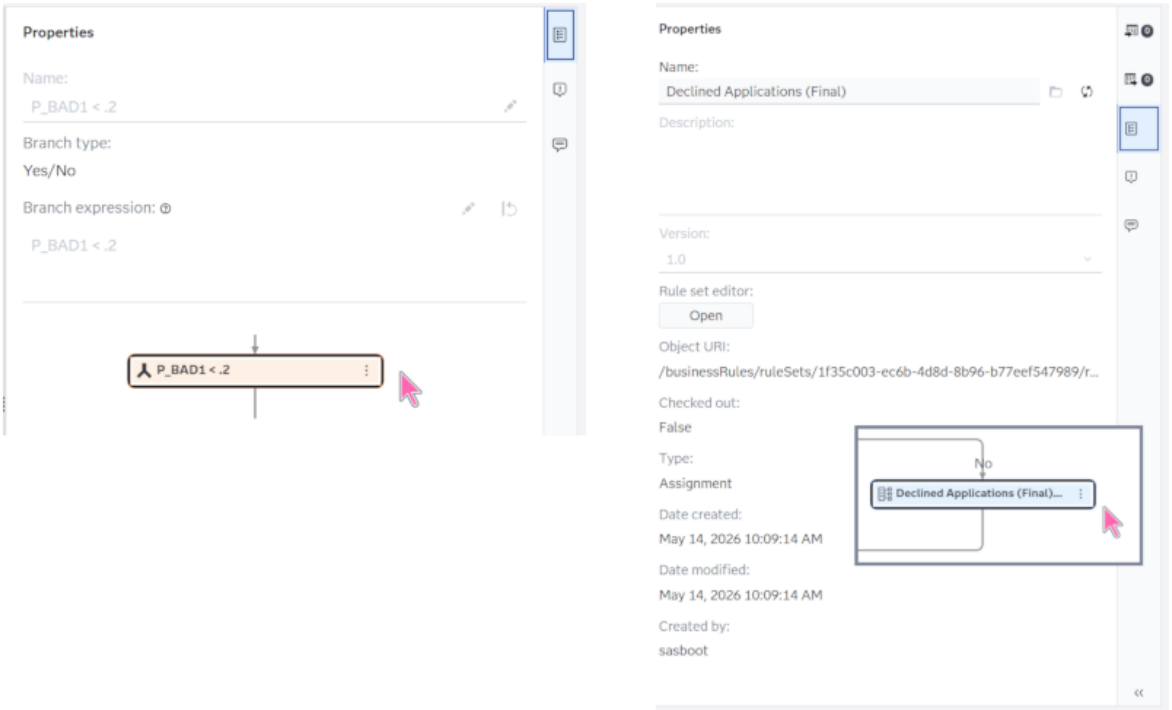


Figure 92: The branch and one rule set node.

9. The blue rule set node in the **Yes** branch has the name **Check Debt to Income (Final)**.
   NOTE: In the rule set for this node the debt to income ratio is checked and only when it is below 40 the loan is approved.
10. The node before the **End** node, is a **Data Query** node named: **Create Email Message (Final)**.

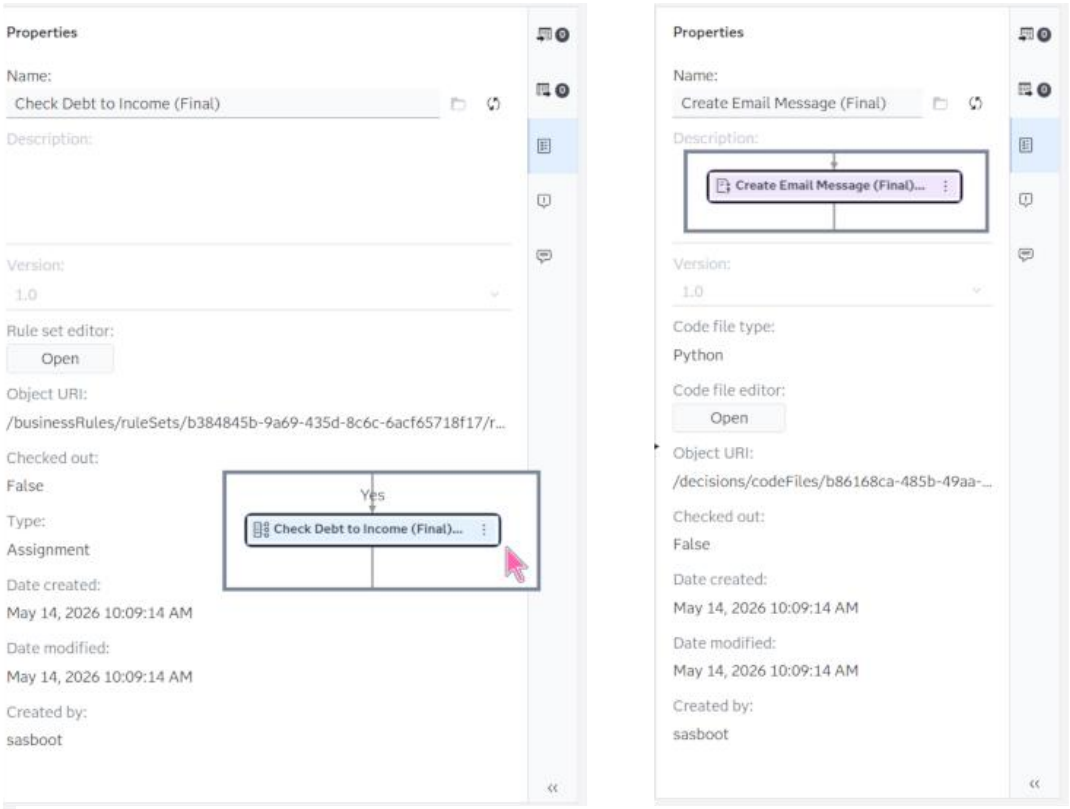


Figure 93: The rule set node for the Yes path and the **Data Query** *node.*

If you would like to practice the creation of a decision flow similar to the one that is shown here based on the model you created and registered, please send an email to emeagapsupport@sas.com with a request for step-by-step instructions referring to this document.

This completes the document. We hope you have enjoyed the learning opportunity. We intend to complete a workbook in the first week of December. It will expand on the use case by providing additional practices for each of the six tasks described here. If you are interested, please send an email to emeagapsupport@sas.com with a request for the workbook referring to this document.

The workbook will contain practices to:

1. Recreate the SAS program for importing and loading data using a low-code/no-code interface.
2. Add additional pages to the SAS Visual Analytics report that:
   a. Allow statistical analysis of variables in order to assist with feature engineering.
   b. Deploy a geographic map;
   c. Use SAS Visual Statistics objects to create and compare models.
3. Enrich and improve the models in SAS Model Studio.
4. Enrich information in SAS Information Catalog for the HOMEEQUITY dataset.
5. Enhance the information in the model card.
6. Create a decision flow using the champion model registered by you.

Appendix

Recommendations for self-study

	e-Learning course	Recommended audience	SAS Skill Builder for Students location
1	SAS Programming 1: Essentials	Anyone starting to write SAS programs.	» Learning Paths by Topic » SAS Programming » SAS Programming 1: Essentials
2	SAS Programming 2: Data Manipulation Techniques	Business analysts and SAS programmers. Together with Course 1, this course provides a foundation for SAS programming.	» Learning Paths by Topic » SAS Programming » SAS Programming 2: Data Manipulation Techniques
3	Managing and Querying Data Using Flows in SAS Studio	Professionals seeking to leverage SAS Studio for data access and preparation.	» Learning Paths by Topic » SAS Programming » Managing and Querying Data Using Flows in SAS Studio
4	SAS Visual Analytics 1 for SAS Viya: Basics	Business analysts who need an introduction to the functionality provided by SAS Visual Analytics.	» Learning Paths by Topic » Data Literacy and Data Visualization » SAS Visual Analytics 1 for SAS Viya: Basics
5	SAS Visual Analytics 2 for SAS Viya: Advanced	Business analysts who need to use advanced functionality provided by SAS Visual Analytics.	» Learning Paths by Topic » Data Literacy and Data Visualization » SAS Visual Analytics 2 for SAS Viya: Advanced
6	Responsible Innovation and Trustworthy AI	Data consumers, IT professionals, managers, analysts, data scientists, and anyone else who uses, designs, consumes information from, or makes decisions based on data and AI.	» Learning Paths by Topic » Statistical Analysis, Predictive Modeling and Machine Learning in SAS Viya » Responsible Innovation and Trustworthy AI
7	Statistics You Need to Know for Machine Learning	Anyone in the field of data science who does not yet have a deep understanding of statistical and machine learning concepts or wants to enhance their knowledge. We recommend this course to everyone who is interested in data science as a starting point for refreshing and extending their knowledge of statistics.	» Learning Paths by Topic » Statistical Analysis, Predictive Modeling and Machine Learning in SAS Viya » Statistics You Need to Know for Machine Learning
8	Machine Learning Using SAS Viya	Business analysts, data analysts, marketing analysts, marketing managers, data scientists, data engineers, financial analysts, data miners, statisticians, mathematicians, and others who work in related fields.	» Learning Paths by Topic » Statistical Analysis, Predictive Modeling and Machine Learning in SAS Viya » Machine Learning Using SAS Viya